

Minkowskian open/closed conformal field theory possibly without vacuum: the Cardy case

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1 Preliminaries on conformal nets

For any Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, we let $\mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ denote the set of bounded linear operators $\mathcal{H}_1 \rightarrow \mathcal{H}_2$, and abbreviate $\mathcal{L}(\mathcal{H}, \mathcal{H})$ as $\mathcal{L}(\mathcal{H})$. Similarly, we let $\mathcal{U}(\mathcal{H}_1, \mathcal{H}_2)$ be the set of unitary maps $\mathcal{H}_1 \rightarrow \mathcal{H}_2$ and write $\mathcal{U}(\mathcal{H}, \mathcal{H})$ as $\mathcal{U}(\mathcal{H})$. For each Hilbert space $\mathcal{H}$, we understand its inner product $\langle \cdot | \cdot \rangle$ to be linear on the right variable $|\cdot\rangle$ and antilinear on the left variable $\langle \cdot |$.

We write $X \Subset Y$ if $X \subset Y$, and if X has compact closure in Y .

1.1 Conformal nets

Let $\text{Diff}^+(\mathbb{S}^1)$ be the orientation-preserving diffeomorphism group of the unit circle $\mathbb{S}^1$. Let $\mathcal{J}$ be the set of (non-empty non-dense open) intervals in $\mathbb{S}^1$. If $I \in \mathcal{J}$, we let I' be the interior of $\mathbb{S}^1 \setminus I$. We let $\text{Diff}_I(\mathbb{S}^1)$ be the subgroup consisting of all $g \in \text{Diff}^+(\mathbb{S}^1)$ fixing points in the closure of I' .

Let $\text{PSU}(1, 1)$ be the subgroup of Möbius transforms preserving $\mathbb{S}^1$, i.e., $\text{PSU}(1, 1)$ consists of $g \in \text{Diff}^+(\mathbb{S}^1)$ of the form

$$z \in \mathbb{S}^1 \mapsto \frac{az + b}{\overline{b}z + \overline{a}}$$

where $a, b \in \mathbb{C}, |a|^2 - |b|^2 = 1$. Let

$$\varrho : \mathbb{R} \rightarrow \text{Diff}^+(\mathbb{S}^1) \quad \varrho(t)z = e^{it}z \tag{1.1}$$

(where $z \in \mathbb{S}^1$) be the one-parameter rotation group.

In this paper, all conformal nets are assumed to be irreducible, defined as follows.

Definition 1.1. An (irreducible non-trivial) **conformal net** denotes a tuple $(\mathcal{A}, \mathcal{H}_0, U_0, \Omega)$ (abbreviated to $(\mathcal{A}, \mathcal{H}_0)$ or simply $\mathcal{A}$) where $\mathcal{H}_0$ is a separable Hilbert space with $\dim \mathcal{H}_0 > 1$, and $\mathcal{A}$ is a map sending each $I \in \mathcal{J}$ to a von Neumann algebra $\mathcal{A}(I)$ on $\mathcal{H}_0$ satisfying the following conditions.

- (a) **(Isotony)** If $I_1 \subset I_2 \in \mathcal{J}$, then $\mathcal{A}(I_1)$ is a von Neumann subalgebra of $\mathcal{A}(I_2)$.
- (b) **(Locality)** If $I_1, I_2 \in \mathcal{J}$ are disjoint, then $[\mathcal{A}(I_1), \mathcal{A}(I_2)] = 0$.
- (c) **(Conformal covariance)** U_0 is a strongly-continuous projectively unitary representation of $\text{Diff}^+(\mathbb{S}^1)$ on $\mathcal{H}_0$ such that the relation

$$V\mathcal{A}(I)V^* = \mathcal{A}(gI)$$

holds for any $g \in \text{Diff}^+(\mathbb{S}^1)$, any $I \in \mathcal{J}$, any unitary $V \in \mathcal{U}(\mathcal{H}_0)$ representing $U_0(g)$. Moreover, if $g \in \text{Diff}_I(\mathbb{S}^1)$ and V represents $U_0(g)$, then $V \in \mathcal{A}(I)$.

- (d) $\Omega \in \mathcal{H}_0$, called the **vacuum vector**, is the unique (up to scalar multiplication) unit vector such that $V\Omega \in \mathbb{C}\Omega$ for each $V \in \mathcal{U}(\mathcal{H}_0)$ representing some $g \in \text{PSU}(1, 1)$. (In particular, U_0 restricts to a strongly-continuous unitary representation U_0 of $\text{PSU}(1, 1)$ on $\mathcal{H}$ satisfying $U_0(g)\Omega = \Omega$ for each $g \in \text{PSU}(1, 1)$.) Moreover, Ω is cyclic under the action of $\bigvee_{I \in \mathcal{J}} \mathcal{A}(I)$.
- (e) (Positive energy) The self-adjoint generator of the one-parameter unitary group $U_0 \circ \varrho$ has spectrum in $\mathbb{R}_{\geq 0}$.

Remark 1.2. A conformal net $\mathcal{A}$ satisfies the following properties, cf. [GL96, Prop. 1.1, 1.2] and the references therein.

- (1) **(Additivity)** $\mathcal{A}(I) = \bigvee_{\alpha} \mathcal{A}(I_{\alpha})$ if (I_{α}) is a collection in $\mathcal{J}$ whose union is $I \in \mathcal{J}$.
- (2) **(Haag duality)** $\mathcal{A}(I)' = \mathcal{A}(I')$ for each $I \in \mathcal{J}$.
- (3) **(Reeh–Schlieder property)** $\mathcal{A}(I)\Omega$ is dense in $\mathcal{H}_0$ for each $I \in \mathcal{J}$.
- (4) **(Irreducibility)** $\bigvee_{I \in \mathcal{J}} \mathcal{A}(I) = \mathfrak{L}(\mathcal{H}_0)$.
- (5) Each $\mathcal{A}(I)$ is a type III factor.

Moreover, by [MTW18], $\mathcal{A}$ satisfies the **split property**, which means that for each $I, J \in \mathcal{J}$ with $I \subseteq J$, there is a type I factor between $\mathcal{A}(I)$ and $\mathcal{A}(J)$. By Haag duality (applied to $I_1 = I$ and $I_2 = J'$), this is equivalent to the statement that for any $I_1, I_2 \in \mathcal{J}$ with disjoint closures, there is a unitary map $\Phi : \mathcal{H}_0 \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$ where the Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$ carry normal representations π_1, π_2 of $\mathcal{A}(I_1)$ and $\mathcal{A}(I_2)$, respectively, such that

$$\text{Ad}_{\Phi}|_{\mathcal{A}(I_1)} = \pi_1 \otimes \text{id}_{\mathcal{H}_2} \quad \text{Ad}_{\Phi}|_{\mathcal{A}(I_2)} = \text{id}_{\mathcal{H}_1} \otimes \pi_2$$

By property (5), one may in fact take $\mathcal{H}_1 = \mathcal{H}_2 = \mathcal{H}_0$, with π_1, π_2 given by the inclusion maps $\mathcal{A}(I_1) \hookrightarrow \mathfrak{L}(\mathcal{H}_0), \mathcal{A}(I_2) \hookrightarrow \mathfrak{L}(\mathcal{H}_0)$, respectively.

Next, we review some useful extensions of $\text{Diff}^+(\mathbb{S}^1)$ and $\text{Diff}_I(\mathbb{S}^1)$.

Definition 1.3. Let

$$\mathcal{G} := \widetilde{\text{Diff}^+(\mathbb{S}^1)} \rightarrow \text{Diff}^+(\mathbb{S}^1) \quad (1.2)$$

be the universal cover of $\text{Diff}^+(\mathbb{S}^1)$. Viewing $\mathbb{S}^1$ as $\mathbb{R}/2\pi\mathbb{Z}$, the group $\mathcal{G}$ consists of smooth functions $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for each $x \in \mathbb{R}$ that

$$F(x + 2\pi) = F(x) + 2\pi \quad F'(x) > 0 \quad (1.3)$$

Definition 1.4. For each $I \in \mathcal{J}$, let $\mathcal{G}(I)$ be the identity component of the preimage of $\text{Diff}_I(\mathbb{S}^1)$ under the map (1.2). Equivalently, $\mathcal{G}(I)$ consists of smooth functions $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying (1.3) and fixing pointwise preimage of I' under the covering map $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$. (In fact, by (1.3), it suffices to assume fixing pointwise any prescribed connected component of the preimage of I' .) See Rem. 1.9 for an alternative description.

Lift the rotation group (1.1) to a one-parameter group

$$\varrho : \mathbb{R} \rightarrow \mathcal{G}$$

Thus, for each $t \in \mathbb{R}$, the function $\varrho(t) : \mathbb{R} \rightarrow \mathbb{R}$ sends x to $x + t$.

The **central extension of $\mathcal{G}$ associated to $\mathcal{A}$** is defined to be the topological group

$$\mathcal{G}_{\mathcal{A}} = \{(g, V) \in \mathcal{G} \times \mathcal{U}(\mathcal{H}_0) : V \text{ represents } U_0(g)\}$$

or more precisely, the surjective group homomorphism $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$ defined by the projection $\mathcal{G} \times \mathcal{U}(\mathcal{H}_0) \rightarrow \mathcal{G}$. The kernel of this homomorphism is $1 \times \mathbb{S}^1 \simeq \mathbb{S}^1$.

Remark 1.5. The topological group $\mathcal{G}_{\mathcal{A}}$ depends only on the central charge of $\mathcal{A}$. Indeed, the net $\text{Vir}_{\mathcal{A}} : I \in \mathcal{J} \mapsto U_0(\text{Diff}_I(\mathbb{S}^1))''$ acting on $\mathcal{K}_0 := \overline{\text{Vir}_{\mathcal{A}}\Omega}$ is a conformal subnet of $\mathcal{A}$; in particular, the representation of $\widetilde{\text{PSU}}(1, 1)$ on $\mathcal{H}_0$ restricts to that on $\mathcal{K}_0$. Therefore, by the irreducibility in Rem. 1.2, $\mathcal{K}_0$ is irreducible as a strongly-continuous projectively-unitary positive-energy representation of $\text{Diff}^+(\mathbb{S}^1)$, and hence is integrated from the unitary module $L(c, 0)$ (for some $c \geq 0$, called the **central charge** of $\mathcal{A}$) of the Virasoro algebra, cf. [Car04, Thm. A.1]. Thus $\text{Vir}_{\mathcal{A}}$ can be identified with the Virasoro net Vir_c with central charge c , and $\mathcal{K}_0$ can be viewed as the Hilbert space completion of $L(c, 0)$. Set

$$\mathcal{G}_c := \mathcal{G}_{\text{Vir}_c}$$

Then the following map defines an isomorphism of topological groups $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$.

$$\mathcal{G}_{\mathcal{A}} \xrightarrow{\simeq} \mathcal{G}_c \quad (g, V) \mapsto (g, V|_{\mathcal{K}_0})$$

We will freely identify $\mathcal{G}_{\mathcal{A}}$ and $\mathcal{G}_c$ throughout the rest of this paper.

Definition 1.6. If $\tilde{g} = (g, V) \in \mathcal{G}_{\mathcal{A}}$, we write $V \in \mathcal{U}(\mathcal{H}_0)$ as $U_0(\tilde{g})$, that is,

$$\tilde{g} = (g, U_0(\tilde{g}))$$

Then U_0 is a strongly-continuous unitary representation of $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ on $\mathcal{H}_0$.

For each $I \in \mathcal{J}$, we let $\mathcal{G}_{\mathcal{A}}(I)$ be the preimage of $\mathcal{G}(I)$ under the map $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$. In particular, we let $\mathcal{G}_c(I) = \mathcal{G}_{\text{Vir}_c}(I)$. By the conformal covariance in Def. 1.1, we have

$$U_0(\mathcal{G}_{\mathcal{A}}(I)) \subset \mathcal{A}(I)$$

Remark 1.7. From the definition of conformal nets, we have a unitary representation U_0 of $\text{PSU}(1, 1)$ rather than merely a projectively unitary one. Thus, we have a one parameter unitary group

$$\varrho_{\mathcal{A}} : \mathbb{R} \rightarrow \mathcal{G}_{\mathcal{A}} \quad \varrho_{\mathcal{A}}(t) = (\varrho(t), U_0(\varrho(t)))$$

called the **rotation group in $\mathcal{G}_{\mathcal{A}}$** . We let ϱ_c denote the rotation group ϱ_{Vir_c} of Vir_c . The isomorphism $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ clearly identifies $\varrho_{\mathcal{A}}$ with ϱ_c .

1.2 Twisted modules of conformal nets

Fix a conformal net $\mathcal{A}$.

Definition 1.8. An **automorphism** of $\mathcal{A}$ denotes a unitary operator $\Phi \in \mathcal{U}(\mathcal{H}_0)$ fixing Ω and satisfying $\Phi \mathcal{A}(I) \Phi^{-1} = \mathcal{A}(I)$ for each $I \in \mathcal{J}$.

We fix a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$ where G is a discrete group, and $\text{Aut}(\mathcal{A})$ is the group of automorphisms of $\mathcal{A}$. We view each $\phi \in G$ as a unitary operator on $\mathcal{H}_0$. Then

$$\phi \mathcal{A}(I) \phi^{-1} = \mathcal{A}(I)$$

Hence G gives rise to a group of automorphisms of each $\mathcal{A}(I)$.

For each $I \in \mathcal{J}$, an **arg function** denotes a continuous function $\arg_I : I \rightarrow \mathbb{R}$ such that $z = e^{i \arg_I(z)}$ for all $z \in I$. An **arg-valued interval** denotes a pair

$$\tilde{I} = (I, \arg_I)$$

where $I \in \mathcal{J}$, and $\arg_I$ is an arg function of I . The set of arg-valued intervals is denoted by $\tilde{\mathcal{J}}$.

Equivalently, $\tilde{I}$ is a connected component of the preimage of I under the covering map $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z} \simeq \mathbb{S}^1$. Therefore, since the universal cover $\mathcal{G}$ acts on $\mathbb{R}$ (cf. (1.3)), it also acts on $\tilde{\mathcal{J}}$. Through the quotient homomorphism $\mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{G}$, the group $\mathcal{G}_{\mathcal{A}} \simeq \mathcal{G}_c$ also acts on $\mathbb{S}^1$, $\mathbb{R}$, and $\tilde{\mathcal{J}}$.

Remark 1.9. Def. 1.4 can be rephrased as follows: For each $I \in \mathcal{J}$,

$$\begin{aligned} \mathcal{G}(I) &= \{g \in \mathcal{G} : g \text{ fixes pointwise } (I', \arg_{I'}) \text{ for each } \arg_{I'}\} \\ &= \{g \in \mathcal{G} : g \text{ fixes pointwise } (I', \arg_{I'}) \text{ for some } \arg_{I'}\} \end{aligned}$$

Therefore,

$$\begin{aligned} \mathcal{G}_{\mathcal{A}}(I) &= \{g \in \mathcal{G}_{\mathcal{A}} : g \text{ fixes pointwise } (I', \arg_{I'}) \text{ for each } \arg_{I'}\} \\ &= \{g \in \mathcal{G}_{\mathcal{A}} : g \text{ fixes pointwise } (I', \arg_{I'}) \text{ for some } \arg_{I'}\} \end{aligned}$$

If $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$, we write $\tilde{I} \subset \tilde{J}$ if $I \subset J$ and $\arg_J|_I = \arg_I$. We say that $\tilde{I}$ and $\tilde{J}$ are **disjoint** if $I \cap J = \emptyset$. (Note that this is not the same as saying that $\tilde{I}$ and $\tilde{J}$ are disjoint as subsets of $\mathbb{R}$.) We say that $\tilde{J}$ is **clockwise** to $\tilde{I}$ (equivalently, that $\tilde{I}$ is **anticlockwise** to $\tilde{J}$) if

$$\arg \zeta < \arg z < \arg \zeta + 2\pi \quad \text{for each } z \in I, \zeta \in J$$

Definition 1.10. The **clockwise complement** $\tilde{I}'$ (resp. the **anticlockwise complement** $\tilde{I}''$) of $\tilde{I} \in \tilde{\mathcal{J}}$ is defined to be the interval I' together with the $\arg$ function so that $\tilde{I}'$ is clockwise to $\tilde{I}$ (resp. $\tilde{I}''$ is anticlockwise to $\tilde{I}$).

Definition 1.11. Let $\phi \in G$. A ϕ -**twisted $\mathcal{A}$ -module** denotes a pair $(\mathcal{H}_i, \pi_i)$ (or simply $\mathcal{H}_i$) where $\mathcal{H}_i$ is a separable Hilbert space, and π_i associates to each $\tilde{I} \in \tilde{\mathcal{J}}$ a normal representation $\pi_{i,\tilde{I}}$ of $\mathcal{A}(I)$ on $\mathcal{H}_i$ satisfying the following properties.

- (a) If $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ and $\tilde{I} \subset \tilde{J}$, then $\pi_{i,\tilde{I}}(x) = \pi_{i,\tilde{J}}(x)$ for each $x \in \mathcal{A}(I)$.
- (b) For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$, we have

$$\pi_{i,\varrho(2\pi)\tilde{I}}(\phi x \phi^{-1}) = \pi_{i,\tilde{I}}(x) \quad (1.4)$$

If condition (b) is dropped, we say that $(\mathcal{H}_i, \pi_i)$ is a **solitonic $\mathcal{A}$ -module**.

Remark 1.12. The above definition of a ϕ -twisted module follows [MS26a, Sec. 3.1], except that the unit circle used there is obtained from ours by reflection across the x -axis. Consequently, the convention in [MS26a, MS26b] uses $\varrho(-2\pi)$ in place of $\varrho(2\pi)$ in (1.4).

Definition 1.13. We let $\mathbf{Rep}^\phi(\mathcal{A})$ be the W^* -category of ϕ -twisted $\mathcal{A}$ -modules. For the identity element e of G we write $\mathbf{Rep}^e(\mathcal{A})$ as $\mathbf{Rep}(\mathcal{A})$. Objects in $\mathbf{Rep}(\mathcal{A})$ are called **untwisted $\mathcal{A}$ -modules**, or simply **$\mathcal{A}$ -modules**.

Remark 1.14. Let $(\mathcal{H}_i, \pi_i)$ be a solitonic $\mathcal{A}$ -module. Assume that $\tilde{J}$ is clockwise to $\tilde{I}$. Then

$$[\pi_{i,\tilde{I}}(\mathcal{A}(I)), \pi_{i,\tilde{J}}(\mathcal{A}(J))] = 0$$

Proof. By the additivity of $\mathcal{A}$ and the normality of $\pi_{i,\tilde{I}}, \pi_{i,\tilde{J}}$, it suffices to treat the case that $\tilde{I}, \tilde{J} \subset \tilde{K}$ for some $\tilde{K} \in \tilde{\mathcal{J}}$. Then the commutativity follows by the locality of $\mathcal{A}$ and part (a) of Def. 1.11. $\square$

Example 1.15. The action of $\mathcal{A}$ on $\mathcal{H}_0$ is clearly an untwisted $\mathcal{A}$ -module, called the **vacuum module** and is denoted by $(\mathcal{H}_0, \pi_0)$ or simply by $\mathcal{H}_0$.

Remark 1.16. Suppose that $x \in \mathcal{A}(I)$ commutes with $\phi \in G$. Then by Def. 1.11-(b), $\pi_{i,\tilde{I}}(x)$ is independent of $\arg_I$. We thus write $\pi_{i,\tilde{I}}(x)$ as $\pi_{i,I}(x)$ in this case.

For example, if $g \in \mathcal{G}_A$, then $U_0(g)$ commutes with any $\phi \in G$; cf. [MS26a, Rem. 3.14]. Hence, if $g \in \mathcal{G}_A(I)$, we can

$$\text{write } \pi_{i,\tilde{I}}(U_0(g)) \text{ as } \pi_{i,I}(U_0(g))$$

As another example, if $(\mathcal{H}_i, \pi_i) \in \text{Rep}(\mathcal{A})$, then $\pi_{i,\tilde{I}}(x)$ can be written as $\pi_{i,I}(x)$ for each $x \in \mathcal{A}(I)$. $\square$

Theorem 1.17. Any ϕ -twisted $\mathcal{A}$ -module $(\mathcal{H}_i, \pi_i)$ is **conformally covariant**, which means that there is a unique strongly-continuous unitary representation U_i of $\mathcal{G}_A \simeq \mathcal{G}_c$ on $\mathcal{H}_i$ satisfying for each $\tilde{I} \in \tilde{\mathcal{J}}, g \in \mathcal{G}_A(I)$ the relation

$$U_i(g) = \pi_{i,I}(U_0(g)) \quad (1.5)$$

Moreover, for each $g \in \mathcal{G}_A$, we have $U_i(g) \in \bigvee_{\tilde{I} \in \tilde{\mathcal{J}}} \pi_{i,\tilde{I}}(\mathcal{A}(I))$.

Proof. This follows from [Hen19], as explained in [Gui21a, Thm. 2.2] (applied to the untwisted Vir_c -module $(\mathcal{H}_i, \pi_i|_{\text{Vir}_c})$) or in [MS26a, Thm. 3.15]. In particular, the uniqueness follows from the following fact. $\square$

Lemma 1.18. Let $\mathcal{I}$ be a collection of intervals covering $\mathbb{S}^1$. Then $\bigcup_{I \in \mathcal{I}} \mathcal{G}(I)$ generates algebraically the group $\mathcal{G}$. Consequently, $\bigcup_{I \in \mathcal{I}} \mathcal{G}_A(I)$ generates algebraically the group $\mathcal{G}_A$.

Proof. This is due to [Hen19, Lem. 17]. $\square$

Remark 1.19. We abbreviate $U_i(g)$ to $U(g)$ or even g when no confusion arises.

Corollary 1.20. For each ϕ -twisted $\mathcal{A}$ -module $\mathcal{H}_i$, $\tilde{I} \in \tilde{\mathcal{J}}, x \in \mathcal{A}(I)$, and $g \in \mathcal{G}_A$, we have

$$U_i(g)\pi_{i,\tilde{I}}(x)U_i(g)^{-1} = \pi_{i,\tilde{I}}(U_0(g)xU_0(g)^{-1}) \quad (1.6)$$

Proof. See [MS26a, Cor. 3.16]. The idea is to check (1.6) using Thm. 1.17 when g is supported in any interval whose union with I is not dense, and then conclude the general case by using Lem. 1.18. $\square$

1.3 The category $\text{Rep}^G(\mathcal{A})$ and the G -action

Fix a conformal net $\mathcal{A}$ with central charge c , a discrete group G , and a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$. Let

$$\text{Rep}^G(\mathcal{A})$$

be the W^* -category whose objects are finite (orthogonal) direct sums of the form $\bigoplus_i \mathcal{H}_i$ where each $\mathcal{H}_i \in \text{Rep}^{\omega_i}(\mathcal{A})$ for some $\omega_i \in G$. Objects in $\text{Rep}^G(\mathcal{A})$ are called **G -twisted $\mathcal{A}$ -modules**. The morphisms in $\text{Rep}^G(\mathcal{A})$ are defined in Def. 1.21.

Definition 1.21. If $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, we let $\mathbf{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$ denote G -twisted $\mathcal{A}$ -module morphisms from $\mathcal{H}_i$ to $\mathcal{H}_j$, that is, elements $T \in \mathcal{L}(\mathcal{H}_i, \mathcal{H}_j)$ satisfying

$$T\pi_{\tilde{I},i}(x) = \pi_{\tilde{I},j}(x)T$$

for each $\tilde{I} \in \tilde{\mathcal{J}}, x \in \mathcal{A}(I)$. Elements of $\mathbf{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$ are called **(homo)morphisms of G -twisted $\mathcal{A}$ -modules** from $\mathcal{H}_i$ to $\mathcal{H}_j$.

Convention 1.22. In this paper, objects of $\text{Rep}^G(\mathcal{A})$ are denoted by symbols such as $\mathcal{H}_i, \mathcal{H}_j, \mathcal{H}_k$, etc. We abbreviate $\pi_{i,\tilde{I}}$ to $\pi_{\tilde{I}}$ when no confusion arises.

Definition 1.23. For each $\phi \in G$ and each G -twisted $\mathcal{A}$ -module $(\mathcal{H}_i, \pi_i)$, we define a G -twisted $\mathcal{A}$ -module

$$(\mathcal{H}_{\phi i}, \pi_{\phi i})$$

abbreviated to $\mathcal{H}_{\phi i}$, where $\mathcal{H}_{\phi i}$ is an abstract Hilbert space unitarily equivalent to $\mathcal{H}_i$ via a prescribed unitary map

$$\Gamma_{\phi}|_{\mathcal{H}_i} : \mathcal{H}_i \xrightarrow{\sim} \mathcal{H}_{\phi i}$$

and $\pi_{\phi i}$ is defined such that for each $\tilde{I} \in \tilde{\mathcal{J}}, x \in \mathcal{A}(I)$, we have

$$\Gamma_{\phi}|_{\mathcal{H}_i} \circ \pi_{i,\tilde{I}}(x) = \pi_{\phi i,\tilde{I}}(\phi x \phi^{-1}) \circ \Gamma_{\phi}|_{\mathcal{H}_i}$$

We abbreviate $\Gamma_{\phi}|_{\mathcal{H}_i}$ to Γ_{ϕ} when no confusion arises. Then the above relation can be abbreviated to

$$\Gamma_{\phi} \circ \pi_{\tilde{I}}(x) = \pi_{\tilde{I}}(\phi x \phi^{-1}) \circ \Gamma_{\phi} \quad (1.7)$$

For each $\omega \in G$, it is clear that $\mathcal{H}_i \in \text{Rep}^{\omega}(\mathcal{A})$ implies $\mathcal{H}_{\phi i} \in \text{Rep}^{\phi\omega\phi^{-1}}(\mathcal{A})$.

Remark 1.24. It is obvious that the pair $(\mathcal{H}_{\phi i}, \Gamma_{\phi}|_{\mathcal{H}_i})$ is uniquely determined by $\mathcal{H}_i$ and ϕ up to unique unitary maps. That is, if $(\tilde{\mathcal{H}}_{\phi i}, \tilde{\Gamma}_{\phi}|_{\mathcal{H}_i})$ satisfies the same property, then there is a unique unitary map (indeed, a unique G -twisted $\mathcal{A}$ -module morphism) $\Psi : \tilde{\mathcal{H}}_{\phi i} \rightarrow \mathcal{H}_{\phi i}$ such that $\Gamma_{\phi}|_{\mathcal{H}_i} = \Psi \circ \tilde{\Gamma}_{\phi}|_{\mathcal{H}_i}$. We call Ψ the (unique) **unitary morphism** $(\tilde{\mathcal{H}}_{\phi i}, \tilde{\Gamma}_{\phi}|_{\mathcal{H}_i}) \rightarrow (\mathcal{H}_{\phi i}, \Gamma_{\phi}|_{\mathcal{H}_i})$.

Example 1.25. For the vacuum module $\mathcal{H}_0$, one can choose $(\tilde{\mathcal{H}}_{\phi 0}, \tilde{\Gamma}_{\phi}|_{\mathcal{H}_0})$ to be $(\mathcal{H}_0, \phi)$ where ϕ is understood as in $\mathcal{U}(\mathcal{H}_0)$. The unitary morphism $(\mathcal{H}_0, \phi) \rightarrow (\mathcal{H}_{\phi 0}, \Gamma_{\phi}|_{\mathcal{H}_0})$ is given by $\Gamma_{\phi}|_{\mathcal{H}_0} \circ \phi^{-1}$.

Remark 1.26. Recall from Rem. 1.16 that Ad_{ϕ} fixes $U_0(g)$ for each $I \in \mathcal{J}$ and $g \in \mathcal{G}_{\mathcal{A}}(I)$. Therefore, by (1.7), $\text{Ad}_{\Gamma_{\phi}}$ sends $\pi_{i,I}(U_0(g))$ to $\pi_{\phi i,I}(U_0(g))$. Thus, by the uniqueness in Thm. 1.17, we have

$$\Gamma_{\phi} \circ U_i(g) = U_{\phi i}(g) \circ \Gamma_{\phi} \quad (1.8)$$

for each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $g \in \mathcal{G}_{\mathcal{A}}$.

Definition 1.27. For each $\phi \in G$, define a $*$ -functor $\mathfrak{T}_\phi : \text{Rep}^G(\mathcal{A}) \rightarrow \text{Rep}^G(\mathcal{A})$ such that

$$\mathfrak{T}_\phi(\mathcal{H}_i) = \mathcal{H}_{\phi i} \quad \mathfrak{T}_\phi(S) = (\Gamma_\phi|_{\mathcal{H}_j}) \circ S \circ (\Gamma_\phi|_{\mathcal{H}_i})^{-1}$$

for each $S \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$. This functor is denoted by T_ϕ in [MS26a].

The following theorem is proved in [MS26a, Sec. 3.6]. Note that, by Rem. 1.12, our balancing differs from that in [MS26a, MS26b]: the latter is defined using $\varrho_{\mathcal{A}}(-2\pi)$, whereas ours is defined using $\varrho_{\mathcal{A}}(2\pi)$.

Theorem 1.28. *Suppose that $G \rightarrow \text{Aut}(\mathcal{A})$ is faithful. Then the W^* -category $\text{Rep}^G(\mathcal{A})$ with grading $\bigoplus_{\phi \in G} \text{Rep}^\phi(\mathcal{A})$ is canonically a G -crossed balanced W^* -tensor category defined by Connes fusion, where the action of G on $\text{Rep}^G(\mathcal{A})$ is defined by $\phi \in G \mapsto \mathfrak{T}_\phi$, and the balancing ϑ is defined by*

$$\vartheta_i = \Gamma_\phi \circ U_i(\varrho_{\mathcal{A}}(2\pi)) : \mathcal{H}_i \rightarrow \mathcal{H}_{\phi i}$$

for each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$.

When $G \rightarrow \text{Aut}(\mathcal{A})$ is not faithful, then $\text{Rep}^G(\mathcal{A})$ does not admit the grading $\bigoplus_{\phi \in G} \text{Rep}^\phi(\mathcal{A})$, since $\text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$ might not be zero for $\mathcal{H}_i \in \text{Rep}^\omega(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\phi(\mathcal{A})$ even when $\omega \neq \phi$ in G . In that case, to apply Thm. 1.28, it suffices to replace G with its image in $\text{Aut}(\mathcal{A})$.

Convention 1.29. Let $\boxtimes$ denote the tensor product bifunctor of $\text{Rep}^G(\mathcal{A})$, and call it the **fusion product** bifunctor. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, we write $\mathcal{H}_{i \boxtimes j} := \mathcal{H}_i \boxtimes \mathcal{H}_j$. According to the definition of G -crossed tensor categories, we have

$$\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A}), \mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A}) \implies \mathcal{H}_i \boxtimes \mathcal{H}_j \in \text{Rep}^{\phi\omega}(\mathcal{A}) \quad (1.9)$$

The braiding operation is denoted by $\mathbb{B}$. More precisely, for each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, the braiding operator (which is a unitary equivalence of G -twisted $\mathcal{A}$ -modules) is denoted by

$$\mathbb{B}_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_i$$

By the coherence conditions for the W^* -tensor category $\text{Rep}^G(\mathcal{A})$, we make the identification

$$(\mathcal{H}_i \boxtimes \mathcal{H}_j) \boxtimes \mathcal{H}_k = \mathcal{H}_i \boxtimes (\mathcal{H}_j \boxtimes \mathcal{H}_k)$$

via the associator, and denote them by $\mathcal{H}_i \boxtimes \mathcal{H}_j \boxtimes \mathcal{H}_k \equiv \mathcal{H}_{i \boxtimes j \boxtimes k}$; we identify

$$\mathcal{H}_0 \boxtimes \mathcal{H}_i = \mathcal{H}_i = \mathcal{H}_i \boxtimes \mathcal{H}_0$$

using the unitors. In other words, we treat $\text{Rep}^G(\mathcal{A})$ as if it is strict. □

Remark 1.30. We describe more explicitly the action of G on $\text{Rep}^G(\mathcal{A})$, which contains the following data besides the association $\phi \mapsto \mathfrak{T}_\phi$. Recall that e is the unit of G

- (a) A unitary isomorphism $\mathbf{A}_i : \mathcal{H}_i \simeq \mathcal{H}_{ei}$ where $e \in G$ is the group identity. This is given by the unique unitary morphism $(\mathcal{H}_i, \text{id}) \rightarrow (\mathcal{H}_{ei}, \Gamma_e|_{\mathcal{H}_i})$. So

$$\mathbf{A}_i = \Gamma_e|_{\mathcal{H}_i}$$

The equivalence $\mathbf{A} : \text{id}_{\text{Rep}^G(\mathcal{A})} \rightarrow \mathfrak{T}_e$ is natural.

$$\begin{array}{ccc} \mathcal{H}_i & \xrightarrow{T} & \mathcal{H}_j \\ \mathbf{A}_i \downarrow & & \downarrow \mathbf{A}_j \\ \mathcal{H}_{ei} & \xrightarrow{\mathfrak{T}_e(T)} & \mathcal{H}_{ej} \end{array}$$

for each $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$, which is straightforward to check.

- (b) A unitary isomorphism $\mathbf{B}_\phi : \mathcal{H}_0 \rightarrow \mathcal{H}_{\phi 0}$ for each $\phi \in G$, which is given by the unique unitary morphism $(\mathcal{H}_0, \phi) \rightarrow (\mathcal{H}_{\phi 0}, \Gamma_\phi|_{\mathcal{H}_0})$ discussed in Exp. 1.25. So

$$\mathbf{B}_\phi = \Gamma_\phi \circ \phi^{-1}|_{\mathcal{H}_0}$$

- (c) A unitary natural isomorphism $\mathfrak{T}_\phi \circ \mathfrak{T}_\psi \rightarrow \mathfrak{T}_{\phi\psi}$ for $\phi, \psi \in G$, where, for each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, the unitary equivalence

$$\mathbf{C}_{\phi, \psi, i} : \mathcal{H}_{\phi(\psi i)} \rightarrow \mathcal{H}_{(\phi\psi)i} \quad \mathbf{C}_{\phi, \psi, i} = \Gamma_{\phi\psi} \Gamma_\psi^{-1} \Gamma_\phi^{-1}|_{\mathcal{H}_{\phi(\psi i)}}$$

is given by the unique unitary morphism $(\mathcal{H}_{\phi(\psi i)}, \Gamma_\phi \circ \Gamma_\psi|_{\mathcal{H}_i}) \rightarrow (\mathcal{H}_{(\phi\psi)i}, \Gamma_{\phi\psi}|_{\mathcal{H}_i})$.

The naturality means that for each $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_j)$,

$$\begin{array}{ccc} \mathcal{H}_{\phi(\psi i)} & \xrightarrow{\mathfrak{T}_\phi \circ \mathfrak{T}_\psi(T)} & \mathcal{H}_{\phi(\psi j)} \\ \mathbf{C}_{\phi, \psi, i} \downarrow & & \downarrow \mathbf{C}_{\phi, \psi, j} \\ \mathcal{H}_{(\phi\psi)i} & \xrightarrow{\mathfrak{T}_{\phi\psi}(T)} & \mathcal{H}_{(\phi\psi)j} \end{array} \quad (1.10a)$$

Moreover, the tuple $(\mathfrak{T}, \mathbf{C}, \mathbf{A})$ is a tensor functor, i.e., $\mathbf{C}$ is a tensorator and $\mathbf{A}$ is a unitor. This means the commutativity of

$$\begin{array}{ccc} \mathcal{H}_{\phi(\psi(\omega i))} & \xrightarrow{\mathfrak{T}_\phi(\mathbf{C}_{\psi, \omega, i})} & \mathcal{H}_{\phi((\psi\omega)i)} \\ \mathbf{C}_{\phi, \psi, \omega i} \downarrow & & \downarrow \mathbf{C}_{\phi, \psi\omega, i} \\ \mathcal{H}_{(\phi\psi)(\omega i)} & \xrightarrow{\mathbf{C}_{\phi\psi, \omega, i}} & \mathcal{H}_{(\phi\psi\omega)i} \end{array} \quad (1.10b)$$

$$\begin{array}{ccc}
\mathcal{H}_{e(\phi i)} & \xrightarrow{\mathbf{C}_{e,\phi,i}} & \mathcal{H}_{\phi i} \\
& \xleftarrow{\mathbf{A}_{\phi i}} & \\
\mathcal{H}_{\phi(ei)} & \xrightarrow{\mathbf{C}_{\phi,e,i}} & \mathcal{H}_{\phi i} \\
& \xleftarrow{\mathfrak{T}_{\phi}(\mathbf{A}_i)} &
\end{array} \quad (1.10c)$$

The commutativity of these diagrams is easy to check.¹

- (d) A unitary (natural) tensorator $\mathfrak{T}_{\phi}(-) \boxtimes \mathfrak{T}_{\phi}(-) \rightarrow \mathfrak{T}_{\phi}(- \boxtimes -)$ for each $\phi \in G$, where, for each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, the unitary equivalence

$$\mathbf{D}_{\phi,i,j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$$

will be described in Def. 1.39. Moreover, $\mathbf{B}_{\phi}$ is a unitor. (So $(\mathfrak{T}_{\phi}, \mathbf{D}_{\phi}, \mathbf{B}_{\phi})$ gives a W^* -tensor automorphism of $\text{Rep}^G(\mathcal{A})$.)

Unraveling the phrases “natural”, “tensorator”, and “unitor” gives the following commutative diagrams, where $S \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_{\tilde{i}})$ and $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_j, \mathcal{H}_{\tilde{j}})$, and $\mathbf{u}_{0,\phi i} : \mathcal{H}_0 \boxtimes \mathcal{H}_{\phi i} \rightarrow \mathcal{H}_{\phi i}$, $\mathbf{u}_{0,i} : \mathcal{H}_0 \boxtimes \mathcal{H}_i \rightarrow \mathcal{H}_i$, $\mathbf{u}_{\phi i,0} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_0 \rightarrow \mathcal{H}_{\phi i}$, and $\mathbf{u}_{i,0} : \mathcal{H}_i \boxtimes \mathcal{H}_0 \rightarrow \mathcal{H}_i$ are the unitors.

$$\begin{array}{ccc}
\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} & \xrightarrow{\mathfrak{T}_{\phi}(S) \boxtimes \mathfrak{T}_{\phi}(T)} & \mathcal{H}_{\tilde{\phi i}} \boxtimes \mathcal{H}_{\tilde{\phi j}} \\
\mathbf{D}_{\phi,i,j} \downarrow & & \downarrow \mathbf{D}_{\phi,\tilde{i},\tilde{j}} \\
\mathcal{H}_{\phi(i \boxtimes j)} & \xrightarrow{\mathfrak{T}_{\phi}(S \boxtimes T)} & \mathcal{H}_{\phi(\tilde{i} \boxtimes \tilde{j})}
\end{array} \quad (1.11a)$$

$$\begin{array}{ccc}
\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_{\phi k} & \xrightarrow{1 \boxtimes \mathbf{D}_{\phi,j,k}} & \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi(j \boxtimes k)} \\
\mathbf{D}_{\phi,i,j} \boxtimes 1 \downarrow & & \downarrow \mathbf{D}_{\phi,i,j \boxtimes k} \\
\mathcal{H}_{\phi(i \boxtimes j)} \boxtimes \mathcal{H}_{\phi k} & \xrightarrow{\mathbf{D}_{\phi,i \boxtimes j,k}} & \mathcal{H}_{\phi(i \boxtimes j \boxtimes k)}
\end{array} \quad (1.11b)$$

$$\begin{array}{ccc}
\mathcal{H}_0 \boxtimes \mathcal{H}_{\phi i} & \xrightarrow{\mathbf{u}_{0,\phi i}} & \mathcal{H}_{\phi i} \\
\mathbf{B}_{\phi} \boxtimes 1 \downarrow & & \uparrow \mathfrak{T}_{\phi}(\mathbf{u}_{0,i}) \\
\mathcal{H}_{\phi 0} \boxtimes \mathcal{H}_{\phi i} & \xrightarrow{\mathbf{D}_{\phi,0,i}} & \mathcal{H}_{\phi(0 \boxtimes i)}
\end{array} \quad (1.11c)$$

$$\begin{array}{ccc}
\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_0 & \xrightarrow{\mathbf{u}_{\phi i,0}} & \mathcal{H}_{\phi i} \\
1 \boxtimes \mathbf{B}_{\phi} \downarrow & & \uparrow \mathfrak{T}_{\phi}(\mathbf{u}_{i,0}) \\
\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi 0} & \xrightarrow{\mathbf{D}_{\phi,i,0}} & \mathcal{H}_{\phi(i \boxtimes 0)}
\end{array} \quad (1.11d)$$

¹In particular, in the first two diagrams, going from the upper-left corner to the lower-right corner along either path yields, respectively, $\Gamma_{\phi\psi} T \Gamma_{\psi}^{-1} \Gamma_{\phi}^{-1}$ and $\Gamma_{\phi\psi\omega} \Gamma_{\omega}^{-1} \Gamma_{\psi}^{-1} \Gamma_{\phi}^{-1}$.

- (e) The natural isomorphism $\mathbf{A} : \text{id}_{\text{Rep}^G(\mathcal{A})} \rightarrow \mathfrak{T}_e$ is tensor natural if the tensorator and unitor of $\text{id}_{\text{Rep}^G(\mathcal{A})}$ are chosen to be the identity, and those of $\mathfrak{T}_e$ are chosen as in (d). This means that the following diagrams commute:

$$\begin{array}{ccc}
 \mathcal{H}_i \boxtimes \mathcal{H}_j & \xrightarrow{\mathbf{A}_i \boxtimes \mathbf{A}_j} & \mathcal{H}_{ei} \boxtimes \mathcal{H}_{ej} \\
 \downarrow = & & \downarrow \mathbf{D}_{e,i,j} \\
 \mathcal{H}_{i \boxtimes j} & \xrightarrow{\mathbf{A}_{i \boxtimes j}} & \mathcal{H}_{e(i \boxtimes j)}
 \end{array} \tag{1.12a}$$

$$\begin{array}{ccc}
 & \xrightarrow{\mathbf{B}_e} & \\
 \mathcal{H}_0 & & \mathcal{H}_{e0} \\
 & \xleftarrow{\mathbf{A}_0} &
 \end{array} \tag{1.12b}$$

- (f) The natural isomorphism $\mathbf{C}_{\phi,\psi} : \mathfrak{T}_\phi \circ \mathfrak{T}_\psi \rightarrow \mathfrak{T}_{\phi \circ \psi}$ is tensor natural if the tensorators and the unitors of $\mathfrak{T}_\phi$, $\mathfrak{T}_\psi$, $\mathfrak{T}_{\phi \circ \psi}$ are chosen as in (d). (In particular, the tensorator and unitor of $\mathfrak{T}_\phi \circ \mathfrak{T}_\psi$ are $\mathfrak{T}_\phi(\mathbf{D}_\psi) \circ \mathbf{D}_\phi$ and $\mathfrak{T}_\phi(\mathbf{B}_\psi) \circ \mathbf{B}_\phi$.) This means that the following diagrams commute:

$$\begin{array}{ccc}
 \mathcal{H}_{\phi(\psi i)} \boxtimes \mathcal{H}_{\phi(\psi j)} & \xrightarrow{\mathbf{C}_{\phi,\psi,i} \boxtimes \mathbf{C}_{\phi,\psi,j}} & \mathcal{H}_{(\phi\psi)i} \boxtimes \mathcal{H}_{(\phi\psi)j} \\
 \downarrow \mathbf{D}_{\phi,\psi,i,\psi j} & & \downarrow \mathbf{D}_{\phi\psi,i,j} \\
 \mathcal{H}_{\phi(\psi i \boxtimes \psi j)} & & \\
 \downarrow \mathfrak{T}_\phi(\mathbf{D}_{\psi,i,j}) & & \downarrow \\
 \mathcal{H}_{\phi(\psi(i \boxtimes j))} & \xrightarrow{\mathbf{C}_{\phi,\psi,i \boxtimes j}} & \mathcal{H}_{(\phi\psi)(i \boxtimes j)}
 \end{array} \tag{1.13a}$$

$$\begin{array}{ccc}
 \mathcal{H}_0 & \xrightarrow{\mathbf{B}_{\phi\psi}} & \mathcal{H}_{(\phi\psi)0} \\
 \downarrow \mathbf{B}_\phi & & \uparrow \mathbf{C}_{\phi,\psi,0} \\
 \mathcal{H}_{\phi 0} & \xrightarrow{\mathfrak{T}_\phi(\mathbf{B}_\psi)} & \mathcal{H}_{\phi(\psi 0)}
 \end{array} \tag{1.13b}$$

The above compatibility conditions in Rem. 1.30 allow the G -action to be treated as strict, in the following sense.

Convention 1.31. Unless otherwise stated, for $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\phi, \psi \in G$, we make the identifications

$$\mathcal{H}_i = \mathcal{H}_{ei} \quad \mathcal{H}_0 = \mathcal{H}_{\phi 0} \quad \mathcal{H}_{\phi(\psi i)} = \mathcal{H}_{(\phi\psi)i} \tag{1.14a}$$

$$\mathcal{H}_{\phi i \boxtimes \phi j} = \mathcal{H}_{\phi(i \boxtimes j)} \tag{1.14b}$$

using the unitary isomorphisms A, B, C, D mentioned in Rem. 1.30. Thus, for $\phi_1, \dots, \phi_n \in G$, we interpret $\mathcal{H}_{\phi_1 \dots \phi_n i}$ with parentheses inserted arbitrarily. Under these identifications, by (a,b,c) of Rem. 1.30, we have

$$\Gamma_e|_{\mathcal{H}_i} = \text{id}_{\mathcal{H}_i} \quad \Gamma_\phi|_{\mathcal{H}_0} = \phi \quad \Gamma_{\phi\psi}|_{\mathcal{H}_i} = \Gamma_\phi \circ \Gamma_\psi|_{\mathcal{H}_i}$$

which we abbreviate as

$$\Gamma_e = \text{id} \quad \Gamma_\phi|_{\mathcal{H}_0} = \phi \quad \Gamma_{\phi\psi} = \Gamma_\phi \circ \Gamma_\psi \quad (1.15)$$

In other words, the assignment $\phi \in G \mapsto \Gamma_\phi$ defines a “categorical group representation” $G \curvearrowright \text{Rep}^G(\mathcal{A})$ extending $G \curvearrowright \mathcal{H}_0$.

As a consequence of (1.15), we have

$$\Gamma_\phi^{-1} = \Gamma_{\phi^{-1}} \quad (1.16)$$

as unitary maps from each $\mathcal{H}_i$ to $\mathcal{H}_{\phi^{-1}i}$.

1.4 G -crossed categorical extensions

Let $\mathcal{A}$ be a conformal net and $G \rightarrow \text{Aut}(\mathcal{A})$ a group homomorphism. In this section, we recall the G -crossed categorical extension for $\mathcal{A}$ established in [MS26a]. Recall Def. 1.10 for the clockwise and anticlockwise complements $\tilde{I}'$ and $\tilde{I}$.

Definition 1.32. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$, let

$$\begin{aligned} \mathbf{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j) &= \{T \in \mathcal{L}(\mathcal{H}_i, \mathcal{H}_j) : T\pi_{i, \tilde{I}}(x) = \pi_{j, \tilde{I}}(x)T \text{ for each } x \in \mathcal{A}(I)\} \\ \mathcal{H}_j(I) &= \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j) \cdot \Omega \end{aligned}$$

where $\mathcal{H}_j(I)$ is independent of $\arg_I$. Note that the Reeh-Schlieder property implies that $\mathcal{H}_j(I)$ is dense in $\mathcal{H}_j$. We write

$$\mathbf{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i) = \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j)$$

We write $\text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j)$ as $\mathbf{Hom}_{\mathcal{A}(I)}(\mathcal{H}_i, \mathcal{H}_j)$ when $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}(\mathcal{A})$.

Explanation. We need to explain why $\mathcal{H}_j(I)$ is independent of $\arg_I$. It suffices to show that for each $\tilde{I} \in \tilde{\mathcal{J}}$, the space $\mathcal{H}_j(I)$ defined by $\tilde{I}$ coincides with the one defined by $\varrho(2\pi)\tilde{I}$, that is,

$$\text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)\Omega = \text{Hom}_{\mathcal{A}(\varrho(2\pi)\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)\Omega$$

Assume without loss of generality that $\mathcal{H}_j \in \text{Rep}^\phi(\mathcal{A})$. Then the above relation follows from the following lemma and the fact that $\phi\Omega = \Omega$. $\square$

Remark 1.33. Note that Haag duality implies $\mathcal{H}_0(I) = \mathcal{A}(I)\Omega$. Note also that if $T \in \text{Hom}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_i, \mathcal{H}_j)$ then

$$T\mathcal{H}_i(I) \subset \mathcal{H}_j(I)$$

Lemma 1.34. Assume that $\phi \in G$ and $\mathcal{H}_j \in \text{Rep}^\phi(\mathcal{A})$. Let $T \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$. Then $T\phi \in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_j)$.

Consequently, if $\xi \in \mathcal{H}_j(I)$, and if $S \in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_j)$ and $T \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$ are the unique maps such that $S\Omega = \xi = T\Omega$, then $S = T\phi$. (The uniqueness is due to the Reeh-Schlieder property for $\mathcal{A}$.)

Proof. By (1.4), for each $x \in \mathcal{A}(I')$ we have

$$\pi_{j,\tilde{I}'}(x)T\phi = \pi_{j,\tilde{I}}(\phi x \phi^{-1})T\phi = T\phi x \phi^{-1}\phi = T\phi x$$

Thus $T\phi \in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_j)$. □

Remark 1.35. By (1.7), we have

$$\Gamma_\phi \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j) \Gamma_\phi^{-1} = \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_{\phi i}, \mathcal{H}_{\phi j}) \quad (1.17)$$

In particular, noting that $\Gamma_\phi^{-1}|_{\mathcal{H}_0}$ equals ϕ^{-1} (due to (1.15)) and hence fixes Ω , we obtain

$$\Gamma_\phi(\mathcal{H}_j(I)) = \mathcal{H}_{\phi j}(I) \quad (1.18)$$

By Cor. 1.20, for each $g \in \mathcal{G}_\mathcal{A}$ we have

$$g \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_j) g^{-1} = \text{Hom}_{\mathcal{A}(g\tilde{I})}(\mathcal{H}_i, \mathcal{H}_k) \quad (1.19)$$

Definition 1.36. Let $A : \mathcal{P} \rightarrow \mathcal{R}$, $B : \mathcal{Q} \rightarrow \mathcal{S}$, $C : \mathcal{P} \rightarrow \mathcal{Q}$, $D : \mathcal{R} \rightarrow \mathcal{S}$ be bounded linear operators between Hilbert spaces. We say that the diagram

$$\begin{array}{ccc} \mathcal{P} & \xrightarrow{C} & \mathcal{Q} \\ A \downarrow & & \downarrow B \\ \mathcal{R} & \xrightarrow{D} & \mathcal{S} \end{array}$$

commutes adjointly if $DA = BC$ and $D^*B = AC^*$.

The following theorem continues to assume Conv. 1.29 but not Conv. 1.31

Theorem 1.37. For each $\tilde{I} \in \tilde{\mathcal{J}}$, $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, $\xi \in \mathcal{H}_i(I)$, we have bounded linear operators

$$\begin{aligned} L(\xi, \tilde{I})|_{\mathcal{H}_k} &\in \text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_k, \mathcal{H}_i \boxtimes \mathcal{H}_k) \\ R(\xi, \tilde{I})|_{\mathcal{H}_k} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_k, \mathcal{H}_k \boxtimes \mathcal{H}_j) \end{aligned} \quad (1.20)$$

associated to each $\mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, called the **L operators** and the **R operators** and abbreviated to $L(\xi, \tilde{I})$, $R(\xi, \tilde{I})$ when no confusion arises, satisfying the following conditions:

(1) (Naturalty) If $\mathcal{H}_k, \mathcal{H}_{k'} \in \text{Rep}^G(\mathcal{A})$, $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_k, \mathcal{H}_{k'})$, $\xi \in \mathcal{H}_i(I)$, and $\chi \in \mathcal{H}_k$, then

$$(\mathbf{1}_i \boxtimes T)L(\xi, \tilde{I})\chi = L(\xi, \tilde{I})T\chi \quad (T \boxtimes \mathbf{1}_i)R(\xi, \tilde{I})\chi = R(\xi, \tilde{I})T\chi \quad (1.21)$$

(2) (State-field correspondence) If $\xi \in \mathcal{H}_i(I)$, then (under the identifications $\mathcal{H}_i = \mathcal{H}_0 \boxtimes \mathcal{H}_i = \mathcal{H}_i \boxtimes \mathcal{H}_0$) we have

$$L(\xi, \tilde{I})\Omega = R(\xi, \tilde{I})\Omega = \xi \quad (1.22)$$

(3) (Density of fusion products) The sets

$$L(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_k \quad R(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_k$$

span dense subspaces of $\mathcal{H}_i \boxtimes \mathcal{H}_k$ and $\mathcal{H}_k \boxtimes \mathcal{H}_i$, respectively.²

(4) (Locality) For any $\mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, any $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$, and any $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$, the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_k & \xrightarrow{R(\eta, \tilde{J})} & \mathcal{H}_k \boxtimes \mathcal{H}_j \\ L(\xi, \tilde{I}) \downarrow & & \downarrow L(\xi, \tilde{I}) \\ \mathcal{H}_i \boxtimes \mathcal{H}_k & \xrightarrow{R(\eta, \tilde{J})} & \mathcal{H}_i \boxtimes \mathcal{H}_k \boxtimes \mathcal{H}_j \end{array} \quad (1.23)$$

(5) (Braiding) If $\phi \in G$ and $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$, the unitary isomorphism $\mathbb{B}_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi j} \boxtimes \mathcal{H}_i$ satisfies

$$\mathbb{B}_{i,j}L(\xi, \tilde{I})\eta = R(\xi, \tilde{I})\Gamma_\phi\eta \quad (1.24)$$

whenever $\xi \in \mathcal{H}_i(I)$ and $\eta \in \mathcal{H}_j$.

(6) (Conformal covariance) If $g \in \mathcal{G}_{\mathcal{A}}$ and $\xi \in \mathcal{H}_i(I)$, there exists a (necessarily unique) element denoted by $g\xi g^{-1} \in \mathcal{H}_i(gI)$ satisfying

$$L(g\xi g^{-1}, g\tilde{I}) = gL(\xi, \tilde{I})g^{-1} \quad R(g\xi g^{-1}, g\tilde{I}) = gR(\xi, \tilde{I})g^{-1} \quad (1.25)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$.

Moreover, assuming the identification (1.14b), we also have:

(7) (G-covariance) For each $\xi \in \mathcal{H}_i(I)$ and $\phi \in G$, noting that $\Gamma_\phi\xi \in \mathcal{H}_{\phi i}(I)$ (cf. Rem. 1.35), we have

$$\Gamma_\phi L(\xi, \tilde{I}) = L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi \quad \Gamma_\phi R(\xi, \tilde{I}) = R(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi \quad (1.26)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ (with targets in $\mathcal{H}_{\phi(i \boxtimes j)} = \mathcal{H}_{\phi i \boxtimes \phi j}$ and $\mathcal{H}_{\phi(j \boxtimes i)} = \mathcal{H}_{\phi j \boxtimes \phi i}$, respectively).

²Indeed, they are equal to the full space $\mathcal{H}_i \boxtimes \mathcal{H}_k$ and $\mathcal{H}_k \boxtimes \mathcal{H}_i$ respectively by the fact that $\mathcal{A}(I)$ is a type III factor.

When we wish to emphasize the dependence on $\mathcal{A}$, we write the operations L, R as $L^{\mathcal{A}}, R^{\mathcal{A}}$.

Remark 1.38. For each $g \in \mathcal{G}_{\mathcal{A}}$, the uniqueness of $g\xi g^{-1}$ in the conformal covariance is due to the state-field correspondence, which implies

$$g\xi g^{-1} = gL(\xi, \tilde{I})g^{-1}\Omega = gR(\xi, \tilde{I})g^{-1}\Omega \quad (1.27)$$

Applying the state-field correspondence again, we obtain

$$g\xi g^{-1} = g\xi \quad \text{if } g^{-1}\Omega = \Omega$$

For example, by the definition of conformal nets, for each $t \in \mathbb{R}$ we have $U_0(\varrho_{\mathcal{A}}(t))\Omega = \Omega$, and hence the **rotation covariance** property: We have

$$\varrho_{\mathcal{A}}(t)\mathcal{H}_i(I) = \mathcal{H}_i(\varrho(t)I) \quad (1.28)$$

and for each $\xi \in \mathcal{H}_i(I)$ we have

$$\varrho_{\mathcal{A}}(t)L(\xi, \tilde{I}) = L(\varrho_{\mathcal{A}}(t)\xi, \varrho(t)\tilde{I})\varrho_{\mathcal{A}}(t) \quad (1.29a)$$

$$\varrho_{\mathcal{A}}(t)R(\xi, \tilde{I}) = R(\varrho_{\mathcal{A}}(t)\xi, \varrho(t)\tilde{I})\varrho_{\mathcal{A}}(t) \quad (1.29b)$$

We are now ready to describe the unitary isomorphism mentioned in Rem. 1.30-(d).

Definition 1.39. The linear map $\mathbf{D}_{\phi, i, j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$ is the unique unitary map (indeed, the unique unitary isomorphism of G -twisted $\mathcal{A}$ -modules) such that

$$\Gamma_{\phi}L(\xi, \tilde{I})\eta = \mathbf{D}_{\phi, i, j} \circ L(\Gamma_{\phi}\xi, \tilde{I})\Gamma_{\phi}\eta \quad \Gamma_{\phi}R(\eta, \tilde{J})\xi = \mathbf{D}_{\phi, i, j} \circ R(\Gamma_{\phi}\eta, \tilde{J})\Gamma_{\phi}\xi \quad (1.30)$$

holds for each $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$, and each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J)$. The uniqueness follows from the density of fusion products in Thm. 1.37.

The existence of such a unitary isomorphism, as well as the compatibility conditions in Rem. 1.30(d), will be explained in Subsec. A.1 and A.2, respectively. The G -covariance property in Thm. 1.37 would then follow automatically.

1.5 Calculus in G -crossed categorical extensions

We continue to fix a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$ for a conformal net $\mathcal{A}$. In this section, we do not assume Convention 1.31.

1.5.1 Conditions (2)-(4) of Thm. 1.37 imply $\text{Rep}^\phi(\mathcal{A}) \boxtimes \text{Rep}^\omega(\mathcal{A}) \subset \text{Rep}^{\phi\omega}(\mathcal{A})$

In [MS26a], the fusion product of two objects in $\text{Rep}^G(\mathcal{A})$ is first constructed as a solitonic $\mathcal{A}$ -module using operators Z^+ , Z^- and path continuations. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$, the operators

$$\begin{aligned} L(\xi, \tilde{I})|_{\mathcal{H}_0} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_i) & L(\xi, \tilde{I})|_{\mathcal{H}_j} &\in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j, \mathcal{H}_i \boxtimes \mathcal{H}_j) \\ R(\eta, \tilde{J})|_{\mathcal{H}_0} &\in \text{Hom}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_0, \mathcal{H}_j) & R(\eta, \tilde{J})|_{\mathcal{H}_i} &\in \text{Hom}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_i, \mathcal{H}_i \boxtimes \mathcal{H}_j) \end{aligned}$$

are likewise constructed using $Z^+(\xi, \tilde{I})$, $Z^-(\eta, \tilde{J})$ and path continuations. The following properties are readily verified:

- (2) $L(\xi, \tilde{I})\Omega = \xi$ and $R(\eta, \tilde{J})\Omega = \eta$.
- (3) Both $L(\mathcal{H}_i(I), \tilde{I})\mathcal{H}_j$ and $R(\mathcal{H}_j(J), \tilde{J})\mathcal{H}_i$ span dense subspaces of $\mathcal{H}_i \boxtimes \mathcal{H}_j$.
- (4) If $\tilde{J}$ is clockwise to $\tilde{I}$, the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_0 & \xrightarrow{R(\eta, \tilde{J})|_{\mathcal{H}_0}} & \mathcal{H}_j \\ L(\xi, \tilde{I})|_{\mathcal{H}_0} \downarrow & & \downarrow L(\xi, \tilde{I})|_{\mathcal{H}_j} \\ \mathcal{H}_i & \xrightarrow{R(\eta, \tilde{J})|_{\mathcal{H}_i}} & \mathcal{H}_i \boxtimes \mathcal{H}_j \end{array}$$

These constructions and properties are, of course, part of those stated in Thm. 1.37. In what follows, we show that they already suffice to imply (1.9).

Lemma 1.40. *If $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, and if $\tilde{J}$ is clockwise to $\tilde{I}$, then for each $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$ and each $x \in \mathcal{A}(I)$, $y \in \mathcal{A}(J)$, we have*

$$\begin{aligned} R(\xi, \tilde{I})y|_{\mathcal{H}_0} &= \pi_{\tilde{J}}(\phi^{-1}y\phi)R(\xi, \tilde{I})|_{\mathcal{H}_0} \\ L(\eta, \tilde{J})x|_{\mathcal{H}_0} &= \pi_{\tilde{I}}(\omega x \omega^{-1})L(\eta, \tilde{J})|_{\mathcal{H}_0} \end{aligned}$$

Proof. Since $\tilde{I}$ is clockwise to $\varrho(2\pi)\tilde{J}$, we have

$$R(\xi, \tilde{I})y|_{\mathcal{H}_0} = \pi_{i, \varrho(2\pi)\tilde{J}}(y)R(\xi, \tilde{I})|_{\mathcal{H}_0} = \pi_{i, \tilde{J}}(\phi^{-1}y\phi)R(\xi, \tilde{I})|_{\mathcal{H}_0}$$

where the last identity is due to the fact that $\mathcal{H}_i$ is ϕ -twisted. This proves the first identity. The second one follows from a similar argument, using the relation $\pi_{j, \varrho(-2\pi)\tilde{I}}(x) = \pi_{j, \tilde{I}}(\omega x \omega^{-1})$. $\square$

Theorem 1.41. *If $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, then the solitonic $\mathcal{A}$ -module $\mathcal{H}_i \boxtimes \mathcal{H}_j$ is $\phi\omega$ -twisted.*

Proof. Choose any $\tilde{K}^+ \in \tilde{\mathcal{J}}$, and let $\tilde{K}^- = \varrho(-2\pi)\tilde{K}^+$. Choose $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ such that $\tilde{J}$ is clockwise to $\tilde{I}$, and $\tilde{I}$ is clockwise to $\tilde{K}^+$.³

Choose any $x \in \mathcal{A}(K)$, $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$. By Lem. 1.40, we have

$$L(\xi, \tilde{I})R(\eta, \tilde{J})x\Omega = L(\xi, \tilde{I})\pi_{\tilde{K}^-}(\omega^{-1}x\omega)R(\eta, \tilde{J})\Omega = \pi_{\tilde{K}^-}(\omega^{-1}x\omega)L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega$$

and

$$R(\eta, \tilde{J})L(\xi, \tilde{I})x\Omega = R(\eta, \tilde{J})\pi_{\tilde{K}^+}(\phi x \phi^{-1})L(\xi, \tilde{I})\Omega = \pi_{\tilde{K}^+}(\phi x \phi^{-1})R(\eta, \tilde{J})L(\xi, \tilde{I})\Omega$$

Therefore, by (4), the operators $\pi_{\tilde{K}^-}(\omega^{-1}x\omega)$ and $\pi_{\tilde{K}^+}(\phi x \phi^{-1})$ are equal when acting on vectors of the form $L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega$ (which equals $L(\xi, \tilde{I})\eta$ by (2)). Thus (3) implies $\pi_{\tilde{K}^-}(\omega^{-1}x\omega) = \pi_{\tilde{K}^+}(\phi x \phi^{-1})$ and hence

$$\pi_{\tilde{K}^-}(x) = \pi_{\tilde{K}^+}(\phi \omega x \omega^{-1} \phi^{-1})$$

when acting on $\mathcal{H}_i \boxtimes \mathcal{H}_j$. Thus $\mathcal{H}_i \boxtimes \mathcal{H}_j$ is $\phi\omega$ -twisted. □

1.5.2 Consequences of (1)-(4) of Thm. 1.37

Proposition 1.42. *Let $\tilde{J}$ be clockwise to $\tilde{I}$, and let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, $\xi \in \mathcal{H}_i(I)$, $\eta \in \mathcal{H}_j(J)$. Then*

$$L(\xi, \tilde{I})\eta = R(\eta, \tilde{J})\xi$$

Proof. By the state-field correspondence and the locality in Thm. 1.37, we have

$$L(\xi, \tilde{I})\eta = L(\xi, \tilde{I})R(\eta, \tilde{J})\Omega = R(\eta, \tilde{J})L(\xi, \tilde{I})\Omega = R(\eta, \tilde{J})\xi$$

□

Corollary 1.43 (Isotony). *If $\tilde{I}_1 \subset \tilde{I}_2 \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I_1)$, then*

$$L(\xi, \tilde{I}_1) = L(\xi, \tilde{I}_2) \quad R(\xi, \tilde{I}_1) = R(\xi, \tilde{I}_2)$$

when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$.

Proof. Choose $\tilde{J} \in \tilde{\mathcal{J}}$ clockwise to $\tilde{I}_2$. By Prop. 1.42, for each $\eta \in \mathcal{H}_j(J)$ we have

$$L(\xi, \tilde{I}_1)\eta = R(\eta, \tilde{J})\xi = L(\xi, \tilde{I}_2)\eta$$

The second identity follows from a similar argument. □

³For example, if $\tilde{K}^+$ is a small neighborhood of $e^{i\pi}$, one can choose $\tilde{I}$ to be a small neighborhood of $e^{i\pi/2}$, and $\tilde{J}$ a small neighborhood of $e^{-i\pi/2}$. Then $\tilde{K}^-$ is a small neighborhood of $e^{-i\pi}$.

Proposition 1.44. Let $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$. Then, when acting on any $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, we have (recalling $\mathcal{H}_0(I) = \mathcal{A}(I)\Omega$)

$$L(x\Omega, \tilde{I}) = R(x\Omega, \tilde{I}) = \pi_{\tilde{I}}(x)$$

Proof. Choose $\tilde{J}$ clockwise to $\tilde{I}$. For each $\eta \in \mathcal{H}_j(J)$, we compute that

$$\begin{aligned} L(x\Omega, \tilde{I})\eta &= L(x\Omega, \tilde{I})R(\eta, \tilde{J})\Omega = R(\eta, \tilde{J})L(x\Omega, \tilde{I})\Omega = R(\eta, \tilde{J})x\Omega \\ &= \pi_{\tilde{I}}(x)R(\eta, \tilde{J})\Omega = \pi_{\tilde{I}}(x)\eta \end{aligned}$$

where the locality and the state-field correspondence in Thm. 1.37 have been used. This proves $L(x\Omega, \tilde{I}) = \pi_{\tilde{I}}(x)$. The second relation follows from a similar argument. $\square$

Proposition 1.45. If $S \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_{i'})$ and $T \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_j, \mathcal{H}_{j'})$ are morphisms in $\text{Rep}^G(\mathcal{A})$, then for each $\xi \in \mathcal{H}_i(I)$ and $\eta \in \mathcal{H}_j$, we have

$$(S \boxtimes T)L(\xi, \tilde{I})\eta = L(S\xi, \tilde{I})T\eta \quad (T \boxtimes S)R(\xi, \tilde{I})\eta = R(S\xi, \tilde{I})T\eta$$

Proof. We prove the second relation, as the first one can be proved in a similar way. It suffices to assume that $\eta \in \mathcal{H}_j(K)$ where $\tilde{K}$ is anticlockwise to $\tilde{I}$. By the naturality in Thm. 1.37, we have

$$(T \boxtimes S)R(\xi, \tilde{I})\eta = (\mathbf{1} \boxtimes S)(T \boxtimes \mathbf{1})R(\xi, \tilde{I})\eta = (\mathbf{1} \boxtimes S)R(\xi, \tilde{I})T\eta$$

By Prop. 1.42, the above expression equals

$$(\mathbf{1} \boxtimes S)L(T\eta, \tilde{K})\xi = L(T\eta, \tilde{K})S\xi = R(S\xi, \tilde{I})T\eta$$

$\square$

We also note the useful fact that, as $\boxtimes$ is a $*$ -bifunctor, we have

$$(S \boxtimes T)^* = S^* \boxtimes T^* \tag{1.31}$$

Proposition 1.46. Let $\mathcal{H}_i, \mathcal{H}_j, \mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$, $\tilde{I} \in \tilde{\mathcal{J}}$, and $\xi \in \mathcal{H}_i(I)$. The following are true.

(a) If $\eta \in \mathcal{H}_j(I)$, then $L(\xi, \tilde{I})\eta \in (\mathcal{H}_i \boxtimes \mathcal{H}_j)(I)$, $R(\xi, \tilde{I})\eta \in (\mathcal{H}_j \boxtimes \mathcal{H}_i)(I)$, and

$$\begin{aligned} L(\xi, \tilde{I})L(\eta, \tilde{I})|_{\mathcal{H}_k} &= L(L(\xi, \tilde{I})\eta, \tilde{I})|_{\mathcal{H}_k} \\ R(\xi, \tilde{I})R(\eta, \tilde{I})|_{\mathcal{H}_k} &= R(R(\xi, \tilde{I})\eta, \tilde{I})|_{\mathcal{H}_k} \end{aligned}$$

(b) If $\psi \in (\mathcal{H}_i \boxtimes \mathcal{H}_j)(I)$ and $\eta \in (\mathcal{H}_j \boxtimes \mathcal{H}_i)(I)$, then $L(\xi, \tilde{I})^*\psi \in \mathcal{H}_j(I)$, $R(\xi, \tilde{I})^*\eta \in \mathcal{H}_j(I)$, and

$$\begin{aligned} L(\xi, \tilde{I})^*L(\psi, \tilde{I})|_{\mathcal{H}_k} &= L(L(\xi, \tilde{I})^*\psi, \tilde{I})|_{\mathcal{H}_k} \\ R(\xi, \tilde{I})^*R(\eta, \tilde{I})|_{\mathcal{H}_k} &= R(R(\xi, \tilde{I})^*\eta, \tilde{I})|_{\mathcal{H}_k} \end{aligned}$$

Proof. We prove the first of (b); part (a) and the second of (b) follows from a similar argument. Part (a) follows as in [Gui21a, Prop. 3.6].

By the state-field correspondence, we have $L(\xi, \tilde{I})^* \psi = L(\xi, \tilde{I})^* L(\psi, \tilde{I}) \Omega$. This implies, noticing $L(\xi, \tilde{I})^* L(\psi, \tilde{I}) \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_j)$, that $L(\xi, \tilde{I})^* \psi \in \mathcal{H}_j(I)$. Choose $\tilde{J}$ clockwise to $\tilde{I}$ and $\chi \in \mathcal{H}_k(J)$. Then

$$\begin{aligned} L(\xi, \tilde{I})^* L(\psi, \tilde{I}) \chi &= L(\xi, \tilde{I})^* L(\psi, \tilde{I}) R(\chi, \tilde{J}) \Omega = R(\chi, \tilde{J}) L(\xi, \tilde{I})^* L(\psi, \tilde{I}) \Omega \\ &= R(\chi, \tilde{J}) L(\xi, \tilde{I})^* \psi = R(\chi, \tilde{J}) L(L(\xi, \tilde{I})^* \psi, \tilde{I}) \Omega = L(L(\xi, \tilde{I})^* \psi, \tilde{I}) R(\chi, \tilde{J}) \Omega \\ &= L(L(\xi, \tilde{I})^* \psi, \tilde{I}) \chi \end{aligned}$$

where the locality and the state-field correspondence in Thm. 1.37 are used. $\square$

1.6 Representations of commutant algebras on fusion products

Fix a conformal net $\mathcal{A}$ and a group homomorphism $G \rightarrow \text{Aut}(\mathcal{A})$. In this section, we only consider objects of $\text{Rep}^G(\mathcal{A})$ that are nonzero.

Definition 1.47. For each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $I \in \mathcal{J}$, a vector $\xi \in \mathcal{H}_i(I)$ is called **I -unitary** if there exists $\arg_I$ such that by setting $\tilde{I} = (I, \arg_I)$, the operator $L(\xi, \tilde{I})|_{\mathcal{H}_0} : \mathcal{H}_0 \rightarrow \mathcal{H}_i$ is unitary. By Lem. A.4, this is equivalent to that $R(\xi, \tilde{I})|_{\mathcal{H}_0} : \mathcal{H}_0 \rightarrow \mathcal{H}_i$ is unitary.

By Isotony (Cor. 1.43), if ξ is I -unitary, then ξ is J -unitary for any $J \in \mathcal{J}$ containing I .

Remark 1.48. I -unitary vectors must exist. Indeed, since $\mathcal{A}(I')$ is a type III factor, there exists a unitary $T \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_0, \mathcal{H}_i)$. Hence $\xi := T\Omega$ is I -unitary.

Proposition 1.49. Suppose that $\xi \in \mathcal{H}_i(I)$ is I -unitary. Then for each $\arg_I$ and each $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, by setting $\tilde{I} = (I, \arg_I)$, the following operators are unitary

$$L(\xi, \tilde{I})|_{\mathcal{H}_j} : \mathcal{H}_j \rightarrow \mathcal{H}_i \boxtimes \mathcal{H}_j \quad R(\xi, \tilde{I})|_{\mathcal{H}_j} : \mathcal{H}_j \rightarrow \mathcal{H}_j \boxtimes \mathcal{H}_i$$

In particular, since we assume that $\mathcal{H}_i, \mathcal{H}_j$ are nonzero, the object $\mathcal{H}_i \boxtimes \mathcal{H}_j$ is also nonzero.

Proof. By Rem. 1.38, if $L(\xi, \tilde{I})|_{\mathcal{H}_0}$ is unitary, so is $L(\xi, \varrho(2\pi)\tilde{I})|_{\mathcal{H}_0}$. Thus, the unitarity of $L(\xi, \tilde{I})|_{\mathcal{H}_0}$ is independent of $\arg_I$. We now fix an arbitrary $\arg_I$. Choose $\tilde{J}$ clockwise to $\tilde{I}$. For each $\eta \in \mathcal{H}_j(J)$, we have

$$\begin{aligned} L(\xi, \tilde{I})^* L(\xi, \tilde{I}) \eta &= L(\xi, \tilde{I})^* L(\xi, \tilde{I}) R(\eta, \tilde{J}) \Omega \\ &= R(\eta, \tilde{J}) L(\xi, \tilde{I})^* L(\xi, \tilde{I}) \Omega = R(\eta, \tilde{J}) \Omega = \eta \end{aligned}$$

Hence $L(\xi, \tilde{I})|_{\mathcal{H}_j}$ is an isometry. Moreover, for $\chi \in \mathcal{H}_0$ we have

$$L(\xi, \tilde{I}) R(\eta, \tilde{J}) \chi = R(\eta, \tilde{J}) L(\xi, \tilde{I}) \chi$$

Thus, by the unitarity of $L(\xi, \tilde{I})|_{\mathcal{H}_0}$, the above vectors span a dense subspace of $R(\mathcal{H}_j, \tilde{J})\mathcal{H}_i = \mathcal{H}_i \boxtimes \mathcal{H}_j$. This proves that $L(\xi, \tilde{I})|_{\mathcal{H}_j}$ is unitary. A similar argument shows that $R(\xi, \tilde{I})|_{\mathcal{H}_j}$ is unitary. $\square$

Proposition 1.50. *Let $\mathcal{H}_i, \mathcal{H}_k \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$. Fix I -unitary vectors $\xi_0 \in \mathcal{H}_i(I), \psi_0 \in \mathcal{H}_k(I)$. Then we have*

$$\begin{aligned} \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_k) &= \{L(\psi, \tilde{I})L(\xi, \tilde{I})^*|_{\mathcal{H}_i} : \xi \in \mathcal{H}_i(I), \psi \in \mathcal{H}_k(I)\} \\ &= \{L(\psi_0, \tilde{I})xL(\xi_0, \tilde{I})^*|_{\mathcal{H}_i} : x \in \mathcal{A}(I)\} \end{aligned}$$

and

$$\begin{aligned} \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_k) &= \{R(\psi, \tilde{I})R(\xi, \tilde{I})^*|_{\mathcal{H}_i} : \xi \in \mathcal{H}_i(I), \psi \in \mathcal{H}_k(I)\} \\ &= \{R(\psi_0, \tilde{I})xR(\xi_0, \tilde{I})^*|_{\mathcal{H}_i} : x \in \mathcal{A}(I)\} \end{aligned}$$

Proof. We prove the first set of equalities; the second follows by the same argument. The inclusions $\supset$ are clear. In particular, the middle set contains the last one, since

$$L(\psi_0, \tilde{I})x|_{\mathcal{H}_0} = L(\psi_0, \tilde{I})L(x\Omega, \tilde{I})|_{\mathcal{H}_0} = L(L(\psi_0, \tilde{I})x\Omega, \tilde{I})|_{\mathcal{H}_0}$$

by Prop. 1.44 and 1.46. It remains to show that the last set contains the first.

Choose any $T \in \text{Hom}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i, \mathcal{H}_k)$. By the locality in Thm. 1.37, the element

$$x := L(\psi_0, \tilde{I})^*TL(\xi_0, \tilde{I})|_{\mathcal{H}_0}$$

belongs to $\text{End}_{\mathcal{A}(I')}(\mathcal{H}_0) = \mathcal{A}(I)$. By the I -unitarity, T equals $L(\psi_0, \tilde{I})xL(\xi_0, \tilde{I})^*|_{\mathcal{H}_i}$. $\square$

Theorem 1.51. *Let $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$. Then there exists an operation $\pi_{\tilde{I}}^L$ associating to each $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ a unique normal unitary representation*

$$\pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} : \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i) \rightarrow \mathfrak{L}(\mathcal{H}_i \boxtimes \mathcal{H}_j) \quad (1.32)$$

commuting adjointly with $R(\eta, \tilde{I}')$ for each $\eta \in \mathcal{H}_j(I')$, that is, for each $A \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_i & \xrightarrow{R(\eta, \tilde{I}')} & \mathcal{H}_i \boxtimes \mathcal{H}_j \\ A \downarrow & & \downarrow \pi_{\tilde{I}}^L(A) \\ \mathcal{H}_i & \xrightarrow{R(\eta, \tilde{I}')} & \mathcal{H}_i \boxtimes \mathcal{H}_j \end{array} \quad (1.33)$$

Moreover, if $x \in \mathcal{A}(I)$, then $\pi_{i, \tilde{I}}(x) \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ and

$$\pi_{\tilde{I}}^L(\pi_{i, \tilde{I}}(x))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = \pi_{i \boxtimes j, \tilde{I}}(x) \quad (1.34)$$

Proof. The uniqueness follows from (1.33) and the density of fusion products (or simply by choosing an I' -unitary vector η). Let us prove the existence.

Fix an I' -unitary vector η_0 , and define the representation $\pi_{\tilde{I}}^L$ on $\mathcal{H}_i \boxtimes \mathcal{H}_j$ by

$$\pi_{\tilde{I}}^L(A)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = R(\eta_0, \tilde{I}') A R(\eta_0, \tilde{I}')^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$$

By Prop. 1.50, each A can be written as $L(\psi_2, \tilde{I}) L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i}$ where $\xi_1, \xi_2 \in \mathcal{H}_i(I)$. The locality in Thm. 1.37 implies that

$$\pi_{\tilde{I}}^L(L(\xi_2, \tilde{I}) L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i})|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = L(\xi_2, \tilde{I}) L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \quad (1.35)$$

Another application of the locality implies the adjoint commutativity of (1.33).

If $x \in \mathcal{A}(I)$, then $\pi_{i, \tilde{I}}(x) \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ by Rem. 1.14, and

$$R(\eta_0, \tilde{I}') \pi_{i, \tilde{I}}(x) R(\eta_0, \tilde{I}')^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = \pi_{i \boxtimes j, \tilde{I}}(x) R(\eta_0, \tilde{I}') R(\eta_0, \tilde{I}')^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = \pi_{i \boxtimes j, \tilde{I}}(x)$$

This proves (1.34). □

Theorem 1.52. Let $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\tilde{J} \in \tilde{\mathcal{J}}$. Then there exists an operation $\pi_{\tilde{J}}^R$ associating to each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ a unique normal unitary representation

$$\pi_{\tilde{J}}^R|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} : \text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j) \rightarrow \mathcal{L}(\mathcal{H}_i \boxtimes \mathcal{H}_j) \quad (1.36)$$

commuting adjointly with $L(\xi, \tilde{J})$ for each $\xi \in \mathcal{H}_i(J')$, that is, for each $B \in \text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)$ the following diagram commutes adjointly.

$$\begin{array}{ccc} \mathcal{H}_j & \xrightarrow{B} & \mathcal{H}_j \\ L(\xi, \tilde{J}) \downarrow & & \downarrow L(\xi, \tilde{J}) \\ \mathcal{H}_i \boxtimes \mathcal{H}_j & \xrightarrow{\pi_{\tilde{J}}^R(B)} & \mathcal{H}_i \boxtimes \mathcal{H}_j \end{array} \quad (1.37)$$

Moreover, if $y \in \mathcal{A}(J)$, then $\pi_{j, \tilde{J}}(y) \in \text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)$ and

$$\pi_{\tilde{J}}^R(\pi_{j, \tilde{J}}(y))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = \pi_{i \boxtimes j, \tilde{J}}(y) \quad (1.38)$$

Proof. This is similar to the proof of Thm. 1.51. □

Remark 1.53. Similar to the proof of (1.39), by the locality in Thm. 1.37, we can prove the more general fact that if $\xi_1, \xi_2 \in \mathcal{H}_i(I)$, $\eta_1, \eta_2 \in \mathcal{H}_j(J)$, and $x \in \mathcal{A}(I)$, $y \in \mathcal{A}(J)$, then

$$\pi_{\tilde{I}}^L(L(\xi_2, \tilde{I}) x L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i})|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = L(\xi_2, \tilde{I}) \pi_{\tilde{I}}(x) L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \quad (1.39a)$$

$$\pi_{\tilde{J}}^R(R(\eta_2, \tilde{J}) y R(\eta_1, \tilde{J})^*|_{\mathcal{H}_j})|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = R(\xi_2, \tilde{J}) \pi_{\tilde{J}}(y) R(\eta_1, \tilde{J})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \quad (1.39b)$$

Remark 1.54. In Thm. 1.51, if $\tilde{I} \subset \tilde{J}$, then $\pi_{\tilde{I}}^L$ restricts to $\pi_{\tilde{I}}^L$ on $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$. Similarly, in Thm. 1.52, if $\tilde{I} \subset \tilde{J}$, then $\pi_{\tilde{I}}^R$ restricts to $\pi_{\tilde{I}}^R$ on $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$.

Proof. Choose any $A \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$. Let $\mu_0 \in \mathcal{H}_j(J')$ by J' -unitary, and let $V = R(\eta_0, \tilde{J}')|_{\mathcal{H}_i}$. Then (1.33) implies $\pi_{\tilde{I}}^L(A) = VAV^{-1} = \pi_{\tilde{I}}^L(A)$. $\square$

Remark 1.55. The representation (1.32) is faithful since, by the adjoint commutativity of (1.33), it is unitarily equivalent to the action of $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ on $\mathcal{H}_i$ via the unitary operator $R(\eta_0, \tilde{I}')|_{\mathcal{H}_i}$, where $\eta_0 \in \mathcal{H}_j(I')$ is I' -unitary. Similarly, the representation (1.36) is faithful. It follows that

$$\pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \text{ and } \pi_{\tilde{J}}^R(\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \text{ are type III factors}$$

since $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ and $\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)$ are so.

Theorem 1.56. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, $g \in \mathcal{G}_A$, and $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J} = g\tilde{I}$. Then

$$\begin{aligned} U_i(g)\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)U_i(g)^* &= \text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_i) \\ U_j(g)\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)U_j(g)^* &= \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j) \end{aligned}$$

Moreover, for each $A \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ and $B \in \text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)$, we have

$$\begin{aligned} U_{i \boxtimes j}(g) \cdot \pi_{\tilde{I}}^L(A)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot U_{i \boxtimes j}(g)^* &= \pi_{\tilde{J}}^L(U_i(g)AU_i(g)^*)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \\ U_{i \boxtimes j}(g) \cdot \pi_{\tilde{I}}^R(B)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot U_{i \boxtimes j}(g)^* &= \pi_{\tilde{J}}^R(U_j(g)BU_j(g)^*)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \end{aligned}$$

Proof. By Prop. 1.50, elements of $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ are precisely of the form

$$A = L(\xi_2, \tilde{I})L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i}$$

where $\xi_1, \xi_2 \in \mathcal{H}_i(\tilde{I})$. By the conformal covariance in Thm. 1.37, for the vectors $g\xi_1g^{-1}, g\xi_2g^{-1} \in \mathcal{H}_i(\tilde{J})$ we have

$$U_i(g)AU_i(g)^* = L(g\xi_2g^{-1}, \tilde{J})L(g\xi_1g^{-1}, \tilde{J})^*|_{\mathcal{H}_i} \quad (\star)$$

This proves that $\text{Ad}_{U_i(g)}$ sends $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ into $\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_i)$. Similarly, $\text{Ad}_{U_j(g^{-1})}$ sends $\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_j)$ into $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j)$. This proves the first identity.

By Rem. 1.53 and the conformal covariance in Thm. 1.37,

$$\begin{aligned} U_{i \boxtimes j}(g) \cdot \pi_{\tilde{I}}^L(A)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot U_{i \boxtimes j}(g)^* &= U_{i \boxtimes j}(g) \cdot L(\xi_2, \tilde{I})L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot U_{i \boxtimes j}(g)^* \\ &= L(g\xi_2g^{-1}, \tilde{J})L(g\xi_1g^{-1}, \tilde{J})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = \pi_{\tilde{I}}^L((\star))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \end{aligned}$$

This proves the third identity. The second and fourth identities are proved in the same way. $\square$

Theorem 1.57. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$. Then the following two von Neumann algebras on $\mathcal{H}_i \boxtimes \mathcal{H}_j$ are commutants of each other.

$$\pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \quad \pi_{\tilde{I}'}^R(\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$$

Proof. Eq. (1.35) and (1.39), together with Prop. 1.50, imply that the above two von Neumann algebras commute with each other.

Suppose that $B \in \mathfrak{L}(\mathcal{H}_i \boxtimes \mathcal{H}_j)$ commutes with $\pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$. Choose an I -unitary $\xi \in \mathcal{H}_i(I)$ and an I' -unitary $\eta \in \mathcal{H}_j(I')$. By Rem. 1.53, B commutes with $L(\xi, \tilde{I})\pi_{\tilde{I}}(\mathcal{A}(I))L(\xi, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$. Thus

$$L(\xi, \tilde{I})^*BL(\xi, \tilde{I})|_{\mathcal{H}_j}$$

commutes with $\pi_{\tilde{I},j}(\mathcal{A}(I))$, and hence (by Prop. 1.50) can be written as $R(\eta, \tilde{I}')yR(\eta, \tilde{I}')^*|_{\mathcal{H}_j}$ for some $y \in \mathcal{A}(I')$. By the locality in Thm. 1.37,

$$\begin{aligned} B &= L(\xi, \tilde{I})R(\eta, \tilde{I}')yR(\eta, \tilde{I}')^*L(\xi, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \\ &= R(\eta, \tilde{I}')\pi_{\tilde{I}'}(y)R(\eta, \tilde{I}')^*L(\xi, \tilde{I})L(\xi, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} = R(\eta, \tilde{I}')\pi_{\tilde{I}'}(y)R(\eta, \tilde{I}')^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \end{aligned}$$

Therefore, by Rem. 1.53, B belongs to $\pi_{\tilde{I}'}^R(\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_j))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$. □

1.7 Conjugates and commutants in fusion products

In this section, we only consider nonzero objects of $\text{Rep}^G(\mathcal{A})$.

Define $\mathfrak{r} : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ to be the conjugation map $z \mapsto \bar{z}$, which can be lifted to the map of the universal cover $\mathfrak{r} : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto -x$. This defines $\mathfrak{r}\tilde{I}$ as an element of $\tilde{\mathcal{J}}$ if $\tilde{I} \in \tilde{\mathcal{J}}$. Let $\tilde{\mathbb{S}}_+^1$ be the upper half circle $\mathbb{S}_+^1 = \{e^{it} : 0 < t < \pi\}$ with arg function ranging in $(0, \pi)$, and let $\tilde{\mathbb{S}}_-^1 = \mathfrak{r}\mathbb{S}_+^1$.

1.7.1 The conjugation in $\mathcal{G}_{\mathcal{A}}$

Remark 1.58. Let $\mathfrak{J}_{\mathcal{A}}$ (or simply $\mathfrak{J}$) be the (involutive) modular conjugation of $\mathcal{A}(\mathbb{S}_+^1)$ with respect to the cyclic separating vector Ω . By the geometric modular theory (cf. [GF93, Sec. II.2]), we have

$$\begin{aligned} \mathfrak{J}\mathcal{A}(I)\mathfrak{J} &= \mathcal{A}(\mathfrak{r}I) & \text{for each } I \in \mathcal{J} \\ \mathfrak{J}U_0(\varrho(t))\mathfrak{J} &= U_0(\varrho(-t)) & \text{for each } t \in \mathbb{R} \end{aligned}$$

Remark 1.59. Each $\varphi \in \text{Aut}(\mathcal{A})$ clearly commutes with the modular S -operator $x\Omega \mapsto x^*\Omega$ where $x \in \mathcal{A}(\mathbb{S}_+^1)$. Therefore, by the polar decomposition, φ commutes with $\mathfrak{J}$.

Definition 1.60. For each $\tilde{g} \in \mathcal{G}_A$, recall that $\tilde{g}$ is of the form (g, V) where $g \in \mathcal{G}$ and $V \in \mathcal{U}(\mathcal{H}_0)$ represents $U_0(g)$. Define $\mathfrak{r}\tilde{g}\mathfrak{r} \in \mathcal{G}_A$ by

$$\mathfrak{r}\tilde{g}\mathfrak{r} = (\mathfrak{r}g\mathfrak{r}, \mathfrak{J}V\mathfrak{J}) \quad (1.40)$$

where, in terms of the description (1.3) of $\mathcal{G}$, the element $\mathfrak{r}g\mathfrak{r} \in \mathcal{G}$ sends each $x \in \mathbb{R}$ to $-g(-x)$. It follows that

$$U_0(\mathfrak{r}\tilde{g}\mathfrak{r}) = \mathfrak{J}U_0(\tilde{g})\mathfrak{J} \quad (1.41)$$

Proof. We need to explain why $\mathfrak{J}V\mathfrak{J}$ represents $U_0(\mathfrak{r}g\mathfrak{r})$. For that purpose, it suffices to show that

$$U'_0 : g \in \text{Diff}^+(\mathbb{S}^1) \mapsto \mathfrak{J}U_0(\mathfrak{r}g\mathfrak{r})\mathfrak{J}$$

agrees with U_0 as projective representations of $\text{Diff}^+(\mathbb{S}^1)$. By [GF93, Thm. 2.19], U_0 agrees with U'_0 when restricted to $\widehat{\text{PSU}}(1, 1)$. Therefore, $\mathcal{A}$ is conformally covariant under U'_0 in the sense of Def. 1.1. By [Wei05, Thm. 6.1.9], a conformal covariance extending a given Möbius covariance of a conformal net is unique. Therefore $U'_0 = U_0$ on $\text{Diff}^+(\mathbb{S}^1)$. $\square$

Remark 1.61. Recall the identification $\mathcal{G}_A = \mathcal{G}_c$ in Rem. 1.5. Then the definition of $\mathfrak{r}\tilde{g}\mathfrak{r}$ for $g \in \mathcal{G}_c$ is independent of the conformal net $\mathcal{A}$ extending Vir_c .

Proof. Let $\mathcal{K}_0 = \overline{\text{Vir}_c\Omega}$. Then $(\text{Vir}_c, \mathcal{K}_0)$ is the Virasoro subnet of $(\mathcal{A}, \mathcal{H}_0)$. Since the unitary representation of $\text{PSU}(1, 1)$ on $\mathcal{H}_0$ restricts to that on $\mathcal{K}_0$, by the Bisognano-Wichmann property [GF93, Thm. 2.19], the modular operators $\Delta_{\mathcal{A}}, \Delta_c$ of $\mathcal{A}(\mathbb{S}^1_+)$ and $\text{Vir}_c(\mathbb{S}^1_+)$ satisfies $\Delta_{\mathcal{A}}^{it}|_{\mathcal{K}_0} = \Delta_c^{it}$. Therefore, by (for instance) the proof of Thm. 4.2 in [Tak03, Sec. IX.4], the modular conjugations $\mathfrak{J}_{\mathcal{A}}, \mathfrak{J}_c$ satisfy $\mathfrak{J}_{\mathcal{A}}|_{\mathcal{K}_0} = \mathfrak{J}_c$. Hence, for each $\tilde{g} = (g, V) \in \mathcal{G}_A$, the operator $\mathfrak{J}_{\mathcal{A}}V\mathfrak{J}_{\mathcal{A}}$ restricts to $\mathfrak{J}_cV\mathfrak{J}_c$ on $\mathcal{K}_0$. $\square$

Example 1.62. Recall the rotation subgroup $\rho_A : \mathbb{R} \rightarrow \mathcal{G}_A$ defined in Rem. 1.7. Then by Rem. 1.58,

$$\mathfrak{r}\rho_A(t)\mathfrak{r} = \rho_A(-t)$$

1.7.2 Conjugate modules

Definition 1.63. For each $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$, define the G -crossed **conjugate $\mathcal{A}$ -module** $\mathcal{H}_{\bar{i}}$ as follows. As a Hilbert space, $\mathcal{H}_{\bar{i}}$ is the complex conjugate of $\mathcal{H}_i$. We denote the conjugation map by $\mathbb{C}_i : \mathcal{H}_i \rightarrow \mathcal{H}_{\bar{i}}$, and abbreviate it to $\mathbb{C}$ when no confusion arises. Thus

$$\mathbb{C}_i\xi = \mathbb{C}\xi = \bar{\xi} \quad \text{for each } \xi \in \mathcal{H}_i$$

For each $\tilde{I} \in \tilde{\mathcal{I}}$ and $x \in \mathcal{A}(I)$, we define

$$\pi_{\bar{i}, \tilde{I}}(x) = \mathbb{C}_i\pi_{i, \mathfrak{r}\tilde{I}}(\mathfrak{J}x\mathfrak{J})\mathbb{C}_i^{-1}$$

noting that $\mathfrak{J}x\mathfrak{J} \in \mathcal{A}(\mathfrak{r}I)$ by Rem. 1.58. In short,

$$\pi_{\tilde{I}}(x) = \mathbb{C}_{\pi_{\mathfrak{r}\tilde{I}}}(\mathfrak{J}x\mathfrak{J})\mathbb{C}^{-1} \quad (1.42)$$

We identify $\mathcal{H}_{\tilde{i}}$ with $\mathcal{H}_i$ canonically, that is, by identifying $\xi \in \mathcal{H}_i$ with $\mathbb{C}_{\tilde{i}}\mathbb{C}_i\xi$ if $\xi \in \mathcal{H}_i$. Then $\mathbb{C}_i^{-1} = \mathbb{C}_{\tilde{i}}$, in short

$$\mathbb{C}^{-1} = \mathbb{C}$$

Remark 1.64. The fact that $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ whenever $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ follows from the fact that if $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ then $\mathcal{H}_{\tilde{i}} \in \text{Rep}^{\phi^{-1}}(\mathcal{A})$.

Proof. Choose any $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$. Then, by Rem. 1.59,

$$\pi_{\tilde{i}, \varrho(-2\pi)\tilde{I}}(\phi x \phi^{-1}) = \mathbb{C}_{\pi_{i, \mathfrak{r}\varrho(-2\pi)\tilde{I}}}(\mathfrak{J}\phi x \phi^{-1}\mathfrak{J})\mathbb{C}^{-1} = \mathbb{C}_{\pi_{i, \varrho(2\pi)\mathfrak{r}\tilde{I}}}(\phi\mathfrak{J}x\mathfrak{J}\phi^{-1})\mathbb{C}^{-1}$$

which equals $\mathbb{C}_{\pi_{i, \mathfrak{r}\tilde{I}}}(\mathfrak{J}x\mathfrak{J})\mathbb{C}^{-1} = \pi_{\tilde{i}, \tilde{I}}(x)$ because $\mathcal{H}_i$ is ϕ -twisted. $\square$

Definition 1.65. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$. A bounded antilinear map $T : \mathcal{H}_i \rightarrow \mathcal{H}_j$ is called an **anti-morphism** of G -crossed $\mathcal{A}$ -modules if one of the following equivalent conditions hold:

(1) For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $x \in \mathcal{A}(I)$, we have

$$T\pi_{i, \tilde{I}}(x) = \pi_{j, \mathfrak{r}\tilde{I}}(\mathfrak{J}x\mathfrak{J})T$$

(2) The map $T \circ \mathbb{C}_{\tilde{i}}^{-1} : \mathcal{H}_{\tilde{i}} \rightarrow \mathcal{H}_j$ is a morphism of G -crossed $\mathcal{A}$ -modules.

(3) The map $\mathbb{C}_j \circ T : \mathcal{H}_i \rightarrow \mathcal{H}_{\tilde{j}}$ is a morphism of G -crossed $\mathcal{A}$ -modules.

The equivalence is clear by (1.42).

Proposition 1.66. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, and let $T \in \mathfrak{L}(\mathcal{H}_i, \mathcal{H}_j)$ be a morphism (resp. anti-morphism) of G -crossed $\mathcal{A}$ -modules. Then for each $g \in \mathcal{G}_{\mathcal{A}}$ we have

$$TU_i(g) = U_j(g)T \quad \text{resp.} \quad TU_i(g) = U_j(\mathfrak{r}g\mathfrak{r})T \quad (1.43)$$

Proof. We treat the case where T is an anti-morphism; the other case is similar and easier. By Lem. 1.18, it suffices to assume that $g \in \mathcal{G}_{\mathcal{A}}(I)$ for some $I \in \mathcal{J}$. Then

$$TU(g) = T\pi_I(U_0(g)) = \pi_{\mathfrak{r}I}(\mathfrak{J}U_0(g)\mathfrak{J})T = \pi_{\mathfrak{r}I}(U_0(\mathfrak{r}g\mathfrak{r}))T = U(\mathfrak{r}g\mathfrak{r})T$$

where Thm. 1.17 and (1.41) are used. $\square$

Corollary 1.67. If $T : \mathcal{H}_i \rightarrow \mathcal{H}_j$ is an anti-morphism of G -crossed $\mathcal{A}$ -modules, then

$$TU_i(\varrho_{\mathcal{A}}(t)) = U_j(\varrho_{\mathcal{A}}(-t))T$$

Proof. This follows immediately from Prop. 1.66 and Exp. 1.62. $\square$

Parallel to Rem. 1.33, we have:

Proposition 1.68. *Let $T : \mathcal{H}_i \rightarrow \mathcal{H}_j$ be an anti-morphism of G -crossed $\mathcal{A}$ -modules, then for each $I \in \mathcal{J}$ we have*

$$T\mathcal{H}_i(I) \subset \mathcal{H}_j(\mathfrak{r}I)$$

Moreover, if T is anti-unitary, then $T\mathcal{H}_i(I) = \mathcal{H}_j(\mathfrak{r}I)$.

Proof. Choose any $\xi \in \mathcal{H}_i(I)$. Note that $(\mathfrak{r}\tilde{I})' = \mathfrak{r}(\tilde{I})$. Therefore, for each $x \in \mathcal{A}((\mathfrak{r}I)')$, we have $\mathfrak{J}x\mathfrak{J} \in \mathcal{A}(\tilde{I})$, and hence

$$TR(\xi, \tilde{I})\mathfrak{J}x|_{\mathcal{H}_0} = T\pi_{\tilde{I}}(\mathfrak{J}x\mathfrak{J})R(\xi, \tilde{I})|_{\mathcal{H}_0} = \pi_{\mathfrak{r}(\tilde{I})}(x)TR(\xi, \tilde{I})\mathfrak{J}|_{\mathcal{H}_0}$$

So $TR(\xi, \tilde{I})\mathfrak{J}|_{\mathcal{H}_0}$ belongs to $\text{Hom}_{\mathcal{A}((\mathfrak{r}\tilde{I})')}(\mathcal{H}_0, \mathcal{H}_i)$, and hence $T\xi = TR(\xi, \tilde{I})\mathfrak{J}\Omega$ belongs to $\mathcal{H}_j(\mathfrak{r}I)$. We have thus finished proving $T\mathcal{H}_i(I) \subset \mathcal{H}_j(\mathfrak{r}I)$. If T is anti-unitary, then similarly we have $T^*\mathcal{H}_i(\mathfrak{J}) \subset \mathcal{H}_j(I)$, and hence $T\mathcal{H}_i(I) = \mathcal{H}_j(\mathfrak{r}I)$. $\square$

Remark 1.69. From the proof of Prop. 1.68, we see that if $\xi \in \mathcal{H}_i(I)$ and $T : \mathcal{H}_i \rightarrow \mathcal{H}_j$ is an anti-morphism of objects of $\text{Rep}^G(\mathcal{A})$, then

$$L(T\xi, \mathfrak{r}\tilde{I})|_{\mathcal{H}_0} = TR(\xi, \tilde{I})\mathfrak{J}|_{\mathcal{H}_0} \quad R(T\xi, \mathfrak{r}\tilde{I})|_{\mathcal{H}_0} = TL(\xi, \tilde{I})\mathfrak{J}|_{\mathcal{H}_0}$$

1.7.3 Conjugates and commutants

Theorem 1.70. *Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$. Then there exists a unique antiunitary map*

$$\Theta_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\bar{j}} \boxtimes \mathcal{H}_{\bar{i}}$$

abbreviated to Θ when no confusion arises, such that for each $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ and $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J)$,

$$\Theta_{i,j}L(\xi, \tilde{I})|_{\mathcal{H}_j} = R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\mathbb{C}|_{\mathcal{H}_{\bar{j}}} \quad (1.44a)$$

$$\Theta_{i,j}R(\eta, \tilde{J})|_{\mathcal{H}_i} = L(\mathbb{C}\eta, \mathfrak{r}\tilde{J})\mathbb{C}|_{\mathcal{H}_{\bar{i}}} \quad (1.44b)$$

Moreover, $\Theta_{i,j}$ is an anti-morphism of G -crossed $\mathcal{A}$ -modules.

Note that by Prop. 1.68, we have $\mathbb{C}\xi \in \mathcal{H}_i(\mathfrak{r}I)$ and $\mathbb{C}\eta \in \mathcal{H}_j(\mathfrak{r}J)$. Thus the RHS of (1.44) can be defined.

Proof. For each $\xi_1, \xi_2 \in \mathcal{H}_i(I)$ and $\eta_1, \eta_2 \in \mathcal{H}_j$, by (A.2) and Rem. 1.69,

$$\begin{aligned} \langle R(\mathbb{C}\xi_2, \mathfrak{r}\tilde{I})\mathbb{C}\eta_2 | R(\mathbb{C}\xi_1, \mathfrak{r}\tilde{I})\mathbb{C}\eta_1 \rangle &= \langle \mathbb{C}\eta_2 | \pi_{\mathfrak{r}\tilde{I}}(R(\mathbb{C}\xi_2, \mathfrak{r}\tilde{I})^* R(\mathbb{C}\xi_1, \mathfrak{r}\tilde{I}) |_{\mathcal{H}_0}) \mathbb{C}\eta_1 \rangle \\ &= \langle \mathbb{C}\pi_{\mathfrak{r}\tilde{I}}(\mathfrak{J}L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})\mathfrak{J} |_{\mathcal{H}_0}) \mathbb{C}\eta_1 | \eta_2 \rangle = \langle \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I}) |_{\mathcal{H}_0}) \eta_1 | \eta_2 \rangle \\ &= \langle L(\xi_1, \tilde{I})\eta_1 | L(\xi_2, \tilde{I})\eta_2 \rangle \end{aligned}$$

where the second last equality is due to Def. 1.63. Therefore, by the density of fusion products in Thm. 1.37, there exists a unique anti-unitary map $\Theta_{i,j}^{\tilde{I}} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\tilde{j}} \boxtimes \mathcal{H}_{\tilde{i}}$ sending each $L(\xi, \tilde{I})\eta$ to $R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\mathbb{C}\eta$. Similar to Step 2 of the proof of Thm. A.3, this map is independent of $\tilde{I}$, and hence can be denoted by $\Theta_{i,j}$.

To show that $\Theta_{i,j}$ is an anti-morphism, we choose any $\tilde{J} \in \tilde{\mathcal{J}}, y \in \mathcal{A}(J)$, let $\tilde{I} = \mathfrak{r}\tilde{J}$ and $\xi \in \mathcal{H}_i(I)$, and compute that

$$\Theta_{i,j}\pi_{\tilde{J}}(y)L(\xi, \tilde{I})\eta = \Theta_{i,j}L(\xi, \tilde{I})\pi_{\tilde{J}}(y)\eta = R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\mathbb{C}\pi_{\tilde{J}}(y)\eta = R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\pi_{\mathfrak{r}\tilde{J}}(\mathfrak{J}y\mathfrak{J})\mathbb{C}\eta$$

Since $\mathfrak{r}\tilde{J}$ is anticlockwise to $\mathfrak{r}\tilde{I}$, we conclude

$$\Theta_{i,j}\pi_{\tilde{J}}(y)L(\xi, \tilde{I})\eta = \pi_{\mathfrak{r}\tilde{J}}(\mathfrak{J}y\mathfrak{J})R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\mathbb{C}\eta = \pi_{\mathfrak{r}\tilde{J}}(\mathfrak{J}y\mathfrak{J})\Theta_{i,j}L(\xi, \tilde{I})\eta$$

Finally, for each $\xi \in \mathcal{H}_i(I)$ where $\tilde{I}$ is anticlockwise to $\tilde{J}$, we have

$$\Theta_{i,j}R(\eta, \tilde{J})\xi = \Theta_{i,j}L(\xi, \tilde{I})\eta = R(\mathbb{C}\xi, \mathfrak{r}\tilde{I})\mathbb{C}\eta = L(\mathbb{C}\eta, \mathfrak{r}\tilde{J})\mathbb{C}\xi$$

due to (1.44a) and Prop. 1.42. This proves (1.44b). $\square$

Remark 1.71. By (1.44), we clearly have

$$\Theta_{\tilde{j},\tilde{i}}\Theta_{i,j} = \text{id}_{\mathcal{H}_i \boxtimes \mathcal{H}_j}$$

In particular, $\Theta_{i,\tilde{i}}$ is an involutive antiunitary operator on $\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{i}}$.

Theorem 1.72. Let $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J} = \mathfrak{r}\tilde{I}$. Then

$$\mathbb{C}_i \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i) \mathbb{C}_i^{-1} = \text{End}_{\mathcal{A}(\mathfrak{r}\tilde{J})}(\mathcal{H}_{\tilde{i}})$$

Moreover, for each $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $A \in \text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$,

$$\Theta_{i,j} \cdot \pi_{\tilde{I}}^L(A) |_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot \Theta_{i,j}^{-1} = \pi_{\tilde{J}}^R(\mathbb{C}_i A \mathbb{C}_i^{-1}) |_{\mathcal{H}_{\tilde{j}} \boxtimes \mathcal{H}_{\tilde{i}}}$$

Proof. By Prop. 1.50, elements of $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i)$ are precisely of the form

$$A = L(\xi_2, \tilde{I})L(\xi_1, \tilde{I})^* |_{\mathcal{H}_i}$$

where $\xi_1, \xi_2 \in \mathcal{H}_i(\tilde{I})$. By Prop. 1.68, we have $\mathbb{C}\xi_1, \mathbb{C}\xi_2 \in \mathcal{H}_{\bar{i}}(\tilde{J})$. By Rem. 1.69, we have

$$\mathbb{C}_i A \mathbb{C}_i^{-1} = R(\mathbb{C}\xi_2, \tilde{J}) R(\mathbb{C}\xi_1, \tilde{J})^*|_{\mathcal{H}_{\bar{i}}} =: B$$

By Prop. 1.50, elements of $\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_{\bar{i}})$ are precisely of the form B . This proves the first identity. By Rem. 1.53 and Thm. 1.70,

$$\begin{aligned} \Theta_{i,j} \cdot \pi_{\tilde{I}}^L(A)|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot \Theta_{i,j}^{-1} &= \Theta_{i,j} \cdot L(\xi_2, \tilde{I}) L(\xi_1, \tilde{I})^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \cdot \Theta_{i,j}^* \\ &= R(\mathbb{C}\xi_2, \tilde{J}) R(\mathbb{C}\xi_1, \tilde{J})^*|_{\mathcal{H}_{\bar{j}} \boxtimes \mathcal{H}_{\bar{i}}} = \pi_{\tilde{J}}^R(B)|_{\mathcal{H}_{\bar{j}} \boxtimes \mathcal{H}_{\bar{i}}} \end{aligned}$$

This proves the second identity. □

Corollary 1.73. *Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J} = \mathfrak{r}\tilde{I}$. Let*

$$\mathcal{M} = \pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_j} \quad \mathcal{N} = \pi_{\tilde{J}}^R(\text{End}_{\mathcal{A}(\tilde{J})}(\mathcal{H}_{\bar{i}}))|_{\mathcal{H}_{\bar{j}} \boxtimes \mathcal{H}_{\bar{i}}}$$

Then $\Theta_{i,j} \mathcal{M} \Theta_{i,j}^{-1} = \mathcal{N}$. Moreover, if $j = \bar{i}$, and if $\tilde{I} = \tilde{\mathbb{S}}_+^1$ (and hence $\tilde{J} = \tilde{\mathbb{S}}_-^1$), then $\Theta_{i,\bar{i}} \mathcal{M} \Theta_{i,\bar{i}} = \mathcal{N} = \mathcal{M}'$.

Proof. Thm. 1.72 implies $\Theta_{i,j} \mathcal{M} \Theta_{i,j}^{-1} = \mathcal{N}$. When $j = \bar{i}$ and $\tilde{I} = \tilde{\mathbb{S}}_+^1$, then $\mathcal{N} = \mathcal{M}'$ follows from Thm. 1.57. □

2 The closed CFT

Fix a conformal net $\mathcal{A}$ with central charge c .

2.1 The closed-string state space $\mathcal{H}_{\text{cl}} = \mathcal{H}_{\sqrt{0}} \boxtimes_{\mathcal{A}^{\otimes 2}} \mathcal{H}_{\sqrt{0}}$

Recall that the tensor product conformal net

$$\mathcal{A}^{\otimes 2} = \mathcal{A} \otimes \mathcal{A}$$

with Hilbert space $\mathcal{H}_0 \otimes_{\mathbb{C}} \mathcal{H}_0 = \mathcal{H}_0 \otimes \mathcal{H}_0$ and vacuum vector $\Omega \otimes \Omega$. By definition, for each $I \in \mathcal{I}$, we have

$$\mathcal{A}^{\otimes 2}(I) = \mathcal{A}(I) \otimes \mathcal{A}(I) := \{x^+ \otimes x^- \in \mathcal{L}(\mathcal{H}_0 \otimes \mathcal{H}_0) : x^{\pm} \in \mathcal{A}(I)\}''$$

The strongly-continuous projectively-unitary representation of $\text{Diff}^+(\mathbb{S}^1)$ on $\mathcal{H}_0 \otimes \mathcal{H}_0$ is defined by

$$g \in \text{Diff}^+(\mathbb{S}^1) \mapsto U_0(g) \otimes U_0(g)$$

2.1.1 The separate conformal structures on untwisted $\mathcal{A}^{\otimes 2}$ -modules

Definition 2.1. Fix $\pm \in \{+, -\}$. Then each $\mathcal{H}_{\mathbb{J}} \in \text{Rep}(\mathcal{A}^{\otimes 2})$ gives rise to an $\mathcal{A}$ -module

$$(\mathcal{H}_{\mathbb{J}}, \pi_{\mathbb{J}}^{\pm})$$

abbreviated to $\pi^{\pm}$ when no confusion arises, such that for each $I \in \mathcal{J}$ and $x \in \mathcal{A}(I)$,

$$\pi_{\mathbb{J},I}^{+}(x) = \pi_{\mathbb{J},I}(x \otimes 1) \quad \pi_{\mathbb{J},I}^{-}(x) = \pi_{\mathbb{J},I}(1 \otimes x)$$

Definition 2.2. By Thm. 1.17, $(\pi_{\mathbb{J}}^{\pm}, \mathcal{H}_{\mathbb{J}})$ admits a canonical strongly-continuous unitary representation of $\mathcal{G}_{\mathcal{A}}$, which we denote by $U_{\mathbb{J}}^{\pm}$, or simply by $U^{\pm}$ when the context is clear. In other words, $U_{\mathbb{J}}^{\pm} : \mathcal{G}_{\mathcal{A}} \rightarrow \mathcal{H}_{\mathbb{J}}$ is the unique strongly-continuous unitary representation such that

$$U_{\mathbb{J}}^{+}(g) = \pi_{\mathbb{J},I}(U_0(g) \otimes 1) \quad U_{\mathbb{J}}^{-}(g) = \pi_{\mathbb{J},I}(1 \otimes U_0(g)) \quad (2.1)$$

for each $I \in \mathcal{J}, g \in \mathcal{G}_{\mathcal{A}}(I)$.

Remark 2.3. Let $\mathcal{H}_{\mathbb{J}} \in \text{Rep}(\mathcal{A}^{\otimes 2})$. Then for each $I, J \in \mathcal{H}$, we have

$$[\pi_{\mathbb{J},I}^{+}(\mathcal{A}(I)), \pi_{\mathbb{J},J}^{-}(\mathcal{A}(J))] = 0$$

Indeed, in the special case that $I \cup J \subset K$ for some $K \in \mathcal{J}$, the commutativity holds because for each $x \in \mathcal{A}(I), y \in \mathcal{A}(J)$, we have

$$\pi_{\mathbb{J},I}^{+}(x)\pi_{\mathbb{J},J}^{-}(y) = \pi_{\mathbb{J},K}(x \otimes y) = \pi_{\mathbb{J},J}^{-}(y)\pi_{\mathbb{J},I}^{+}(x)$$

The general case reduces to this special case by the additivity of $\mathcal{A}$.

It thus follows from Thm. 1.17 that

$$[U^{+}(g^{+}), U^{-}(g^{-})] = 0 \quad [U^{\pm}(g^{\pm}), \pi_{\mathbb{J},I}^{\mp}(x)] = 0$$

for each $\pm \in \{+, -\}, g^{\pm} \in \mathcal{G}_{\mathcal{A}}, I \in \mathcal{J}$, and $x \in \mathcal{A}(I)$. □

2.1.2 The σ -twisted $\mathcal{A}^{\otimes 2}$ -module $\mathcal{H}_{\sqrt{i}}$

Definition 2.4. For each $\tilde{I} \in \tilde{\mathcal{J}}$, define elements $\sqrt{\tilde{I}} = (\sqrt{I}, \arg_{\sqrt{I}})$ and $-\sqrt{\tilde{I}} = (-\sqrt{I}, \arg_{-\sqrt{I}})$ of $\tilde{\mathcal{J}}$ by

$$\begin{aligned} \sqrt{I} &= \{e^{\frac{i\theta}{2}} : \theta \in \arg_{\tilde{I}}(\tilde{I})\} & \arg_{\sqrt{I}}(e^{\frac{i\theta}{2}}) &= \frac{\theta}{2} \\ -\sqrt{I} &= \{-e^{\frac{i\theta}{2}} : \theta \in \arg_{\tilde{I}}(\tilde{I})\} & \arg_{-\sqrt{I}}(-e^{\frac{i\theta}{2}}) &= \frac{\theta}{2} - \pi \end{aligned}$$

for each $\theta \in \arg_{\tilde{I}}(\tilde{I})$. In other words, by viewing $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$ and writing $\tilde{I} = (a, b)$, we have $\sqrt{\tilde{I}} = (\frac{a}{2}, \frac{b}{2})$ and $-\sqrt{\tilde{I}} = (\frac{a}{2} - \pi, \frac{b}{2} - \pi)$.

We warn the readers that the notation $\sqrt{I}, -\sqrt{I}$ depends not only on I , but also on $\arg_I$.

Definition 2.5. For each $\tilde{I} \in \tilde{\mathcal{J}}$, choose $g_+, g_- \in \text{Diff}^+(\mathbb{S}^1)$ such that

$$g_+(e^{i\theta}) = e^{\frac{i\theta}{2}} \quad g_-(e^{i\theta}) = -e^{\frac{i\theta}{2}} \quad \text{for each } \theta \in \arg_{\tilde{I}}(\tilde{I})$$

Choose representing elements of $U_0(g_{\pm})$ (which we still denote by $U_0(g_{\pm})$), and define (spatial) isomorphisms of von Neumann algebras

$$\begin{aligned} \mathbf{Q}_{\sqrt{\tilde{I}}} : \mathcal{A}(I) &\rightarrow \mathcal{A}(\sqrt{I}) & x &\mapsto U_0(g_+)xU_0(g_+)^{-1} \\ \mathbf{Q}_{-\sqrt{\tilde{I}}} : \mathcal{A}(I) &\rightarrow \mathcal{A}(-\sqrt{I}) & x &\mapsto U_0(g_-)xU_0(g_-)^{-1} \end{aligned}$$

This definition is independent of the choice of $g_{\pm}$.

Explanation. To prove the independence of $\mathbf{Q}_{\sqrt{\tilde{I}}}$ on g_+ (and similarly the independence of $\mathbf{Q}_{-\sqrt{\tilde{I}}}$ on g_-), assume that h_+ satisfies the same property as g_+ . Then $h_+^{-1}g_+ \in \text{Diff}_{I'}(\mathbb{S}^1)$, and hence $U_0(h_+)^{-1}U_0(g_+)$ is a unitary element in $\mathcal{A}(I')$ and hence commutes with $\mathcal{A}(I)$. $\square$

Definition 2.6. Throughout this paper, we let $\sigma \in \text{Aut}(\mathcal{A}^{\otimes 2})$ be defined by

$$\sigma : \mathcal{H}_0 \otimes \mathcal{H}_0 \rightarrow \mathcal{H}_0 \otimes \mathcal{H}_0 \quad \xi \otimes \eta \mapsto \eta \otimes \xi$$

Definition 2.7. For each $\mathcal{H}_i \in \text{Rep}(\mathcal{A})$, define $\mathcal{H}_{\sqrt{i}} \in \text{Rep}^{\sigma}(\mathcal{A}^{\otimes 2})$ as follows. As a Hilbert space, $\mathcal{H}_{\sqrt{i}}$ is identical to $\mathcal{H}_i$. For each $\tilde{I} \in \tilde{\mathcal{J}}$, $\pi_{\sqrt{i}, \tilde{I}} : \mathcal{A}^{\otimes 2}(I) \rightarrow \mathfrak{L}(\mathcal{H}_i)$ is the unique normal unitary representation such that

$$\pi_{\sqrt{i}, \tilde{I}}(x^+ \otimes 1) = \pi_{i, \sqrt{I}} \circ \mathbf{Q}_{\sqrt{\tilde{I}}}(x^+) \quad \pi_{\sqrt{i}, \tilde{I}}(1 \otimes x^-) = \pi_{i, -\sqrt{I}} \circ \mathbf{Q}_{-\sqrt{\tilde{I}}}(x^-) \quad (2.2)$$

for each $x^{\pm} \in \mathcal{A}(I)$.

Proof. We need to prove the existence of such $\mathcal{H}_{\sqrt{i}}$; the uniqueness is obvious. By the split property (cf. Rem. 1.2), there exists a unitary $\Phi : \mathcal{H}_0 \rightarrow \mathcal{H}_0 \otimes \mathcal{H}_0$ such that

$$\Phi x^+ \Phi^{-1} = x^+ \otimes 1 \quad \Phi x^- \Phi^{-1} = 1 \otimes x^- \quad \text{for each } x^{\pm} \in \mathcal{A}(\pm\sqrt{I})$$

Let $g_{\pm}$ be as in Def. 2.5. Define $\pi_{\sqrt{0}, \tilde{I}} : \mathcal{A}^{\otimes 2}(I) \rightarrow \mathfrak{L}(\mathcal{H}_0)$ by

$$\pi_{\sqrt{0}, \tilde{I}} = \text{Ad}_{\Phi^{-1}} \circ \text{Ad}_{U_0(g_+) \otimes U_0(g_-)}$$

Then, for each $x^{\pm} \in \mathcal{A}(I)$, we have

$$\text{Ad}_{U_0(g_+) \otimes U_0(g_-)}(x^+ \otimes 1) = \mathbf{Q}_{\sqrt{\tilde{I}}}(x^+) \otimes 1 \quad \text{Ad}_{U_0(g_+) \otimes U_0(g_-)}(1 \otimes x^-) = 1 \otimes \mathbf{Q}_{-\sqrt{\tilde{I}}}(x^-)$$

and hence

$$\pi_{\sqrt{0}, \tilde{I}}(x^+ \otimes 1) = \mathbf{Q}_{\sqrt{\tilde{I}}}(x^+) \quad \pi_{\sqrt{0}, \tilde{I}}(1 \otimes x^-) = \mathbf{Q}_{-\sqrt{\tilde{I}}}(x^-)$$

In particular, $\pi_{\sqrt{0}, \tilde{I}}$ ranges in $\mathcal{A}(\sqrt{\tilde{I}}) \vee \mathcal{A}(-\sqrt{\tilde{I}})$. Choose any $K \in \mathcal{J}$ containing $\sqrt{\tilde{I}}$ and $-\sqrt{\tilde{I}}$. Then $\pi_{\sqrt{i}, \tilde{I}}$ can be defined by

$$\pi_{\sqrt{i}, \tilde{I}} = \pi_{i, K} \circ \pi_{\sqrt{0}, \tilde{I}}$$

We have thus constructed $(\mathcal{H}_{\sqrt{i}}, \pi_{\sqrt{i}})$ as a solitonic $\mathcal{A}$ -modules. By using (2.2), one easily checks that it is σ -twisted. $\square$

2.1.3 The separate conformal structures on $\mathcal{H}_{\sqrt{i}}$

Definition 2.8. For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $g \in \text{Diff}_I(\mathbb{S}^1)$, define $\sqrt{g} \in \text{Diff}_{\sqrt{\tilde{I}}}(\mathbb{S}^1)$ and $-\sqrt{g} \in \text{Diff}_{-\sqrt{\tilde{I}}}(\mathbb{S}^1)$ by

$$\sqrt{g} = g_+ \cdot g \cdot g_+^{-1} \quad -\sqrt{g} = g_- \cdot g \cdot g_-^{-1}$$

where $g_{\pm}$ are as in Def. 2.5. This definition is independent of the choice $g_{\pm}$. Indeed, $\sqrt{g}$ is the unique element in $\text{Diff}_{\sqrt{\tilde{I}}}(\mathbb{S}^1)$ whose restriction to $\sqrt{\tilde{I}}$ is the pullback of $g|_I : I \rightarrow I$ along the squaring map $\sqrt{\tilde{I}} \rightarrow I$, and $-\sqrt{g}$ is the unique element in $\text{Diff}_{-\sqrt{\tilde{I}}}(\mathbb{S}^1)$ whose restriction to $-\sqrt{\tilde{I}}$ is the pullback of $g|_I : I \rightarrow I$ along the squaring map $-\sqrt{\tilde{I}} \rightarrow I$.

One easily checks that $-\sqrt{g}$ is the pullback of $\sqrt{g}$ along the map $z \mapsto -z$, i.e.,

$$(-\sqrt{g})(z) = -(\sqrt{g}(-z)) \quad \text{for each } z \in \mathbb{S}^1 \quad (2.3)$$

Definition 2.9. For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $g \in \mathcal{G}(I)$, the elements $\sqrt{g} \in \mathcal{G}(\sqrt{\tilde{I}})$ and $-\sqrt{g} \in \mathcal{G}(-\sqrt{\tilde{I}})$ are defined by lifting the elements $\pm\sqrt{\kappa(g)} \in \text{Diff}_{\pm\sqrt{\tilde{I}}}(\mathbb{S}^1)$ to $\mathcal{G}(\pm\sqrt{\tilde{I}})$ via the covering map $\kappa : \mathcal{G} \rightarrow \text{Diff}^+(\mathbb{S}^1)$.

Definition 2.10. For each $\tilde{I} \in \tilde{\mathcal{J}}$ and $\tilde{g} = (g, V)$ in $\mathcal{G}_{\mathcal{A}}(I)$, we define $\sqrt{\tilde{g}} \in \mathcal{G}_{\mathcal{A}}(\sqrt{\tilde{I}})$ and $-\sqrt{\tilde{g}} \in \mathcal{G}_{\mathcal{A}}(-\sqrt{\tilde{I}})$ by

$$\sqrt{\tilde{g}} = (\sqrt{g}, \mathbf{Q}_{\sqrt{\tilde{I}}}(V)) \quad -\sqrt{\tilde{g}} = (-\sqrt{g}, \mathbf{Q}_{-\sqrt{\tilde{I}}}(V))$$

In particular, we have

$$U_0(\sqrt{\tilde{g}}) = \mathbf{Q}_{\sqrt{\tilde{I}}}(U_0(\tilde{g})) \quad U_0(-\sqrt{\tilde{g}}) = \mathbf{Q}_{-\sqrt{\tilde{I}}}(U_0(\tilde{g}))$$

Note that both $\pm\sqrt{g}$ and $\pm\sqrt{\tilde{g}}$ depend on $\arg_I$. On the other hand, it is clear that under the identification $\mathcal{G}_{\mathcal{A}} = \mathcal{G}_c$, the definition of $\pm\sqrt{\tilde{g}}$ is independent of the conformal net $\mathcal{A}$ extending Vir_c .

Proposition 2.11. Let $\tilde{I} \in \tilde{\mathcal{I}}$ and $\tilde{g} \in \mathcal{G}_{\mathcal{A}}(I)$. Then for each $\mathcal{H}_i \in \text{Rep}(\mathcal{A})$ we have

$$U_i(\sqrt{\tilde{g}}) = \pi_{\sqrt{i}, \tilde{I}}(U_0(\tilde{g}) \otimes 1) \quad U_i(-\sqrt{\tilde{g}}) = \pi_{\sqrt{i}, \tilde{I}}(1 \otimes U_0(\tilde{g}))$$

Proof. By Thm. 1.17 and Def. 2.10,

$$U_i(\sqrt{\tilde{g}}) = \pi_{i, \sqrt{I}}(U_0(\sqrt{\tilde{g}})) = \pi_{i, \sqrt{I}}(\mathbf{Q}_{\sqrt{\tilde{I}}}(U_0(\tilde{g}))) \stackrel{\text{Def. 2.7}}{=} \pi_{\sqrt{i}, \tilde{I}}(U_0(\tilde{g}) \otimes 1)$$

A similar argument proves the second identity. □

2.1.4 The state space of the closed CFT

Definition 2.12. Let $\mathbb{Z}_2 \leq \text{Aut}(\mathcal{A}^{\otimes 2})$ be generated by σ . Using the fusion product in $\text{Rep}^{\mathbb{Z}_2}(\mathcal{A}^{\otimes 2})$, we define the untwisted $\mathcal{A}^{\otimes 2}$ -module

$$\mathcal{H}_{\text{cl}} := \mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\overline{\sqrt{0}}}$$

where $\mathcal{H}_{\overline{\sqrt{0}}}$ is the $\mathbb{Z}_2$ -crossed conjugate module of $\mathcal{H}_{\sqrt{0}}$.

Remark 2.13. The σ -twisted module $\mathcal{H}_{\overline{\sqrt{0}}}$ is indeed unitarily isomorphic to $\mathcal{H}_{\sqrt{0}}$, as one can check (by applying Def. 1.60 to $g_{\pm}$) that $\mathfrak{J}_{\mathcal{A}}$ gives an anti-unitary anti-automorphism of $\mathcal{H}_{\sqrt{0}}$. However, we will not need this fact in the paper.

2.2 Double cones in the Einstein cylinder $\mathbb{R}_{\text{cl}}^{1,1}$

2.2.1 The Minkowski spaces

Recall that $\mathbb{R}^{1,1}$ is the two dimensional Minkowski space, i.e., $\mathbb{R}^{1,1} = \mathbb{R} \times \mathbb{R}$ with Minkowski metric $(dt)^2 - (dx)^2$, and x, t denote the first and second components of $\mathbb{R}^{1,1}$. Let

$$\mathbb{R}_{\text{cl}}^{1,1} = (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{R} \quad \mathbb{R}_{(-\pi, \pi)}^{1,1} = (-\pi, \pi) \times \mathbb{R}$$

where $\mathbb{R}_{\text{cl}}^{1,1}$ is viewed as a quotient space of $\mathbb{R}^{1,1}$, and $\mathbb{R}_{(-\pi, \pi)}^{1,1}$ is viewed as an open subset of both $\mathbb{R}^{1,1}$ and $\mathbb{R}_{\text{cl}}^{1,1}$. For each $x \in \mathbb{R}$, let

$$\mathcal{L}_x = (x + 2\pi\mathbb{Z}) \times \mathbb{R} \tag{2.4}$$

which is a straight line in $\mathbb{R}_{\text{cl}}^{1,1}$. Then

$$\mathbb{R}_{(-\pi, \pi)}^{1,1} \simeq \mathbb{R}_{\text{cl}}^{1,1} \setminus \mathcal{L}_{\pi} = \mathbb{R}_{\text{cl}}^{1,1} \setminus \mathcal{L}_{-\pi}$$

2.2.2 The u^+u^- -coordinates

Definition 2.14. We introduce a new set of coordinates (u^+, u^-) on $\mathbb{R}^{1,1}$ (and hence on $\mathbb{R}_{(-\pi, \pi)}^{1,1}$) by

$$\begin{cases} u^+ = t + x \\ u^- = t - x \end{cases} \quad \text{equivalently} \quad \begin{cases} x = \frac{u^+ - u^-}{2} \\ t = \frac{u^+ + u^-}{2} \end{cases}$$

In this coordinate, (u^+, u^-) and $(u^+ + 2n\pi, u^- - 2n\pi)$ correspond to the same point of $\mathbb{R}_{\text{cl}}^{1,1}$ when $n \in \mathbb{Z}$, that is,

$$\mathbb{R}_{\text{cl}}^{1,1} = \mathbb{R}^{1,1} / (2\pi, -2\pi)\mathbb{Z} \quad \text{in the } u^+u^- \text{ coordinates} \quad (2.5)$$

Definition 2.15. Two points p_1, p_2 of $\mathbb{R}_{\text{cl}}^{1,1}$ are called **spacelike separated** if, in the u^+u^- -coordinates, they can be represented by (u_1^+, u_1^-) and (u_2^+, u_2^-) such that

$$0 < u_2^+ - u_1^+ < 2\pi \quad 0 < u_1^- - u_2^- < 2\pi \quad (2.6a)$$

equivalently (by (2.5)),

$$0 < u_1^+ - u_2^+ < 2\pi \quad 0 < u_2^- - u_1^- < 2\pi \quad (2.6b)$$

For each $O \subset \mathbb{R}_{\text{cl}}^{1,1}$, the interior

$$O' := \text{Int}\{q \in \mathbb{R}_{\text{cl}}^{1,1} : q \text{ is spacelike separated with any } p \in O\}$$

is called the **spacelike complement** of O .

Remark 2.16. Suppose that $p_1, p_2 \in \mathbb{R}_{(-\pi, \pi]}^{1,1} = (-\pi, \pi] \times \mathbb{R}$ are spacelike separated. Then precisely one of (2.6) holds true. In fact, (2.6a) holds iff $x_1 < x_2$, and (2.6b) holds iff $x_2 < x_1$.

Proof. Clearly $x_1 \neq x_2$, otherwise $u_2^+ - u_1^+$ and $u_2^- - u_1^-$ would have the same sign, contradicting (2.6).

Suppose $x_1 < x_2$. Choose $(w_2^+, w_2^-) \in (u_2^+, u_2^-) + (2\pi, -2\pi)\mathbb{Z}$ such that

$$0 < w_2^+ - u_1^+ < 2\pi \quad 0 < u_1^- - w_2^- < 2\pi \quad (2.7)$$

Let $\tilde{x}_2$ be the x -coordinate of (w_2^+, w_2^-) . (Note that (w_2^+, w_2^-) and (u_2^+, u_2^-) have the same t -coordinate t_2 .) Then $x_1 < \tilde{x}_2 < x_1 + 2\pi$, because

$$(w_2^+ - w_2^-) - (u_1^+ - u_1^-) = (w_2^+ - u_1^+) + (u_1^- - w_2^-)$$

is > 0 and $< 4\pi$.

Since $x_1 > -\pi$, we have $\tilde{x}_2 > -\pi$. We claim that $\tilde{x}_2 \leq \pi$. If not, then $\pi < \tilde{x}_2 < x_1 + 2\pi$, and hence

$$-\pi < \tilde{x}_2 - 2\pi < x_1$$

In particular, $-\pi < \tilde{x}_2 - 2\pi \leq \pi$. Thus, the unique representation of p_2 in $\mathbb{R}_{(-\pi, \pi]}^{1,1}$ is given by $(\tilde{x}_2 - 2\pi, t_2)$. Therefore $x_2 = \tilde{x}_2 - 2\pi$, and hence $x_2 < x_1$, contradicting the assumption $x_1 < x_2$.

We have thus proved that $-\pi < \tilde{x}_2 \leq \pi$. Then $(\tilde{x}_2, t_2)$ is the unique representation of p_2 in $\mathbb{R}_{(-\pi, \pi]}^{1,1}$. Thus $w_2^+ = u_2^+$ and $w_2^- = u_2^-$, and hence (2.6a) follows from (2.7). A similar argument shows that if $x_2 < x_1$ then (2.6b) follows. $\square$

2.2.3 Double cones

Definition 2.17. For each element (u^+, u^-) of $\mathbb{R}^{1,1}$ (in the u^+u^- coordinate), its image in $\mathbb{R}_{\text{cl}}^{1,1}$ is called a point in the **closed string worldsheet**, and the ordered pair of arg-valued points (e^{iu^+}, e^{iu^-}) with arguments u^+, u^- is called the corresponding pair in the **double circle picture**.

Definition 2.18. A subset $O \subset \mathbb{R}_{\text{cl}}^{1,1}$ is called a **double cone** if it is the image of an open rectangle $K^+ \times K^-$ (in the u^+u^- coordinate) under the quotient map $\mathbb{R}^{1,1} \rightarrow \mathbb{R}_{\text{cl}}^{1,1}$, where the lengths of the open intervals $K^\pm \subset \mathbb{R}$ satisfy

$$0 < |K^+|, |K^-| < 2\pi$$

Equivalently, a double cone is an open subset of the closed string worldsheet corresponding to a pair $(\tilde{I}, \tilde{J})$ with $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ in the double circle picture.

Definition 2.19. The **top**, **bottom**, **left**, and **right corners** of O are understood in the usual sense. That is, if in the u^+u^- -coordinates we have $O = K^+ \times K^-$, by setting $a^\pm = \inf K^\pm$ and $b^\pm = \sup K^\pm$, these four corners are respectively the images of the following points under the covering map $\mathbb{R}^{1,1} \rightarrow \mathbb{R}_{\text{cl}}^{1,1}$.

$$(b^+, b^-) \quad (a^+, a^-) \quad (a^+, b^-) \quad (b^+, a^-)$$

Remark 2.20. Note that infinitely many pairs $(\tilde{I}, \tilde{J})$ correspond to a given double cone. However, if O is a double cone in $\mathbb{R}_{(-\pi, \pi]}^{1,1}$, since the u^+u^- -coordinates on $\mathbb{R}_{(-\pi, \pi]}^{1,1}$ are genuine coordinates, O corresponds to a unique pair $(\tilde{I}, \tilde{J})$ in the double circle picture, i.e., the one satisfying

$$|\arg_I z - \arg_J \zeta| < 2\pi \quad \text{for each } z \in I, \zeta \in J \quad (2.8)$$

Conversely, any $(\tilde{I}, \tilde{J})$ satisfying (2.8) corresponds to a double cone $O \subset \mathbb{R}_{(-\pi, \pi]}^{1,1}$.

Convention 2.21. Unless otherwise specified, the pair $(\tilde{I}, \tilde{J})$ corresponding to a double cone $O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ is chosen to satisfy (2.8).

Example 2.22. Let $O \subset \mathbb{R}_{\text{cl}}^{1,1}$ be a double cone. Then O' is also a double cone. Indeed, if $(\tilde{I}, \tilde{J})$ corresponds to O , then both $(\tilde{I}', \tilde{J})$ and $(\tilde{I}, \tilde{J}')$ correspond to O' .

Definition 2.23. A double cone $O \subset \mathbb{R}_{\text{cl}}^{1,1}$ is called **equilateral** if it corresponds to some $(\tilde{I}, \tilde{J})$ in the double circle picture with $|\tilde{I}| = |\tilde{J}|$. A double cone is called **standard** if it corresponds to $(\tilde{I}, \tilde{I})$ for a (necessarily unique) $\tilde{I} \in \tilde{\mathcal{J}}$.

Standard double cones must be compactly contained in $\mathbb{R}_{(-\pi, \pi)}^{1,1}$. Indeed, one checks easily that, in the xt -coordinate, a standard double cone is precisely double cone $O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ whose top and bottom corners lie in the t -axis.

Remark 2.24. Let $O \subset \mathbb{R}_{\text{cl}}^{1,1}$ be a double cone corresponding to $(\tilde{I}, \tilde{J})$. Then

$$O \Subset \mathbb{R}_{(-\pi, \pi)}^{1,1} \iff \text{there exists } \tilde{K} \in \tilde{\mathcal{J}} \text{ containing } \tilde{I}, \tilde{J}$$

In other words, $O \Subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ iff O is contained in some standard double cone.

Proof. This is due to (2.8), which implies that when $\tilde{I}, \tilde{J}$ are viewed as intervals in $\mathbb{R}$, the property $O \Subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ holds iff the diameter of $\tilde{I} \cup \tilde{J}$ is $< 2\pi$. $\square$

Definition 2.25. We let

$$\mathcal{O} = \{\text{double cones } O \subset \mathbb{R}_{\text{cl}}^{1,1}\} \quad \mathcal{O}_{(-\pi, \pi)} = \{\text{double cones } O \Subset \mathbb{R}_{(-\pi, \pi)}^{1,1}\}$$

2.2.4 Conformal symmetry

Definition 2.26. We define a group action of $\mathcal{G} \times \mathcal{G}$ on $\mathbb{R}_{\text{cl}}^{1,1}$, which lifts to a group action of $\mathcal{G}_c \times \mathcal{G}_c$ on $\mathbb{R}_{\text{cl}}^{1,1}$. For each $g^+, g^- \in \mathcal{G}$, by viewing them as diffeomorphisms of $\mathbb{R}$ (cf. (1.3)), we let

$$(g^+, g^-)(u^+, u^-) = (g^+(u^+), g^-(u^-))$$

Thanks to (2.5) and the relation $g^\pm(u + 2\pi) = g^\pm(u) + 2\pi$, this defines a well-defined action of $\mathcal{G} \times \mathcal{G}$ on $\mathbb{R}_{\text{cl}}^{1,1}$.

Remark 2.27. Although we will not use this fact, we note that $\mathcal{G} \times \mathcal{G}$ is the universal cover of the group $\text{Cf}(\mathbb{R}_{\text{cl}}^{1,1})$ of diffeomorphisms of $\mathbb{R}_{\text{cl}}^{1,1}$ preserving both the orientation of the manifold $\mathbb{R}_{\text{cl}}^{1,1}$ and the direction of t . Moreover, the kernel of the covering map $\mathcal{G} \times \mathcal{G} \rightarrow \text{Cf}(\mathbb{R}_{\text{cl}}^{1,1})$ is isomorphic to $\mathbb{Z}$, generated by $(\varrho(2\pi), \varrho(-2\pi))$.

2.3 The bulk net $\mathcal{B}_{\text{cl}}$ acting on $\mathcal{H}_{\text{cl}}$

Definition 2.28. For each $I, J \in \mathcal{J}$ whose closures are disjoint, let

$$\mathfrak{A}(I, J) = \mathcal{A}(I^\vee)' \cap \mathcal{A}(J^\vee)'$$

where $I^\vee, J^\vee$ are the two components of the interior of $\mathbb{S}^1 \setminus (I \cup J)$.

For example, if $\tilde{K} \in \tilde{\mathcal{J}}$ and $I = \sqrt{K}, J = -\sqrt{K}$, then $I^\vee, J^\vee$ can be chosen to be $\sqrt{K'}, -\sqrt{K'}$.

2.3.1 The net $\mathcal{B}_{\text{cl}}$ defined on $\mathcal{O}_{(-\pi, \pi)}$

Recall Thm. 1.51 for the normal representation π_I^L . For each $\tilde{I} \in \tilde{\mathcal{J}}$, by the construction of $\mathcal{H}_{\sqrt{0}}$ in Subsec. 2.1.2, we have

$$\text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I}')}(\mathcal{H}_{\sqrt{0}}) = \mathcal{A}(\sqrt{I'})' \cap \mathcal{A}(-\sqrt{I'})' = \mathfrak{A}(\sqrt{I}, -\sqrt{I}) \quad (2.9)$$

Consequently, if $\tilde{I}, \tilde{J} \subset \tilde{K}$, then

$$\mathfrak{A}(\sqrt{I}, -\sqrt{J}) \subset \text{End}_{\mathcal{A}^{\otimes 2}(\tilde{K}')}(\mathcal{H}_{\sqrt{0}}) \quad (2.10)$$

where $\sqrt{I}$ and $-\sqrt{J}$ are defined using the clockwise complements $\tilde{I}$ and $\tilde{J}$, respectively.

Definition 2.29. Let $O \in \mathcal{O}_{(-\pi, \pi)}$, which corresponds uniquely to some $(\tilde{I}, \tilde{J})$ satisfying (2.8). By Rem. 2.24, there exists a standard double cone D containing O . Choose $\tilde{K} \in \tilde{\mathcal{J}}$ such that $(\tilde{K}, \tilde{K})$ represents D , and so $\tilde{I}, \tilde{J} \subset \tilde{K}$. Define the von Neumann algebra $\mathcal{B}_{\text{cl}}(O)$ on $\mathcal{H}_{\text{cl}}$ by

$$\mathcal{B}_{\text{cl}}(O) = \pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}}$$

In particular, if O is standard (and hence $\tilde{I} = \tilde{J}$), then

$$\mathcal{B}_{\text{cl}}(O) = \pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I}')}(\mathcal{H}_{\sqrt{0}}))|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}} \quad (2.11)$$

Explanation. This definition is independent of the choice of D containing O . In fact, for two such double cones D_1, D_2 corresponding to $(\tilde{K}_1, \tilde{K}_1)$ and $(\tilde{K}_2, \tilde{K}_2)$ respectively, choose $\tilde{K}_0 \in \tilde{\mathcal{J}}$ containing $\tilde{I}, \tilde{J}$ and contained in $\tilde{K}_1, \tilde{K}_2$. By Rem. 1.54,

$$\pi_{\tilde{K}_i}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}} = \pi_{\tilde{K}_0}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}}$$

for $i = 1, 2$. □

2.3.2 The net $\mathcal{B}_{\text{cl}}$ defined on $\mathcal{O}$

Definition 2.30. The net of von Neumann algebras $\mathcal{B}_{\text{cl}} : \mathcal{O} \in \mathcal{O} \mapsto \mathcal{B}_{\text{cl}}(\mathcal{O}) \subset \mathfrak{L}(\mathcal{H}_{\text{cl}})$ is defined as follows. If $\mathcal{O} \in \mathcal{O}_{(-\pi, \pi)}$, then $\mathcal{B}_{\text{cl}}(\mathcal{O})$ is defined as in Def. 2.29. If $\mathcal{O} \notin \mathcal{O}_{(-\pi, \pi)}$, that is, if the line $\mathcal{L}_\pi$ defined in (2.4) intersects the closure of $\mathcal{O}$, by viewing $\mathbb{R}_{(-\pi, \pi)}^{1,1} = \mathbb{R}_{\text{cl}}^{1,1} \setminus \mathcal{L}_\pi$, we let

$$\mathcal{B}_{\text{cl}}(\mathcal{O}) = \left(\bigcup_{D \in \mathcal{O}_{(-\pi, \pi)}, D \subset \mathcal{O}} \mathcal{B}_{\text{cl}}(D) \right)'' \quad (2.12)$$

Definition 2.31. For each $g^+, g^- \in \mathcal{G}_c$, define $U_{\text{cl}}(g^+, g^-) \in \mathcal{U}(\mathcal{H}_{\text{cl}})$ by

$$U_{\text{cl}}(g^+, g^-) = U_{\text{cl}}^+(g^+) U_{\text{cl}}^-(g^-)$$

where $U_{\text{cl}}^+, U_{\text{cl}}^- : \mathcal{G}_c \rightarrow \mathcal{U}(\mathcal{H}_{\text{cl}})$ are defined in Def. 2.2. Then

$$U_{\text{cl}} : \mathcal{G}_c \times \mathcal{G}_c \rightarrow \mathcal{U}(\mathcal{H}_{\text{cl}})$$

is clearly a strongly-continuous unitary representation.

The following observation will be crucial to the proof of the conformal covariance of $\mathcal{B}_{\text{cl}}$.

Remark 2.32. By Def. 2.2, if $I \in \mathcal{J}$ and $g^+, g^- \in \mathcal{G}_c(I)$, then

$$U_{\text{cl}}(g^+, g^-) = \pi_{\sqrt{0} \boxtimes \sqrt{0}, I} (U_0(g^+) \otimes U_0(g^-)) \quad (2.13)$$

By Prop. 2.11, if $\tilde{I} \in \tilde{\mathcal{J}}$ and $g^+, g^- \in \mathcal{G}_c(I)$, noting that $\sqrt{g^+} \in \mathcal{G}_c(\sqrt{I})$ and $-\sqrt{g^-} \in \mathcal{G}_c(-\sqrt{I})$ are defined in Def. 2.10 and rely on $\arg_I$, we have

$$U_0(\sqrt{g^+} \cdot (-\sqrt{g^-})) = \pi_{\sqrt{0}, \tilde{I}} (U_0(g^+) \otimes U_0(g^-)) \quad (2.14)$$

Recall that τ acts on the universal cover $\mathbb{R}$ of $\mathbb{S}^1$ by $u \mapsto -u$. Thus (τ, τ) acts on $\mathbb{R}^{1,1}$ and hence acts on $\mathbb{R}_{\text{cl}}^{1,1}$:

$$(\tau, \tau) : \mathbb{R}_{\text{cl}}^{1,1} \rightarrow \mathbb{R}_{\text{cl}}^{1,1} \quad (u^+, u^-) \mapsto (-u^+, -u^-) \quad (2.15)$$

Theorem 2.33. The net $\mathcal{B}_{\text{cl}}$ satisfies the following properties for each $\mathcal{O}, \mathcal{O}_1, \mathcal{O}_2 \in \mathcal{O}$.

- (1) (Isotony) If $\mathcal{O}_1 \subset \mathcal{O}_2$, then $\mathcal{B}_{\text{cl}}(\mathcal{O}_1) \subset \mathcal{B}_{\text{cl}}(\mathcal{O}_2)$.
- (2) (Locality) If $\mathcal{O}_1$ and $\mathcal{O}_2$ are spacelike separated, then $\mathcal{B}_{\text{cl}}(\mathcal{O}_1)$ commutes with $\mathcal{B}_{\text{cl}}(\mathcal{O}_2)$.
- (3) (Extension property) If $\mathcal{O}$ corresponds to $(\tilde{I}, \tilde{J})$, then

$$\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I)) \vee \pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J)) \subset \mathcal{B}_{\text{cl}}(\mathcal{O})$$

(4) (Conformal covariance) For each $g^+, g^- \in \mathcal{G}_c$, we have

$$U_{\text{cl}}(g^+, g^-) \mathcal{B}_{\text{cl}}(O) U_{\text{cl}}(g^+, g^-)^* = \mathcal{B}_{\text{cl}}((g^+, g^-)O)$$

Moreover, if O corresponds (non-uniquely) to some $(\tilde{I}, \tilde{J})$ in the double circle picture, and if $g^+ \in \mathcal{G}_c(I), g^- \in \mathcal{G}_c(J)$, then

$$U_{\text{cl}}(g^+, g^-) \in \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I)) \vee \pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J))$$

and hence $U_{\text{cl}}(g^+, g^-) \in \mathcal{B}_{\text{cl}}(O)$ by the extension property.

(5) (PCT symmetry) There exists an anti-unitary operator $\mathfrak{J}_{\text{cl}}$ on $\mathcal{H}_{\text{cl}}$ satisfying

$$\begin{aligned} \mathfrak{J}_{\text{cl}}^2 &= 1 & \mathfrak{J}_{\text{cl}} U_{\text{cl}}(g^+, g^-) \mathfrak{J}_{\text{cl}} &= U_{\text{cl}}(\mathfrak{r}g^+ \mathfrak{r}, \mathfrak{r}g^- \mathfrak{r}) \\ \mathfrak{J}_{\text{cl}} \mathcal{B}_{\text{cl}}(O) \mathfrak{J}_{\text{cl}} &= \mathcal{B}_{\text{cl}}((\mathfrak{r}, \mathfrak{r})O) \end{aligned}$$

for each $g^+, g^- \in \mathcal{G}_c$ and $O \in \mathcal{O}$.

(6) (Positive energy) The self-adjoint generators of the one-parameter groups $u \mapsto U_{\text{cl}}(\varrho_c(u), 0)$ and $u \mapsto U_{\text{cl}}(0, \varrho_c(u))$ are positive.

(7) (Irreducibility) $\mathcal{B}_{\text{cl}}(O)$ is a type III factor. Moreover, $\bigcup_{O \in \mathcal{O}} \mathcal{B}_{\text{cl}}(O)$ generates the von Neumann algebra $\mathfrak{L}(\mathcal{H}_{\text{cl}})$.

Proof. Isotony: If $O_1, O_2 \in \mathcal{O}_{(-\pi, \pi)}$, the inclusion is obvious. If $O_1 \in \mathcal{O}_{(-\pi, \pi)}, O_2 \notin \mathcal{O}_{(-\pi, \pi)}$, this follows by choosing $O = O_2, D = O_1$ in (2.12). The case $O_1, O_2 \notin \mathcal{O}_{(-\pi, \pi)}$ follows from (2.12) and the case $O_1, O_2 \in \mathcal{O}_{(-\pi, \pi)}$.

Locality: See Thm. 2.43 in Sec. 2.6.

Extension property: See Cor. 2.56 in Sec. 2.8.

Conformal covariance: See Thm. 2.55 and Cor. 2.57 in Sec. 2.8.

PCT symmetry: See Thm. 2.60 in Sec. 2.9.

Positive energy: By [Wei06], for each (normal) representation of $\mathcal{A}$ the generator of rotation is positive. Apply this result to $\pi_{\text{cl}}^\pm$.

Irreducibility: In the special case that O is standard and hence corresponds to $(\tilde{I}, \tilde{I})$ for some $\tilde{I} \in \tilde{\mathcal{J}}$, $\mathcal{B}_{\text{cl}}(O)$ is described by (2.11), and hence is a type III factor by Rem. 1.55. The general case reduces to this special case by the conformal covariance mentioned above. In particular, since $\mathcal{B}_{\text{cl}}(O)$ is a factor, $\mathcal{B}_{\text{cl}}(O) \cup \mathcal{B}_{\text{cl}}(O)'$ generates $\mathfrak{L}(\mathcal{H}_{\text{cl}})$. By the Haag duality (Thm. 2.62), $\mathcal{B}_{\text{cl}}(O) \cup \mathcal{B}_{\text{cl}}(O')$ generates $\mathfrak{L}(\mathcal{H}_{\text{cl}})$. Thus $\bigcup_{O \in \mathcal{O}} \mathcal{B}_{\text{cl}}(O)$ generates $\mathfrak{L}(\mathcal{H}_{\text{cl}})$. $\square$

2.4 Local conformal covariance for $\mathcal{B}_{\text{cl}}|_{\mathcal{O}_{(-\pi, \pi)}}$

Lemma 2.34. Let $O \in \mathcal{O}_{(-\pi, \pi)}$ be contained in some standard double cone $\hat{O}$. Assume that $\hat{O}$ corresponds to $(\tilde{K}, \tilde{K})$ in the double circle picture. Then for each $g^+, g^- \in \mathcal{G}_{\mathcal{A}}(K)$ we have

$$U_{\text{cl}}(g^+, g^-) \mathcal{B}_{\text{cl}}(O) U_{\text{cl}}(g^+, g^-)^* = \mathcal{B}_{\text{cl}}((g^+, g^-)O)$$

Proof. Assume that O corresponds to $(\tilde{I}, \tilde{J})$. Then $\tilde{I}, \tilde{J} \subset \tilde{K}$. Let $\sqrt{g^+}, -\sqrt{g^-}$ be defined by $\tilde{K}$. By Rem. 2.32,

$$\begin{aligned} \text{Ad}_{U_{\text{cl}}(g^+, g^-)}(\mathcal{B}_{\text{cl}}(O)) &= \text{Ad}_{U_{\text{cl}}(g^+, g^-)}\left(\pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right) \\ &= \text{Ad}_{\pi_{\sqrt{0} \boxtimes \sqrt{0}, K}(U_0(g^+) \otimes U_0(g^-))}\left(\pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right) \end{aligned}$$

By (1.34), the above equals

$$\begin{aligned} &\text{Ad}_{\pi_{\tilde{K}}^L(\pi_{\sqrt{0}, \tilde{K}}(U_0(g^+) \otimes U_0(g^-)))}\left(\pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right) \\ &= \pi_{\tilde{K}}^L\left(\text{Ad}_{\pi_{\sqrt{0}, \tilde{K}}(U_0(g^+) \otimes U_0(g^-))}(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right) \\ &= \pi_{\tilde{K}}^L\left(\text{Ad}_{U_0(\sqrt{g^+} \cdot (-\sqrt{g^-}))}(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right) \end{aligned}$$

where Rem. 2.32 is used in the last equality. Note that $\sqrt{I} \subset \sqrt{K}$, $-\sqrt{J} \subset -\sqrt{K}$, and $\sqrt{g^+} \cdot (-\sqrt{g^-})$ fixes points outside $\sqrt{K} \cup (-\sqrt{K})$.

Moreover, if $O_1 := (g^+, g^-)O$ corresponds to $(\tilde{I}_1, \tilde{J}_1)$, then we have $\tilde{I}_1, \tilde{J}_1 \subset \tilde{K}$, $g^+ \tilde{I} = \tilde{I}_1$, and $g^- \tilde{J} = \tilde{J}_1$. Thus

$$\sqrt{g^+} \sqrt{I} = \sqrt{I_1} \quad (-\sqrt{g^-})(-\sqrt{J}) = -\sqrt{J_1}$$

Therefore, by the conformal covariance of $\mathcal{A}$, we have

$$\text{Ad}_{U_0(\sqrt{g^+} \cdot (-\sqrt{g^-}))}(\mathfrak{A}(\sqrt{I}, -\sqrt{J})) = \mathfrak{A}(\sqrt{I_1}, -\sqrt{J_1})$$

Inserting the right hand side above into $\pi_{\tilde{K}}^L$ yields $\mathcal{B}_{\text{cl}}(O_1)$. This finishes the proof. $\square$

Lemma 2.35. *Let $O \in \mathcal{O}_{(-\pi, \pi)}$. Let $a, b > 0$ such that the interval $(-a, b)$ is the connected component containing 0 of the set*

$$\{s \in \mathbb{R} : (\varrho(s), \varrho(-s))O \in \mathbb{R}_{(-\pi, \pi)}^{1,1}\}$$

Then for each $-a < s < b$ we have

$$U_{\text{cl}}(\varrho_c(s), \varrho_c(-s)) \cdot \mathcal{B}_{\text{cl}}(O) \cdot U_{\text{cl}}(\varrho_c(s), \varrho_c(-s))^* = \mathcal{B}_{\text{cl}}((\varrho(s), \varrho(-s))O)$$

Note that $(\varrho(s), \varrho(-s))$ is the translation in the x -axis by s units.

Proof. We treat the case $s > 0$; the case $s < 0$ is similar. It suffices to prove the lemma for s satisfying the extra condition that there exists a standard cone $\hat{O}_1$ such that

$$\overline{\bigcup_{0 \leq v \leq s} (\varrho(v), \varrho(-v))O} \subset \hat{O}_1$$

Then applying this result repeatedly proves the lemma for general s .

Choose standard double cones $\hat{O}_2, \hat{O}_3$ such that $\hat{O}_1 \subseteq \hat{O}_2 \subseteq \hat{O}_3$. Let $\hat{O}_i, O$ correspond to $(\tilde{K}_i, \tilde{K}_i)$ and $(\tilde{I}, \tilde{J})$ respectively. Then, by viewing $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$, we have

$$\tilde{K}_1 \subseteq \tilde{K}_2 \subseteq \tilde{K}_3 \quad \overline{\bigcup_{0 \leq v \leq s} \varrho(v)\tilde{I}} \subset \tilde{K}_1 \quad \overline{\bigcup_{0 \leq v \leq s} \varrho(-v)\tilde{J}} \subset \tilde{K}_1$$

We now consider two vector fields on $\mathbb{R}$ as the universal cover of $\mathbb{S}^1$. The first one is ∂_u , where u denotes the standard coordinate of $\mathbb{R}$. The second one is $f\partial_u$ where $f : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth 2π -periodic function which equals 1 on $\tilde{K}_2$, which vanishes outside $E \setminus \tilde{K}_3$ where $E \subset \mathbb{R}$ is any interval of length 2π containing $\tilde{K}_3$. Then, in view of the description (1.3) of $\mathcal{G}$, the flow on $\mathbb{R}$ generated by ∂_u defines the rotation subgroup ϱ of $\mathcal{G}$, and the flow on $\mathbb{R}$ generated by $f\partial_u$ defines a one-parameter subgroup $\gamma : \mathbb{R} \rightarrow \mathcal{G}$. We choose an arbitrary lift of γ to $\gamma_c : \mathbb{R} \rightarrow \mathcal{G}_c$.

Clearly, there exists $\varepsilon > 0$ such that the each $-\varepsilon \leq v \leq \varepsilon$, the translation of $\tilde{K}_1$ by v units remains inside $\tilde{K}_2$. We now fix $0 \leq v \leq \varepsilon$. Then

$$g^+ := \gamma_c(v)^{-1}\varrho_c(v) \in \mathcal{G}_c(K'_1) \quad g^- := \gamma_c(-v)^{-1}\varrho_c(-v) \in \mathcal{G}_c(K'_1)$$

since $\gamma(\pm v)^{-1}\varrho(\pm v) : \mathbb{R} \rightarrow \mathbb{R}$ fixes points of $\tilde{K}_1 + 2\pi\mathbb{Z}$. Thus, by Rem. 2.32,

$$U_{\text{cl}}(g^+, g^-) = \pi_{\sqrt{0}\boxtimes\sqrt{0}, K'}(U_0(g^+) \otimes U_0(g^-)) \stackrel{(1.38)}{=} \pi_{\tilde{K}'}^R(\pi_{\sqrt{0}, \tilde{K}'}(U_0(g^+) \otimes U_0(g^-)))$$

Since $\pi_{\tilde{K}}^L$ commutes with $\pi_{\tilde{K}'}^R$ (cf. Thm. 1.57), the expression

$$\text{Ad}_{U_{\text{cl}}(g^+, g^-)}(\mathcal{B}_{\text{cl}}(O)) = \text{Ad}_{U_{\text{cl}}(g^+, g^-)}\left(\pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))\right)$$

equals $\pi_{\tilde{K}}^L(\mathfrak{A}(\sqrt{I}, -\sqrt{J}))$, and hence

$$\text{Ad}_{U_{\text{cl}}(g^+, g^-)}(\mathcal{B}_{\text{cl}}(O)) = \mathcal{B}_{\text{cl}}(O)$$

On the other hand, since $\gamma(\pm v) \in \mathcal{G}(K_3)$, we have $\gamma_c(\pm v) \in \mathcal{G}_c(K_3)$, and hence

$$\text{Ad}_{U_{\text{cl}}(\gamma_c(v), \gamma_c(-v))}(\mathcal{B}_{\text{cl}}(O)) = \mathcal{B}_{\text{cl}}((\gamma(v), \gamma(-v))O) = \mathcal{B}_{\text{cl}}((\varrho(v), \varrho(-v))O)$$

by Lem. 2.34. Combining the above two relations together yields

$$\text{Ad}_{U_{\text{cl}}(\varrho_c(v), \varrho_c(-v))}(\mathcal{B}_{\text{cl}}(O)) = \mathcal{B}_{\text{cl}}((\varrho(v), \varrho(-v))O)$$

for each $0 \leq v \leq \varepsilon$. Replacing O with $(\varrho(\varepsilon), \varrho(-\varepsilon))O$, the above relation holds for $\varepsilon \leq v \leq 2\varepsilon$. Repeating this procedure, we obtain the above relation for all $0 \leq v \leq s$. $\square$

Lemma 2.36. *Let $O \in \mathcal{O}$. Then for each $s \in \mathbb{R}$ we have*

$$U_{\text{cl}}(\varrho_c(s), \varrho_c(s)) \cdot \mathcal{B}_{\text{cl}}(O) \cdot U_{\text{cl}}(\varrho_c(s), \varrho_c(s))^* = \mathcal{B}_{\text{cl}}((\varrho(s), \varrho(s))O)$$

Note that $(\varrho(s), \varrho(s))$ is the translation in the t -axis by s units. Therefore, $(\varrho(s), \varrho(s))O$ belongs to $\mathbb{R}_{(-\pi, \pi)}^{1,1}$ for each $s \in \mathbb{R}$.

Proof. In the case $O \in \mathcal{O}_{(-\pi, \pi)}$, this is similar to the proof of Lem. 2.35, and hence is omitted. The case $O \notin \mathcal{O}_{(-\pi, \pi)}$ reduces easily to the case $O \in \mathcal{O}_{(-\pi, \pi)}$ by Def. 2.30. $\square$

2.5 Some additivity properties

Proposition 2.37. *Let $G \rightarrow \text{Aut}(\mathcal{A})$ be a group homomorphism. Let $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$. Let $(\tilde{I}_\alpha)_{\alpha \in \mathcal{A}}$ be a collection in $\tilde{\mathcal{J}}$ such that $\tilde{I}_\alpha \subset \tilde{I}$ and $\bigcup_\alpha \tilde{I}_\alpha = \tilde{I}$. Then*

$$\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i) = \left(\bigcup_\alpha \text{End}_{\mathcal{A}(\tilde{I}_\alpha)}(\mathcal{H}_i) \right)''$$

Clearly, the same relation holds with $\tilde{I}', \tilde{I}'_\alpha$ replaced by $\tilde{I}, \tilde{I}_\alpha$.

Proof. The relation $\supset$ is obvious. Let us prove $\subset$.

Step 1. We first consider the case that there exists $\tilde{I}_0 \in \tilde{\mathcal{J}}$ such that $\tilde{I}_0 \subset \tilde{I}_\alpha$ for all α . Fix an I_0 -unitary $\xi_0 \in \mathcal{H}_i(\tilde{I}_0)$, and let $V = L(\xi_0, \tilde{I})|_{\mathcal{H}_0}$. By Prop. 1.50, we have

$$\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}_i) = V \mathcal{A}(\tilde{I}) V^* \quad \text{End}_{\mathcal{A}(\tilde{I}_\alpha)}(\mathcal{H}_i) = V \mathcal{A}(\tilde{I}_\alpha) V^*$$

This the proposition follows from the additivity of $\mathcal{A}$.

Step 2. We consider the case where $\mathcal{A}$ is a finite set, and prove the proposition by induction on $|\mathcal{A}|$. The case $|\mathcal{A}| = 1$ is obvious, and the case $|\mathcal{A}| = n$ can be reduced to the case $|\mathcal{A}| = n - 1$ by using Step 1 and the fact that at least two distinct members of (I_α) overlap (otherwise I would be disconnected).

Step 3. We consider the case where $(\tilde{I}_\alpha)$ is an increasing sequence. But this case follows directly from Step 1.

Step 4. We consider the general case. Choose an increasing sequence $(\tilde{K}_n)$ in $\mathcal{J}$ such that $\tilde{K}_n \subset \tilde{I}$, $K_n \subseteq I$, and $\bigcup_n K_n = I$. By Step 3, $\text{End}_{\mathcal{A}(\tilde{I})}(\mathcal{H}')$ is generated by all $\text{End}_{\mathcal{A}(\tilde{K}'_n)}(\mathcal{H}')$. Since K_n has compact closure in I , there exists a finite set $\mathcal{A}_n \subset \mathcal{A}$ such that $K_n \subset \bigcup_{\alpha \in \mathcal{A}_n} I_\alpha$. By Step 2, $\text{End}_{\mathcal{A}(\tilde{K}'_n)}(\mathcal{H}')$ is contained in the von Neumann algebra generated by $\text{End}_{\mathcal{A}(\tilde{I}_\alpha)}(\mathcal{H}')$ for all $\alpha \in \mathcal{A}_n$. This proves the proposition. $\square$

Definition 2.38. Let $O \in \mathcal{O}_{(-\pi, \pi)}$. We consider the xt -coordinates. A subset $\Lambda \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ is called an **arc between the top and bottom corners** of O if the following property holds: By a translation and a rotation so that the top, bottom, left, right corners of O become the points $(0, b)$, $(a, 0)$, $(0, 0)$, (a, b) with $a, b > 0$, we have

$$\Lambda = \{(x, f(x)) : 0 < x < a\}$$

where f is a smooth function from the open interval $\{x : -\varepsilon < x < a + \varepsilon\}$ (where $\varepsilon > 0$) to $\mathbb{R}$ satisfying

$$f(0) = b \quad f(a) = 0 \quad f' < 0$$

Proposition 2.39. *Let $O \in \mathcal{O}_{(-\pi, \pi)}$, and let Λ be an arc between the top and bottom corners of O . Let $(O_\alpha)_{\alpha \in \mathcal{A}}$ be a collection in $\mathcal{O}_{(-\pi, \pi)}$ such that $O_\alpha \subset O$ for each α , and $\Lambda \subset \bigcup_\alpha O_\alpha$. Then*

$$\mathcal{B}_{\text{cl}}(O) = \left(\bigcup_\alpha \mathcal{B}_{\text{cl}}(O_\alpha) \right)'' \quad (2.16)$$

Proof. The relation $\supset$ is obvious. Let us prove $\subset$.

Choose a standard double cone corresponding to some $(\tilde{K}, \tilde{K})$ and containing O . By using the function f in Def. 2.38, one finds some $g \in \mathcal{G}(K)$ such that $g(\tilde{J}) = \tilde{I}$, and that $(1, g)\Lambda$ is the vertical open interval between the top and bottom corners of the standard double cone $(1, g)O$. (This standard cone corresponds to $(\tilde{I}, \tilde{I})$.) Therefore, by Lem. 2.34, one may assume at the beginning that O is standard, and Λ is the open interval between the top and bottom corners of O . In particular, Λ lies in the t -axis.

For each $p \in \Lambda$, we choose a standard double cone $O_p \in \mathcal{O}_{(-\pi, \pi)}$ containing p and contained in O_α for some α . Then it suffices to prove

$$\mathcal{B}_{\text{cl}}(O) = \left(\bigcup_{p \in \Lambda} \mathcal{B}_{\text{cl}}(O_p) \right)''$$

Let $(\tilde{I}, \tilde{I})$ correspond to O and $(\tilde{I}_p, \tilde{I}_p)$ correspond to O_p . Then $\tilde{I}_p \subset \tilde{I}$, and $\bigcup_p \tilde{I}_p = \tilde{I}$ by the fact that $\Lambda \subset \bigcup_p O_p$. By (2.9) and Prop. 2.37, we have

$$\begin{aligned} \mathfrak{A}(\sqrt{I}, -\sqrt{I}) &= \text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I})}(\mathcal{H}_{\sqrt{0}}) \\ &= \left(\bigcup_p \text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I}_p)}(\mathcal{H}_{\sqrt{0}}) \right)'' = \left(\bigcup_p \mathfrak{A}(\sqrt{I_p}, -\sqrt{I_p}) \right)'' \end{aligned}$$

Applying the normal representation $\pi_I^L(\cdots)|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}}$ to both sides yields the desired relation. $\square$

Recall (2.4) for the meaning of $\mathcal{L}_x$.

Corollary 2.40. *Let $O \in \mathcal{O}$, let E be a (possibly empty) finite subset of $[-\pi, \pi]$, and let $\mathcal{L}_E = \bigcup_{x \in E} \mathcal{L}_x$. Then for each collection $(O_\alpha)_{\alpha \in \mathfrak{A}}$ in $\mathcal{O}$ satisfying $O \setminus \mathcal{L}_E = \bigcup_\alpha O_\alpha$, we have*

$$\mathcal{B}_{\text{cl}}(O) = \left(\bigcup_{\alpha \in \mathfrak{A}} \mathcal{B}_{\text{cl}}(O_\alpha) \right)''$$

Proof. The only nontrivial part is to prove $\subset$. We first consider the case $O \in \mathcal{O}_{(-\pi, \pi)}$. By Prop. 2.39, $\mathcal{B}_{\text{cl}}(O)$ is generated by $\mathcal{B}_{\text{cl}}(D)$ for all equilateral double cones $D \subset O$. Let $D_\alpha = D \cap O_\alpha$, and ignore α if $D_\alpha = \emptyset$. Let Λ be an arc between the top and bottom corners of D such that $\Lambda \cap \mathcal{L}_E = \emptyset$. Then $\Lambda \subset \bigcup_\alpha D_\alpha$, and hence $\mathcal{B}_{\text{cl}}(D)$ is generated by all $\mathcal{B}_{\text{cl}}(D_\alpha)$ due to Prop. 2.39. Thus $\mathcal{B}_{\text{cl}}(O)$ is generated by all $\mathcal{B}_{\text{cl}}(O_\alpha)$.

Next, assume $O \notin \mathcal{O}_{(-\pi, \pi)}$. Def. 2.30, $\mathcal{B}_{\text{cl}}(O)$ is generated by $\mathcal{B}_{\text{cl}}(D)$ for all $D \in \mathcal{O}_{(-\pi, \pi)}$, $D \subset O$. By the above paragraph, $\mathcal{B}_{\text{cl}}(D)$ is generated by all $\mathcal{B}_{\text{cl}}(D_\alpha)$, where $D_\alpha := D \cap O_\alpha$ can be viewed as in $\mathcal{O}_{(-\pi, \pi)}$ because $D \in \mathcal{O}_{(-\pi, \pi)}$. Thus, by the isotony in Thm. 2.33, $\mathcal{B}_{\text{cl}}(O)$ is again generated by all $\mathcal{B}_{\text{cl}}(O_\alpha)$. $\square$

As an immediate application, we obtain the following improvement of Lem. 2.35.

Lemma 2.41. Let $O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ be a double cone. Let $a, b > 0$ such that the interval $[-a, b]$ is the connected component containing 0 of the set

$$\{s \in \mathbb{R} : (\varrho(s), \varrho(-s))O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}\}$$

Then for each $-a \leq s \leq b$ we have

$$U_{\text{cl}}(\varrho_c(s), \varrho_c(-s)) \cdot \mathcal{B}_{\text{cl}}(O) \cdot U_{\text{cl}}(\varrho_c(s), \varrho_c(-s))^* = \mathcal{B}_{\text{cl}}((\varrho(s), \varrho(-s))O)$$

Proof. This is an immediate consequence of Lem. 2.35 and Cor. 2.40. □

2.6 Locality

Lemma 2.42. Suppose that $O_1, O_2 \in \mathcal{O}_{(-\pi, \pi)}$ have spacelike separated closures. Then there exist finite collections $(O_{1,\alpha})_{\alpha \in \mathcal{A}}$ and $(O_{2,\beta})_{\beta \in \mathcal{B}}$ of double cones in $\mathcal{O}_{(-\pi, \pi)}$ such that

$$O_1 = \bigcup_{\alpha \in \mathcal{A}} O_{1,\alpha} \quad O_2 = \bigcup_{\beta \in \mathcal{B}} O_{2,\beta}$$

and that for each $\alpha \in \mathcal{A}, \beta \in \mathcal{B}$, the sets $O_{1,\alpha}$ and $O_{2,\beta}$ are compactly contained in a common standard double cone $\hat{O}_{\alpha,\beta}$.

Proof. Step 1. We show that if $p_1, p_2 \in \mathbb{R}_{(-\pi, \pi)}^{1,1}$ are spacelike separated, then p_1, p_2 are contained in a common standard double cone. In the u^+u^- -coordinate, write $p_1 = (a_1, b_1)$ and $p_2 = (a_2, b_2)$. Let R be the diameter of $\{a_1, b_1, a_2, b_2\}$. If we can show that $R < 2\pi$, then there exists an open interval in $\mathbb{R}$ of length $< 2\pi$ containing a_1, b_1, a_2, b_2 . This interval corresponds to some $\tilde{I} \in \tilde{\mathcal{J}}$, and p_1, p_2 belong to the double cone corresponding to $(\tilde{I}, \tilde{I})$.

Since the x -coordinates of p_1, p_2 are $> -\pi$ and $< \pi$, we have

$$|a_1 - b_1| < 2\pi \quad |a_2 - b_2| < 2\pi$$

By Rem. 2.16, we have

$$|a_1 - a_2| < 2\pi \quad |b_1 - b_2| < 2\pi$$

We claim that

$$|a_1 - b_2| < R \quad |a_2 - b_1| < R$$

which immediately implies $R < 2\pi$.

Suppose that $|a_1 - b_2| = R$. Then either $a_1 \leq \min\{a_2, b_1\} \leq \max\{a_2, b_1\} \leq b_2$, or $b_2 \leq \min\{a_2, b_1\} \leq \max\{a_2, b_1\} \leq a_1$. In both cases, $a_2 - a_1$ and $b_2 - b_1$ have the same sign, contradicting Rem. 2.16. So we must have $|a_1 - b_2| < R$. A similar argument shows $|a_2 - b_1| < R$.

Step 2. Choose spacelike separated $D_1, D_2 \in \mathcal{O}_{(-\pi, \pi)}$ such that $O_1 \subseteq D_1$ and $O_2 \subseteq D_2$. By Step 1 and the compactness of $\overline{O_2}$ (together with the Lebesgue number lemma), for each $p \in \overline{O_1}$, there exists a double cone $D_{1,p} \subset D_1$ containing p , and some $\delta_p > 0$, such that for each double cone $\Delta_2 \subset D_2$ intersecting $\overline{O_2}$ with diameter $< \delta_p$, then $D_{1,p}$ and Δ_2 are compactly contained in a common standard double cone.

Therefore, by the compactness of $\overline{O_1}$ and $\overline{O_2}$, there exist finite collections $(D_{1,\alpha})_{\alpha \in \mathcal{A}}$ and $(D_{2,\beta})_{\beta \in \mathcal{B}}$ of equilateral double cones in $\mathcal{O}_{(-\pi, \pi)}$ such that

$$\overline{O_1} \subset \bigcup_{\alpha \in \mathcal{A}} D_{1,\alpha} \quad \overline{O_2} \subset \bigcup_{\beta \in \mathcal{B}} D_{2,\beta}$$

and that for each $\alpha \in \mathcal{A}, \beta \in \mathcal{B}$, we have $D_{1,\alpha} \subset D_1, D_{2,\beta} \subset D_2$, and $D_{1,\alpha}$ and $D_{2,\beta}$ are compactly contained in a common standard double cone $\hat{D}_{\alpha,\beta}$. The proof is finished by taking

$$O_{1,\alpha} = D_{1,\alpha} \cap O_1 \quad O_{2,\beta} = D_{2,\beta} \cap O_2$$

if $O_{1,\alpha}$ (resp. $O_{2,\beta}$) is non-empty; in the case that $O_{1,\alpha}$ is empty, define $O_{1,\alpha}$ to be O_{1,α_0} for some α_0 such that $O_{1,\alpha_0} \neq \emptyset$, and define $O_{2,\beta}$ in a similar way. $\square$

Theorem 2.43. *Suppose that $O_1, O_2 \in \mathcal{O}$ are spacelike separated. Then $[\mathcal{B}_{\text{cl}}(O_1), \mathcal{B}_{\text{cl}}(O_2)] = 0$.*

Proof. By Def. 2.30, it suffices to consider the case that $O_1, O_2 \in \mathcal{O}_{(-\pi, \pi)}$. By Cor. 2.40 and Lem. 2.42, it suffices to assume that O_1, O_2 are contained in some standard double cone $\hat{O}$ corresponding to some $(\tilde{K}, \tilde{K})$, and that $\overline{O_1}$ and $\overline{O_2}$ are spacelike separated. In fact, we will prove the commutativity without assuming spacelike separation of the closures; only the spacelike separation of O_1, O_2 will be used.

Let O_1, O_2 correspond to $(\tilde{I}_1, \tilde{J}_1)$ and $(\tilde{I}_2, \tilde{J}_2)$. Then $\tilde{I}_1, \tilde{J}_1, \tilde{I}_2, \tilde{J}_2$ are contained in $\tilde{K}$. By Rem. 2.16 and possibly a change of subscripts, we may assume that $\tilde{I}_2$ is clockwise to $\tilde{I}_1$ and $\tilde{J}_2$ is anticlockwise to $\tilde{J}_1$. Therefore, the following arg-valued intervals are in clockwise order:

$$\sqrt{\tilde{I}_1} \quad \sqrt{\tilde{I}_2} \quad -\sqrt{\tilde{J}_2} \quad -\sqrt{\tilde{J}_1}$$

Moreover, by enlarging $\tilde{I}_1$ and $\tilde{J}_1$ (and hence enlarging O_1), we assume in addition that by viewing $\mathbb{S}^1 = \mathbb{R}/2\pi\mathbb{Z}$, we have $\inf \tilde{I}_1 = \sup \tilde{I}_2$ and $\inf \tilde{J}_2 = \sup \tilde{J}_1$.

Applying the following Lem. 2.44 to the case that $E_3 \setminus \overline{E_2}$ is the disjoint union $\sqrt{\tilde{I}_1} \sqcup -\sqrt{\tilde{J}_1}$, and $E_2 \setminus \overline{E_1}$ is the disjoint union $\sqrt{\tilde{I}_2} \sqcup -\sqrt{\tilde{J}_2}$, we see that $\mathfrak{A}(\sqrt{\tilde{I}_1}, -\sqrt{\tilde{J}_1})$ commutes with $\mathfrak{A}(\sqrt{\tilde{I}_2}, -\sqrt{\tilde{J}_2})$. Applying $\pi_{\tilde{K}}^L(\cdots)|_{\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}'}}$, we see that $\mathcal{B}_{\text{cl}}(O_1)$ commutes with $\mathcal{B}_{\text{cl}}(O_2)$. $\square$

Lemma 2.44. *Let $E_1, E_2, E_3 \in \mathcal{J}$ such that $E_1 \subseteq E_2 \subseteq E_3$. Then $\mathcal{A}(E_3) \cap \mathcal{A}(E_2)'$ commutes with $\mathcal{A}(E_2) \cap \mathcal{A}(E_1)'$*

Proof. This is obvious. $\square$

2.7 PCT symmetry and Haag duality for $O_{\leftarrow}$ and $O_{\rightarrow}$

Definition 2.45. Recall from Rem. 1.71 that $\Theta_{\sqrt{0}, \sqrt{0}}$ is an involutive antiunitary anti-automorphism of the $\mathcal{A}^{\otimes 2}$ -module $\mathcal{H}_{\sqrt{0}} \boxtimes \mathcal{H}_{\sqrt{0}}$. Define

$$\mathfrak{J}_{\text{cl}} = U_{\text{cl}}^{-}(\varrho_c(-\pi)) \cdot \Theta_{\sqrt{0}, \sqrt{0}} \cdot U_{\text{cl}}^{-}(\varrho_c(\pi))$$

called the **PCT operator** of the closed CFT $(\mathcal{B}_{\text{cl}}, \mathcal{H}_{\text{cl}})$. Due to the following lemma, we have

$$\mathfrak{J}_{\text{cl}} = U_{\text{cl}}^{-}(\varrho_c(-2\pi)) \cdot \Theta_{\sqrt{0}, \sqrt{0}} = \Theta_{\sqrt{0}, \sqrt{0}} \cdot U_{\text{cl}}^{-}(\varrho_c(2\pi))$$

Lemma 2.46. For each $\pm \in \{+, -\}$ and $s \in \mathbb{R}$, we have

$$U_{\text{cl}}^{\pm}(\varrho_c(s)) \Theta_{\sqrt{0}, \sqrt{0}} = \Theta_{\sqrt{0}, \sqrt{0}} U_{\text{cl}}^{\pm}(\varrho_c(-s))$$

Proof. Since $\Theta_{\sqrt{0}, \sqrt{0}}$ is an anti-morphism of $\mathcal{A}^{\otimes 2}$ -modules, and since the modular conjugation for

$$\mathcal{A}^{\otimes 2}(\mathbb{S}_+^1) = \mathcal{A}(\mathbb{S}_+^1) \otimes \mathcal{A}(\mathbb{S}_+^1)$$

and the cyclic separating vector $\Omega \otimes \Omega$ is $\mathfrak{J}_{\mathcal{A}} \otimes \mathfrak{J}_{\mathcal{A}}$, we see from Def. 1.65 that $\Theta_{\sqrt{0}, \sqrt{0}}$ is an anti-automorphism of the $\mathcal{A}$ -module $(\mathcal{H}_{\text{cl}}, \pi_{\text{cl}}^{\pm})$ (where $\pi_{\text{cl}}^{\pm}$ is defined in Def. 2.1). The desired relation then follows from Cor. 1.67. $\square$

Theorem 2.47. Let $O_{\uparrow}$, $O_{\rightarrow}$, and $O_{\leftarrow}$ be the equilateral double cones corresponding to

$$(\tilde{\mathbb{S}}_+^1, \tilde{\mathbb{S}}_+^1) \quad (\tilde{\mathbb{S}}_+^1, \tilde{\mathbb{S}}_-^1) \quad (\tilde{\mathbb{S}}_-^1, \tilde{\mathbb{S}}_+^1)$$

Then

$$\text{Ad}_{\mathfrak{J}_{\text{cl}}}(\mathcal{B}_{\text{cl}}(O_{\rightarrow})) = \mathcal{B}_{\text{cl}}(O_{\rightarrow})' = \mathcal{B}_{\text{cl}}(O_{\leftarrow}) \quad (2.17)$$

Note that $O_{\uparrow}$ is standard, and hence compactly contained in $\mathbb{R}_{(-\pi, \pi)}^{1,1}$. Although $O_{\rightarrow}$, $O_{\leftarrow}$ are contained in $\mathbb{R}_{(-\pi, \pi)}^{1,1}$, they are not compactly contained in $\mathbb{R}_{(-\pi, \pi)}^{1,1}$.

Proof. Step 1. We write $\Theta_{\sqrt{0}, \sqrt{0}} = \Theta$ and $U_{\text{cl}}^{\pm}(\varrho_c(s)) = V^{\pm}(s)$ for simplicity. Clearly

$$(1, \varrho(\pi))O_{\rightarrow} = O_{\uparrow} = (\varrho(\pi), 1)O_{\leftarrow}$$

Therefore, by Lem. 2.36 and 2.41, we have

$$\mathcal{B}_{\text{cl}}(O_{\rightarrow}) = \text{Ad}_{V^{-}(-\pi)}(\mathcal{B}_{\text{cl}}(O_{\uparrow})) \quad \mathcal{B}_{\text{cl}}(O_{\leftarrow}) = \text{Ad}_{V^{+}(-\pi)}(\mathcal{B}_{\text{cl}}(O_{\uparrow})) \quad (2.18)$$

By Def. 2.29, we have

$$\mathcal{B}_{\text{cl}}(O_{\uparrow}) = \pi_{\tilde{\mathbb{S}}_+^1}^L(\text{End}_{\mathcal{A}^{\otimes 2}(\tilde{\mathbb{S}}_-^1)}(\mathcal{H}_{\sqrt{0}}))|_{\mathcal{H}_{\text{cl}}}$$

It follows from Cor. 1.73 that

$$\mathrm{Ad}_\Theta(\mathcal{B}_{\mathrm{cl}}(O_\uparrow)) = \mathcal{B}_{\mathrm{cl}}(O_\uparrow)' \quad (2.19)$$

Combining (2.19) with (2.18), we obtain

$$\mathrm{Ad}_{V^-(-\pi)\Theta V^-(\pi)}(\mathcal{B}_{\mathrm{cl}}(O_\rightarrow)) = \mathcal{B}_{\mathrm{cl}}(O_\rightarrow)' \quad (2.20a)$$

$$\mathrm{Ad}_{V^+(-\pi)\Theta V^+(\pi)}(\mathcal{B}_{\mathrm{cl}}(O_\leftarrow)) = \mathcal{B}_{\mathrm{cl}}(O_\leftarrow)' \quad (2.20b)$$

This proves the first half of (2.17).

Step 2. By (2.20) and the locality of $\mathcal{B}_{\mathrm{cl}}$ (cf. Thm. 2.43), we obtain that

$$\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \subset \mathrm{Ad}_{V^-(-\pi)\Theta V^-(\pi)}(\mathcal{B}_{\mathrm{cl}}(O_\rightarrow)) \quad (2.21a)$$

$$\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) \subset \mathrm{Ad}_{V^+(-\pi)\Theta V^+(\pi)}(\mathcal{B}_{\mathrm{cl}}(O_\leftarrow)) \quad (2.21b)$$

and that the inclusion in (2.21a) is an equality iff $\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) = \mathcal{B}_{\mathrm{cl}}(O_\rightarrow)'$.

By Lem. 2.46 and $\Theta^2 = 1$, we have

$$V^-(-\pi)\Theta V^-(\pi)V^+(-\pi)\Theta V^+(\pi) = V^+(2\pi)V^-(-2\pi)$$

$$V^+(-\pi)\Theta V^+(\pi)V^-(-\pi)\Theta V^-(\pi) = V^+(-2\pi)V^-(2\pi)$$

Combining this with (2.21), we get

$$\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \subset \mathrm{Ad}_{V^+(2\pi)V^-(-2\pi)}\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \quad (2.22a)$$

$$\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) \subset \mathrm{Ad}_{V^+(-2\pi)V^-(2\pi)}\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) \quad (2.22b)$$

and that if the inclusion in (2.22a) is an equality, then $\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) = \mathcal{B}_{\mathrm{cl}}(O_\rightarrow)'$.

By (2.18), we have

$$\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) = \mathrm{Ad}_{V^+(\pi)V^-(-\pi)}\mathcal{B}_{\mathrm{cl}}(O_\leftarrow)$$

Therefore, (2.22) is equivalent to

$$\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \subset \mathrm{Ad}_{V^+(\pi)V^-(-\pi)}\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) \quad (2.23a)$$

$$\mathcal{B}_{\mathrm{cl}}(O_\rightarrow) \subset \mathrm{Ad}_{V^+(-\pi)V^-(\pi)}\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \quad (2.23b)$$

Combining these two inclusion relations together, we get

$$\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) \subset \mathcal{B}_{\mathrm{cl}}(O_\leftarrow)$$

where the inclusion is an equality iff the two inclusions in (2.23) are both equalities. But the above inclusion is clearly an equality. Therefore, the inclusion in (2.23a) (hence the one in (2.22a)) is an equality, and thus $\mathcal{B}_{\mathrm{cl}}(O_\leftarrow) = \mathcal{B}_{\mathrm{cl}}(O_\rightarrow)'$. $\square$

2.8 Conformal covariance

Recall Thm. 2.47 for the meanings of $O_{\rightarrow}$ and $O_{\leftarrow}$.

Lemma 2.48. *We have*

$$U_{\text{cl}}(\varrho_c(\pi), \varrho_c(-\pi)) \cdot \mathcal{B}_{\text{cl}}(O_{\rightarrow}) \cdot U_{\text{cl}}(\varrho_c(\pi), \varrho_c(-\pi))^* = \mathcal{B}_{\text{cl}}(O_{\leftarrow})$$

Proof. At the end of the proof of Thm. 2.47, we have seen that the inclusion in (2.23a) is indeed the equality. $\square$

2.8.1 A computation of relative commutants

Lemma 2.49. *Let $E_1, E_2, E_3 \in \mathcal{J}$ such that $E_1 \subseteq E_2 \subseteq E_3$. Then the relative commutant of $\mathcal{A}(E_2) \cap \mathcal{A}(E_1)'$ in $\mathcal{A}(E_3) \cap \mathcal{A}(E_1)'$ is $\mathcal{A}(E_3) \cap \mathcal{A}(E_2)'$.*

Proof. By the split property (cf. Rem. 1.2), the inclusion of $\mathcal{A}(E_2) \subset \mathcal{A}(E_3)$ contains a type I intermediate factor, and hence is unitarily equivalent to

$$\mathbb{C} \otimes \mathcal{A}(E_2) \subset \mathcal{A}_3 \otimes \mathfrak{L}(\mathcal{H}_0) \quad \curvearrowright \mathcal{K}_3 \otimes \mathcal{H}_0$$

where $\mathcal{A}_3$ is a von Neumann algebra acting on a Hilbert space $\mathcal{K}_3$. Similarly, the inclusion $\mathcal{A}(E_1) \subset \mathcal{A}(E_2)$ is unitarily equivalent to

$$\mathbb{C} \otimes \mathcal{A}(E_1) \subset \mathcal{A}_2 \otimes \mathfrak{L}(\mathcal{H}_0) \quad \curvearrowright \mathcal{K}_2 \otimes \mathcal{H}_0$$

where $\mathcal{A}_2$ is a von Neumann algebra acting on a Hilbert space $\mathcal{K}_2$. Set $\mathcal{K}_1 = \mathcal{H}_0$ and $\mathcal{A}_1 = \mathcal{A}(E_1)$. Then the chain $\mathcal{A}(E_1) \subset \mathcal{A}(E_2) \subset \mathcal{A}(E_3)$ is unitarily equivalent to

$$\mathbb{C} \otimes \mathbb{C} \otimes \mathcal{A}_1 \subset \mathbb{C} \otimes \mathcal{A}_2 \otimes \mathfrak{L}(\mathcal{K}_1) \subset \mathcal{A}_3 \otimes \mathfrak{L}(\mathcal{K}_2) \otimes \mathfrak{L}(\mathcal{K}_1) \quad \curvearrowright \mathcal{K}_3 \otimes \mathcal{K}_2 \otimes \mathcal{K}_1$$

It follows that

$$\begin{aligned} \mathcal{A}(E_2) \cap \mathcal{A}(E_1)' &= \mathbb{C} \otimes \mathcal{A}_2 \otimes \mathcal{A}_1' \\ \mathcal{A}(E_3) \cap \mathcal{A}(E_1)' &= \mathcal{A}_3 \otimes \mathfrak{L}(\mathcal{K}_2) \otimes \mathcal{A}_1' \\ \mathcal{A}(E_3) \cap \mathcal{A}(E_2)' &= \mathcal{A}_3 \otimes \mathcal{A}_2' \otimes \mathbb{C} \end{aligned}$$

Taking the commutant of the first equality, we get

$$(\mathcal{A}(E_2) \cap \mathcal{A}(E_1))' = \mathfrak{L}(\mathcal{K}_3) \otimes \mathcal{A}_2' \otimes \mathcal{A}_1$$

Using the fact that $\mathcal{A}_1 = \mathcal{A}(E_1)$ is a (type III) factor and hence $\mathcal{A}_1 \cap \mathcal{A}_1' = \mathbb{C}$, we obtain

$$(\mathcal{A}(E_3) \cap \mathcal{A}(E_1))' \cap (\mathcal{A}(E_2) \cap \mathcal{A}(E_1))' = \mathcal{A}_3 \otimes \mathcal{A}_2' \otimes \mathbb{C}$$

which equals $\mathcal{A}(E_3) \cap \mathcal{A}(E_2)'$. $\square$

Proposition 2.50. *Let $O, O_1 \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ be double cones with $O_1 \subset O$. Assume that O and O_1 have the same right corner, but have different top, bottom, and left corners. Note that $O_2 := O \cap O_1'$ is also a double cone. Then*

$$\mathcal{B}_{\text{cl}}(O_1) = \mathcal{B}_{\text{cl}}(O) \cap \mathcal{B}_{\text{cl}}(O_2)' \quad \mathcal{B}_{\text{cl}}(O_2) = \mathcal{B}_{\text{cl}}(O) \cap \mathcal{B}_{\text{cl}}(O_1)'$$

Proof. By Lem. 2.41, we may assume in addition that $O \subseteq \mathbb{R}_{(-\pi, \pi)}^{1,1}$, and hence O is contained in a standard double cone corresponding to some $(\tilde{K}, \tilde{K})$. Let O, O_i correspond to $(\tilde{I}, \tilde{J})$ and $(\tilde{I}_i, \tilde{J}_i)$. Then $\tilde{I}_2$ is clockwise to $\tilde{I}_1$, $\tilde{J}_1$ is clockwise to $\tilde{J}_2$, $I_1 \sqcup I_2$ is dense in I , and $J_1 \sqcup J_2$ is dense in J . Applying Lem. 2.49 to the case where $E_2 \setminus \overline{E_1} = \sqrt{I_2} \sqcup -\sqrt{J_2}$ and $E_3 \setminus \overline{E_2} = \sqrt{I_1} \sqcup -\sqrt{J_1}$, we obtain

$$\mathfrak{A}(\sqrt{I_1}, -\sqrt{J_1}) = \mathfrak{A}(\sqrt{I}, -\sqrt{J}) \cap \mathfrak{A}(\sqrt{I_2}, -\sqrt{J_2})'$$

Since the representation $\pi_{\tilde{K}}^L(\cdot \cdots)|_{\mathcal{H}_{\text{cl}}}$ on $\mathfrak{A}(\sqrt{K}, -\sqrt{K}) = \text{End}_{\mathcal{A}^{\otimes 2}(\tilde{K}')}(\mathcal{H}_{\sqrt{0}})$ is faithful (cf. Rem. 1.55), applying this representation to the above identity yields the first relation. The second relation can be proved in the same manner (by choosing different E_1, E_2, E_3). $\square$

2.8.2 Proof of the conformal covariance

Lemma 2.51. *Let $O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ be an equilateral double cone such that in the xt -coordinates, the right corner of O is $(\pi, 0)$, and the diameter r of O satisfies $r < \pi$. Then*

$$U_{\text{cl}}(\varrho_c(r), \varrho_c(-r)) \cdot \mathcal{B}_{\text{cl}}(O) \cdot U_{\text{cl}}(\varrho_c(r), \varrho_c(-r))^* = \mathcal{B}_{\text{cl}}((\varrho(r), \varrho(-r))O)$$

Proof. We write $U_{\text{cl}}(\varrho_c(s), \varrho_c(-s)) = V(s)$ and $\delta(s) = (\varrho(s), \varrho(-s))$ for simplicity. Since $\delta(r - \pi)O$ is contained in $O_{\rightarrow}$, Lem. 2.41 implies

$$\text{Ad}_{V(r-\pi)}(\mathcal{B}_{\text{cl}}(O)) \subset \mathcal{B}_{\text{cl}}(O_{\rightarrow})$$

and hence

$$\text{Ad}_{V(r)}(\mathcal{B}_{\text{cl}}(O)) = \text{Ad}_{V(\pi)}\text{Ad}_{V(r-\pi)}(\mathcal{B}_{\text{cl}}(O)) \subset \text{Ad}_{V(\pi)}(\mathcal{B}_{\text{cl}}(O_{\rightarrow})) = \mathcal{B}_{\text{cl}}(O_{\leftarrow})$$

where the last equality is due to Lem. 2.48.

Let

$$O_2 := \delta(r)O \quad O_1 = O_{\leftarrow} \cap O_2'$$

Since O and $\delta(-r)O_1$ are spacelike separated (because they are contained in $O_{\rightarrow}$ and $O_{\leftarrow}$, respectively), by the locality (Thm. 2.43), $\mathcal{B}_{\text{cl}}(O)$ commutes with $\mathcal{B}_{\text{cl}}(\delta(-r)O_1)$, and hence commutes with $\text{Ad}_{V(-r)}(\mathcal{B}_{\text{cl}}(O_1))$ by Lem. 2.41. Thus

$$\text{Ad}_{V(r)}(\mathcal{B}_{\text{cl}}(O)) \subset \mathcal{B}_{\text{cl}}(O_1)'$$

By Prop. 2.50 and $O_2 = O_{\leftarrow} \cap O'_1$, we obtain

$$\text{Ad}_{V(r)}(\mathcal{B}_{\text{cl}}(O)) \subset \mathcal{B}_{\text{cl}}(O_2)$$

A similar argument shows $\text{Ad}_{V(-r)}(\mathcal{B}_{\text{cl}}(O_2)) \subset \mathcal{B}_{\text{cl}}(O)$. Thus $\text{Ad}_{V(r)}(\mathcal{B}_{\text{cl}}(O)) = \mathcal{B}_{\text{cl}}(O_2)$. $\square$

Lemma 2.52. *Let $O \subset \mathbb{R}_{(-\pi, \pi)}^{1,1}$ be an equilateral double cone whose diameter in the xt -axis is $r < \pi$. Choose $s \in \mathbb{R}$ such that $(\rho(s), \rho(-s))O$ does not intersect $\mathcal{L}_{\pm\pi}$ (and hence can be viewed as contained in $\mathbb{R}_{(-\pi, \pi)}^{1,1}$). Then*

$$U_{\text{cl}}(\varrho_c(s), \varrho_c(-s)) \cdot \mathcal{B}_{\text{cl}}(O) \cdot U_{\text{cl}}(\varrho_c(s), \varrho_c(-s))^* = \mathcal{B}_{\text{cl}}((\varrho(s), \varrho(-s))O) \quad (2.24)$$

Proof. The case $s < 0$ follows easily from the case $s \geq 0$. So we assume $s \geq 0$. Then the translation of O appearing in this lemma can be decomposed into translations of the types considered in Lem. 2.41 and Lem. 2.51. The result therefore follows from those two lemmas. $\square$

Lemma 2.53. *For each $O \in \mathcal{O}$ and $s \in \mathbb{R}$, the relation (2.24) holds.*

Proof. Write $\delta(s) = (\varrho(s), \varrho(-s))$. Choose finite sets $E, F \subset (-\pi, \pi]$ containing π such that $\delta(s)\mathcal{L}_E = \mathcal{L}_F$. Let (O_α) be a collection of equilateral double cones whose union is $O \setminus \mathcal{L}_E$, and whose diameters in the xt -coordinates are $< \pi$. Then each O_α and $\delta(s)O_\alpha$ can be viewed as subsets of $\mathbb{R}_{(-\pi, \pi)}^{1,1}$. By Lem. 2.52, (2.24) holds when O is replaced by O_α .

By Cor. 2.40, $\mathcal{B}_{\text{cl}}(O)$ is generated by all $\mathcal{B}_{\text{cl}}(O_\alpha)$. Since $(\delta(s)O) \setminus \mathcal{L}_F = \bigcup_\alpha \delta(s)O_\alpha$, Cor. 2.40 also implies that $\mathcal{B}_{\text{cl}}(\delta(s)O)$ is generated by all $\mathcal{B}_{\text{cl}}(\delta(s)O_\alpha)$. This finishes the proof. $\square$

Lemma 2.54. *Let $O \in \mathcal{O}$ be corresponding to some $(\tilde{I}, \tilde{J})$. Let $E, F \in \mathcal{J}$. Suppose that there exist $\tilde{K}, \tilde{L} \in \mathcal{J}$ satisfying*

$$\tilde{I} \subset \tilde{K} \quad E \subset K \quad \tilde{J} \subset \tilde{L} \quad F \subset L$$

Then for each $g^+ \in \mathcal{G}_c(E)$ and $g^- \in \mathcal{G}_c(F)$ we have

$$U_{\text{cl}}(g^+, g^-) \mathcal{B}_{\text{cl}}(O) U_{\text{cl}}(g^+, g^-)^* = \mathcal{B}_{\text{cl}}((g^+, g^-)O)$$

Proof. Note that $g^+ \in \mathcal{G}_c(K)$ and $g^- \in \mathcal{G}_c(L)$. By enlarging either $\tilde{K}$ or $\tilde{L}$, we assume they have the same length. Choose $s \in \mathbb{R}$ such that $\varrho(s)\tilde{L} = \tilde{K}$. Let

$$h^- = \varrho_c(s)g^- \varrho_c(-s) \quad D = (1, \varrho(s))O$$

Then $h^- \in \mathcal{G}_c(K)$, and D is contained in the standard double cone corresponding to $(\tilde{K}, \tilde{K})$. Therefore, by Lem. 2.34,

$$U_{\text{cl}}(g^+, h^-) \mathcal{B}_{\text{cl}}(D) U_{\text{cl}}(g^+, h^-)^* = \mathcal{B}_{\text{cl}}((g^+, h^-)D)$$

Thus, writing $U_{\text{cl}}^-(\varrho_c(s))$ as V , and applying Ad_V^* to the above identity, we have

$$U_{\text{cl}}(g^+, g^-)V^*\mathcal{B}_{\text{cl}}(D)VU_{\text{cl}}(g^+, g^-)^* = V^*\mathcal{B}_{\text{cl}}((g^+, h^-)D)V$$

By Lem. 2.36 and 2.53, we have $V^*\mathcal{B}_{\text{cl}}(D)V = \mathcal{B}_{\text{cl}}(O)$ and

$$\begin{aligned} V^*\mathcal{B}_{\text{cl}}((g^+, h^-)D)V &= \mathcal{B}_{\text{cl}}((g^+, \varrho_c(-s)h^-)D) \\ &= \mathcal{B}_{\text{cl}}((g^+, g^- \varrho_c(-s))D) = \mathcal{B}_{\text{cl}}((g^+, g^-)O) \end{aligned}$$

This finishes the proof. $\square$

Theorem 2.55. *For each $O \in \mathcal{O}$ and $g^+, g^- \in \mathcal{G}_c$, we have*

$$U_{\text{cl}}(g^+, g^-)\mathcal{B}_{\text{cl}}(O)U_{\text{cl}}(g^+, g^-)^* = \mathcal{B}_{\text{cl}}((g^+, g^-)O)$$

Proof. By Lem. 1.18, $\mathcal{G}_c$ is algebraically generated by $\mathcal{G}_c(E)$ for all $E \in \mathcal{J}$ whose length is $< \pi$. Therefore, it suffices to prove the desired relation for $g^+ \in \mathcal{G}_c(E)$, $g^- \in \mathcal{G}_c(F)$ where $E, F \in \mathcal{J}$ have lengths $< \pi$. Moreover, by Cor. 2.40, it suffices to consider the case where the arg-valued intervals $(\tilde{I}, \tilde{J})$ corresponding to O have lengths $< \pi$. Then one can find $\tilde{K}, \tilde{L} \in \tilde{\mathcal{J}}$ satisfying the requirement of Lem. 2.54. The conclusion now follows from that lemma. $\square$

2.8.3 Consequences of conformal covariance

Corollary 2.56. *If $O \in \mathcal{O}$ corresponds to $(\tilde{I}, \tilde{J})$, then*

$$\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I)) \subset \mathcal{B}_{\text{cl}}(O) \quad \pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J)) \subset \mathcal{B}_{\text{cl}}(O)$$

Proof. We first consider the case $\tilde{I} = \tilde{J}$. Let $x \in \mathcal{A}(I)$. By (1.34), we have

$$\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(x) = \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}(x \otimes 1) = \pi_{\tilde{I}}^L(\pi_{\sqrt{0}, \tilde{I}}(x \otimes 1))|_{\mathcal{H}_{\sqrt{0} \boxtimes \sqrt{0}}}$$

where $\pi_{\sqrt{0}, \tilde{I}}(x \otimes 1)$ belongs to $\text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I})}(\mathcal{H}_{\sqrt{0}})$. Therefore $\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(x)$ belongs to $\mathcal{B}_{\text{cl}}(O) = \pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}^{\otimes 2}(\tilde{I})}(\mathcal{H}_{\sqrt{0}}))|_{\mathcal{H}_{\sqrt{0} \boxtimes \sqrt{0}}}$ (cf. Def. 2.29). This proves the first inclusion. Similarly, the second inclusion follows from

$$\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^-(x) = \pi_{\tilde{I}}^R(\pi_{\sqrt{0}, \tilde{I}}(1 \otimes x))|_{\mathcal{H}_{\sqrt{0} \boxtimes \sqrt{0}}}$$

Now we consider the general case. Choose $\gamma \in \mathcal{G}_c$ such that $\gamma\tilde{J} = \tilde{I}$. Let $O \in \mathcal{O}_{(-\pi, \pi)}$ be corresponding to $(\tilde{I}, \tilde{J})$. By Def. 2.31, we have $U_{\text{cl}}(1, \gamma) = U_{\text{cl}}^-(\gamma)$. Thus, Thm. 2.55 implies

$$\text{Ad}_{U_{\text{cl}}^-(\gamma)}(\mathcal{B}_{\text{cl}}(O)) = \mathcal{B}_{\text{cl}}(D)$$

By Rem. 2.3, $\text{Ad}_{U_{\text{cl}}^-(\gamma)}$ sends $\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I))$ to itself. By Cor. 1.20, $\text{Ad}_{U_{\text{cl}}^-(\gamma)}$ sends $\pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J))$ to $\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^-(\mathcal{A}(I))$. Thus, the general case reduces to the special case $\tilde{I} = \tilde{J}$. $\square$

Corollary 2.57. If $O \in \mathcal{O}$ corresponds to $(\tilde{I}, \tilde{J})$ and $g^+ \in \mathcal{G}_c(I), g^- \in \mathcal{G}_c(J)$, then

$$U_{\text{cl}}(g^+, g^-) \in \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I)) \vee \pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J))$$

and hence $U_{\text{cl}}(g^+, g^-) \in \mathcal{B}_{\text{cl}}(O)$ by Cor. 2.56.

It follows that for each $m, n \in \mathbb{Z}$, we have

$$U_{\text{cl}}(g^+, g^-) \in \mathcal{B}_{\text{cl}}((\varrho(2m\pi), \varrho(2n\pi))O) \quad (2.25)$$

because $(\varrho(2m\pi), \varrho(2n\pi))O$ also corresponds to arg-valued intervals whose underlying intervals are I and J .

Proof. We have

$$U_{\text{cl}}^+(g^+) = U_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(g^+) \stackrel{\text{Def. 1.17}}{=} \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(U_0(g^+)) \in \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I))$$

and, similarly, $U_{\text{cl}}^-(g^-)$ belongs to $\pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J))$. Thus $U_{\text{cl}}(g^+, g^-) = U_{\text{cl}}^+(g^+)U_{\text{cl}}^-(g^-)$ belongs to $\pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\mathcal{A}(I)) \vee \pi_{\sqrt{0} \boxtimes \sqrt{0}, J}^-(\mathcal{A}(J))$. $\square$

2.8.4 An enhanced version of the additivity property

Definition 2.58. Let $O \in \mathcal{O}$. A subset $\Lambda \subset O$ is called an **arc between the top and bottom corners** of O if, after choosing $s \in \mathbb{R}$ such that $D := (\varrho(s), \varrho(-s))O$ belongs to $\mathcal{O}_{(-\pi, \pi)}$, the set $(\varrho(s), \varrho(-s))\Lambda$ is an arc between the top and bottom corners of D in the sense of Def. 2.38.

Theorem 2.59. Let $O \in \mathcal{O}$, and let Λ be an arc between the top and bottom corners of O . Let $(O_\alpha)_{\alpha \in \mathcal{A}}$ be a collection in $\mathcal{O}_{(-\pi, \pi)}$ such that $O_\alpha \subset O$ for each α , and $\Lambda \subset \bigcup_\alpha O_\alpha$. Then

$$\mathcal{B}_{\text{cl}}(O) = \left(\bigcup_\alpha \mathcal{B}_{\text{cl}}(O_\alpha) \right)'' \quad (2.26)$$

Proof. This is an easy consequence of Prop. 2.39 and Thm. 2.55. $\square$

2.9 PCT symmetry

Recall the PCT operator $\mathfrak{J}_{\text{cl}}$ defined in Def. 2.45. Recall the reflection $(\mathfrak{r}, \mathfrak{r})$ of $\mathbb{R}_{\text{cl}}^{1,1}$ described in (2.15). Recall Def. 1.60 for the definition of $\mathfrak{r}g\mathfrak{r}$ where $g \in \mathcal{G}_c$.

Theorem 2.60. The anti-unitary operator $\mathfrak{J}_{\text{cl}}$ is involutive (i.e. $\mathfrak{J}_{\text{cl}}^2 = 1$), and satisfies

$$\mathfrak{J}_{\text{cl}} U_{\text{cl}}(g^+, g^-) \mathfrak{J}_{\text{cl}} = U_{\text{cl}}(\mathfrak{r}g^+ \mathfrak{r}, \mathfrak{r}g^- \mathfrak{r}) \quad (2.27a)$$

$$\mathfrak{J}_{\text{cl}} \mathcal{B}_{\text{cl}}(O) \mathfrak{J}_{\text{cl}} = \mathcal{B}_{\text{cl}}((\mathfrak{r}, \mathfrak{r})O) \quad (2.27b)$$

for each $g^+, g^- \in \mathcal{G}_c$ and $O \in \mathcal{O}$.

Proof. We write $\Theta_{\sqrt{0}, \sqrt{0}} = \Theta$ and $U_{\text{cl}}^{\pm}(\varrho_c(s)) = V^{\pm}(s)$ for simplicity. Then $\mathfrak{J}_{\text{cl}} = V^{-}(-2\pi)\Theta = \Theta V^{-}(2\pi)$. By Lem. 2.46, we have

$$\mathfrak{J}_{\text{cl}}^2 = \Theta V^{-}(2\pi)\Theta V^{-}(2\pi) = \Theta^2 V^{-}(-2\pi)V^{-}(2\pi) = 1$$

From the proof of Lem. 2.46, we see that Θ is anti-automorphism of the $\mathcal{A}$ -module $(\mathcal{H}_{\text{cl}}, \pi_{\text{cl}}^{\pm})$. Therefore, by Prop. 1.66, we have

$$\Theta U_{\text{cl}}^{+}(g^{+})\Theta = U_{\text{cl}}^{+}(\mathfrak{r}g^{+}\mathfrak{r}) \quad \Theta U_{\text{cl}}^{-}(g^{-})\Theta = U_{\text{cl}}^{-}(\mathfrak{r}g^{-}\mathfrak{r})$$

Multiplying these two identities yields

$$\Theta U_{\text{cl}}(g^{+}, g^{-})\Theta = U_{\text{cl}}(\mathfrak{r}g^{+}\mathfrak{r}, \mathfrak{r}g^{-}\mathfrak{r})$$

For each $\pm \in \{+, -\}$, by Cor. 1.20, $V^{\pm}(2\pi)$ is an automorphism of the $\mathcal{A}$ -module $(\mathcal{H}_{\text{cl}}, \pi_{\text{cl}}^{\pm})$. So it commutes with $U_{\text{cl}}^{\pm}(g^{\pm})$ by Prop. 1.66. By Rem. 2.3, $V^{\pm}(2\pi)$ commutes with $U_{\text{cl}}^{\mp}(g^{\mp})$. Thus (2.27a) is proved.

Choose $g^{+}, g^{-} \in \mathcal{G}_c$ such that $O_{\rightarrow} = (g^{+}, g^{-})O$. By Thm. 2.55, we have

$$\mathcal{B}_{\text{cl}}(O_{\rightarrow}) = \text{Ad}_{U_{\text{cl}}(g^{+}, g^{-})}(\mathcal{B}_{\text{cl}}(O))$$

and hence

$$\text{Ad}_{\mathfrak{J}_{\text{cl}}}(\mathcal{B}_{\text{cl}}(O_{\rightarrow})) = \text{Ad}_{\mathfrak{J}_{\text{cl}}}\text{Ad}_{U_{\text{cl}}(g^{+}, g^{-})}(\mathcal{B}_{\text{cl}}(O)) \stackrel{(2.27a)}{=} \text{Ad}_{U_{\text{cl}}(\mathfrak{r}g^{+}\mathfrak{r}, \mathfrak{r}g^{-}\mathfrak{r})}\text{Ad}_{\mathfrak{J}_{\text{cl}}}(\mathcal{B}_{\text{cl}}(O))$$

By Thm. 2.47, the above expression equals $\mathcal{B}_{\text{cl}}(O_{\leftarrow})$. Since $O_{\leftarrow} = (\mathfrak{r}, \mathfrak{r})O_{\rightarrow}$, we have

$$O_{\leftarrow} = (\mathfrak{r}, \mathfrak{r})(g^{+}, g^{-})O = (\mathfrak{r}g^{+}\mathfrak{r}, \mathfrak{r}g^{-}\mathfrak{r})(\mathfrak{r}, \mathfrak{r})O$$

and hence, by Thm. 2.55,

$$\mathcal{B}_{\text{cl}}(O_{\leftarrow}) = \text{Ad}_{U_{\text{cl}}(\mathfrak{r}g^{+}\mathfrak{r}, \mathfrak{r}g^{-}\mathfrak{r})}\mathcal{B}_{\text{cl}}((\mathfrak{r}, \mathfrak{r})O)$$

This finishes the proof. □

2.10 Haag duality for multi double-cones

Definition 2.61. For $O_1, \dots, O_n \in \mathcal{O}$ whose closures are mutually spacelike separated, let

$$\mathcal{B}_{\text{cl}}(O_1 \cup \dots \cup O_n) := (\mathcal{B}_{\text{cl}}(O_1) \cup \dots \cup \mathcal{B}_{\text{cl}}(O_n))''$$

Note that $(O_1 \cup \dots \cup O_n)'$ is also the disjoint union of some $D_1, \dots, D_n \in \mathcal{O}$ whose closures are mutually spacelike separated.

Theorem 2.62. Let $n \geq 1$, and let $O_1, \dots, O_n$ and $D_1, \dots, D_n$ be as in Def. 2.61. Then

$$(\mathcal{B}_{\text{cl}}(O_1 \cup \dots \cup O_n))' = \mathcal{B}_{\text{cl}}(D_1 \cup \dots \cup D_n)$$

Proof. When $n = 1$, this follows from Thm. 2.47 and 2.55. Now assume $n \geq 2$, and let $\Delta = O'_n$. Then

$$O_1 \cup \cdots \cup O_{n-1} \subset \Delta \quad \Delta \cap O'_1 \cap \cdots \cap O'_{n-1} = D_1 \sqcup \cdots \sqcup D_n$$

The proved special case implies $\mathcal{B}_{\text{cl}}(O_n)' = \mathcal{B}_{\text{cl}}(\Delta)$. Thus, it suffices to show

$$\mathcal{B}_{\text{cl}}(\Delta) \cap \mathcal{B}_{\text{cl}}(O_1 \cup \cdots \cup O_{n-1})' = \mathcal{B}_{\text{cl}}(D_1 \cup \cdots \cup D_n) \quad (2.28)$$

By Thm. 2.55, it suffices to assume that Δ is standard, and hence corresponding to $(\tilde{\Gamma}, \tilde{\Gamma})$ for some $\tilde{\Gamma} \in \tilde{\mathcal{J}}$. By changing the subscripts, we assume that the following double subcones of Δ are listed from right to left:

$$D_1, O_1, D_2, O_2, \dots, O_{n-1}, D_n$$

Let O_i correspond to $(\tilde{I}_i, \tilde{J}_i)$ and D_j to $(\tilde{K}_j, \tilde{L}_j)$, where all arg-valued intervals are contained in $\tilde{\Gamma}$. Then, the following arg-valued intervals are listed in the clockwise order:

$$\begin{aligned} & \sqrt{\tilde{K}_1}, \sqrt{\tilde{I}_1}, \sqrt{\tilde{K}_2}, \sqrt{\tilde{I}_2}, \dots, \sqrt{\tilde{I}_{n-1}}, \sqrt{\tilde{K}_n} \\ & -\sqrt{\tilde{L}_n}, -\sqrt{\tilde{J}_{n-1}}, -\sqrt{\tilde{L}_{n-1}}, -\sqrt{\tilde{J}_{n-2}}, \dots, -\sqrt{\tilde{J}_1}, -\sqrt{\tilde{L}_1} \end{aligned}$$

Applying the following Lem. 2.63 to the case

$$F_j \setminus \overline{E_j} = \sqrt{\tilde{K}_j} \sqcup -\sqrt{\tilde{L}_j} \quad E_i \setminus \overline{F_{i+1}} = \sqrt{\tilde{I}_i} \sqcup -\sqrt{\tilde{J}_i}$$

(where $i \leq n-1, j \leq n$), and using the faithfulness of the normal representation $\pi_{\tilde{\Gamma}}^L(\cdots)|_{\mathcal{H}_{\text{cl}}}$ as in the proof of Prop. 2.50, we obtain (2.28). $\square$

Lemma 2.63. *Let $F_1, E_1, \dots, F_n, E_n \in \mathcal{J}$ such that*

$$F_1 \supseteq E_1 \supseteq F_2 \supseteq E_2 \supseteq \cdots \supseteq F_n \supseteq E_n$$

Then the relative commutant of

$$\bigvee_{i=1}^{n-1} (\mathcal{A}(E_i) \cap \mathcal{A}(F_{i+1})')$$

in

$$\mathcal{A}(F_1) \cap \mathcal{A}(E_n)'$$

is equal to

$$\bigvee_{j=1}^n (\mathcal{A}(F_j) \cap \mathcal{A}(E_j)')$$

Proof. Similar to the proof of Lem. 2.49, the chain $\mathcal{A}(F_1) \supset \mathcal{A}(E_1) \supset \cdots \supset \mathcal{A}(E_n)$ is unitarily equivalent to

$$\begin{aligned}
& \mathcal{B}_1 \otimes \mathfrak{L}(\mathcal{H}_1) \otimes \mathfrak{L}(\mathcal{K}_2) \otimes \mathfrak{L}(\mathcal{H}_2) \otimes \cdots \otimes \mathfrak{L}(\mathcal{K}_n) \otimes \mathfrak{L}(\mathcal{H}_n) \\
& \supset \mathbb{C} \otimes \mathcal{A}_1 \otimes \mathfrak{L}(\mathcal{K}_2) \otimes \mathfrak{L}(\mathcal{H}_2) \otimes \cdots \otimes \mathfrak{L}(\mathcal{K}_n) \otimes \mathfrak{L}(\mathcal{H}_n) \\
& \supset \mathbb{C} \otimes \mathbb{C} \otimes \mathcal{B}_2 \otimes \mathfrak{L}(\mathcal{H}_2) \otimes \cdots \otimes \mathfrak{L}(\mathcal{K}_n) \otimes \mathfrak{L}(\mathcal{H}_n) \\
& \supset \mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C} \otimes \mathcal{A}_2 \otimes \cdots \otimes \mathfrak{L}(\mathcal{K}_n) \otimes \mathfrak{L}(\mathcal{H}_n) \\
& \supset \cdots \\
& \supset \mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C} \otimes \cdots \otimes \mathcal{B}_n \otimes \mathfrak{L}(\mathcal{H}_n) \\
& \supset \mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C} \otimes \mathbb{C} \otimes \cdots \otimes \mathbb{C} \otimes \mathcal{A}_n
\end{aligned}$$

where each $\mathcal{A}_i$ is a von Neumann algebra on $\mathcal{H}_i$, and $\mathcal{B}_j$ is a von Neumann algebra on $\mathcal{K}_j$. Then our goal reduces to showing that the relative commutant of

$$\mathbb{C} \otimes \mathcal{A}_1 \otimes \mathcal{B}'_2 \otimes \mathcal{A}_2 \otimes \mathcal{B}'_3 \otimes \cdots \otimes \mathcal{A}_{n-1} \otimes \mathcal{B}'_n \otimes \mathbb{C}$$

in

$$\mathcal{B}_1 \otimes \mathfrak{L}(\mathcal{H}_1) \otimes \mathfrak{L}(\mathcal{K}_2) \otimes \mathfrak{L}(\mathcal{H}_2) \otimes \mathfrak{L}(\mathcal{K}_3) \otimes \cdots \otimes \mathfrak{L}(\mathcal{H}_{n-1}) \otimes \mathfrak{L}(\mathcal{K}_n) \otimes \mathcal{A}'_n$$

is equal to

$$\mathcal{B}_1 \otimes \mathcal{A}'_1 \otimes \mathcal{B}_2 \otimes \mathcal{A}'_2 \otimes \mathcal{B}_3 \otimes \cdots \otimes \mathcal{A}'_{n-1} \otimes \mathcal{B}_n \otimes \mathcal{A}'_n$$

But this is obvious. □

3 The open CFT

Fix a conformal net $\mathcal{A}$ with central charge c . In this chapter, all objects of $\text{Rep}(\mathcal{A})$ are assumed to be nonzero.

3.1 Boundary intervals and double cones in $\mathbb{R}_{\text{op}}^{1,1}$

3.1.1 Boundary intervals in $\mathbb{R}_{\text{op}}^{1,1}$

Definition 3.1. In the xt -coordinates, define subsets of $\mathbb{R}^{1,1}$:

$$\begin{aligned}
\mathbb{R}_{\text{op}}^{1,1} &= \{(x, t) \in \mathbb{R}^{1,1} : 0 \leq x \leq \pi\} & \mathbb{R}_{(0,\pi)}^{1,1} &= \{(x, t) \in \mathbb{R}^{1,1} : 0 < x < \pi\} \\
\partial_+ \mathbb{R}_{\text{op}}^{1,1} &= \{0\} \times \mathbb{R} & \partial_- \mathbb{R}_{\text{op}}^{1,1} &= \{\pi\} \times \mathbb{R}
\end{aligned}$$

also viewed as subsets of $\mathbb{R}_{\text{cl}}^{1,1}$. Two points $p_1, p_2 \in \mathbb{R}_{\text{op}}^{1,1}$ are called **spacelike separated** if they are so as elements of $\mathbb{R}_{\text{cl}}^{1,1}$, and hence can be described by Rem. 2.16.

Definition 3.2. The group action of $\mathcal{G}$ on $\mathbb{R}_{\text{op}}^{1,1}$, which lifts to a group action of $\mathcal{G}_c$ on $\mathbb{R}_{\text{op}}^{1,1}$, is defined as follows. Consider the u^+u^- -coordinates. By viewing each $g \in \mathcal{G}$ as a diffeomorphism of $\mathbb{R}$ (cf. (1.3)), for each (u^+, u^-) in $\mathbb{R}_{\text{op}}^{1,1}$, let

$$g(u^+, u^-) = (g(u^+), g(u^-))$$

Remark 3.3. The above action of g on $\mathbb{R}_{\text{op}}^{1,1}$ is well-defined. In particular, note that in the u^+u^- -coordinates we have

$$\partial_+ \mathbb{R}_{\text{op}}^{1,1} = \{(u, u) : u \in \mathbb{R}\} \quad \partial_- \mathbb{R}_{\text{op}}^{1,1} = \{(u + 2\pi, u) : u \in \mathbb{R}\} \quad (3.1)$$

Then clearly $g(\partial_+ \mathbb{R}_{\text{op}}^{1,1}) = \partial_+ \mathbb{R}_{\text{op}}^{1,1}$. Using the fact that $g(u + 2\pi) = g(u) + 2\pi$, cf. (1.3), one immediately obtains $g(\partial_- \mathbb{R}_{\text{op}}^{1,1}) = \partial_- \mathbb{R}_{\text{op}}^{1,1}$.

Definition 3.4. For each $\pm \in \{+, -\}$, let

$$\tilde{\mathcal{J}}_{\pm} = \{\text{non-empty open intervals in } \partial_{\pm} \mathbb{R}_{\text{op}}^{1,1} \text{ of lengths } < 2\pi\}$$

Unless otherwise stated, we assume the equivalences

$$\tilde{\mathcal{J}}_+ \simeq \tilde{\mathcal{J}} \simeq \tilde{\mathcal{J}}_- \quad (3.2)$$

defined in the xt -coordinates by

$$\{0\} \times \mathbb{R} \simeq \mathbb{R} \simeq \{\pi\} \times \mathbb{R} \quad (0, t) \simeq t \simeq (\pi, \pi + t) \quad (3.3a)$$

equivalently, defined in the u^+u^- -coordinates (cf. (3.1)) by

$$(u, u) \simeq u \simeq (u + 2\pi, u) \quad (3.3b)$$

For example, $\tilde{\mathbb{S}}_+^1$ corresponds in the xt -coordinates to $\{(0, t) : 0 < t < \pi\}$ in $\tilde{\mathcal{J}}_+$, and $\tilde{\mathbb{S}}_-^1$ corresponds in the xt -coordinates to $\{(\pi, t) : 0 < t < \pi\}$ in $\tilde{\mathcal{J}}_-$.

Definition 3.5. By Def. 3.2 and Rem. 3.3, the group actions of $\mathcal{G}, \mathcal{G}_c$ on subsets of $\mathbb{R}_{\text{op}}^{1,1}$ restrict to group actions of $\mathcal{G}, \mathcal{G}_c$ on $\tilde{\mathcal{J}}_+$ and $\tilde{\mathcal{J}}_-$, both equivalent to the actions of $\mathcal{G}, \mathcal{G}_c$ on $\tilde{\mathcal{J}}$ via the equivalence (3.2).

3.1.2 Double cones in $\mathbb{R}_{(0,\pi)}^{1,1}$

Definition 3.6. For each double cone $O \subset \mathbb{R}_{(-\pi,\pi)}^{1,1}$, by choosing unique $(\tilde{I}, \tilde{J})$ satisfying (2.8) and corresponding to O , we define

$$\sqrt{O} = \text{the double cone corresponding to } (\sqrt{\tilde{I}}, -\sqrt{\tilde{J}})$$

In other words, in the u^+u^- -coordinates,

$$\sqrt{O} = \left\{ \left(\frac{u^+}{2}, \frac{u^-}{2} - \pi \right) : (u^+, u^-) \in O \right\}$$

Then $O \mapsto \sqrt{O}$ gives a one-to-one correspondence between double cones inside $\mathbb{R}_{(-\pi,\pi)}^{1,1}$ and $\mathbb{R}_{(0,\pi)}^{1,1}$.

Definition 3.7. Recall Def. 2.25 for the meaning of $\mathcal{O}_{(-\pi,\pi)}$. Let

$$\mathcal{O}_{(\mathbf{0},\pi)} = \{\text{double cones } O \subseteq \mathbb{R}_{(0,\pi)}^{1,1}\}$$

Then we have a bijective map

$$\mathcal{O}_{(-\pi,\pi)} \xrightarrow{\cong} \mathcal{O}_{(0,\pi)} \quad O \mapsto \sqrt{O}$$

Remark 3.8. One easily checks that double cones in $\mathbb{R}_{(0,\pi)}^{1,1}$ are precisely those corresponding to some $(\tilde{I}, \tilde{J})$ satisfying

$$0 < \arg_I z - \arg_J \zeta < 2\pi \quad \text{for each } z \in I, \zeta \in J$$

Therefore, $\mathcal{O}_{(0,\pi)}$ consists precisely of double cones corresponding to some $(\tilde{I}, \tilde{J})$ such that $\tilde{J}$ is clockwise to $\tilde{I}$, and that $\tilde{I}, \tilde{J}$ can be contained in some $\tilde{K} \in \tilde{\mathcal{J}}$.

3.2 Left boundary net $\mathcal{B}_{\text{op}(i)}$ and right boundary net $\mathcal{B}'_{\text{op}(i)}$ for the boundary condition i

Definition 3.9. For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}(\mathcal{A})$, the **state space with boundary conditions i, j** is defined to be

$$\mathcal{H}_{\text{op}(i,j)} = \mathcal{H}_i \boxtimes \mathcal{H}_{\bar{j}}$$

where the fusion product is in $\text{Rep}(\mathcal{A})$.

Definition 3.10. For each $\mathcal{H}_i \in \text{Rep}(\mathcal{A})$, the **left resp. right boundary nets with boundary condition i** are collections of von Neumann algebras

$$\mathcal{B}_{\text{op}(i)} = (\mathcal{B}_{\text{op}(i)}(\tilde{I}))_{\tilde{I} \in \tilde{\mathcal{J}}} \quad \text{resp.} \quad \mathcal{B}'_{\text{op}(i)} = (\mathcal{B}'_{\text{op}(i)}(\tilde{I}))_{\tilde{I} \in \tilde{\mathcal{J}}} \quad (3.4)$$

on $\mathcal{H}_{\text{op}(i,i)} = \mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}$ defined by

$$\mathcal{B}_{\text{op}(i)}(\tilde{I}) = \pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(I')}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}} \quad (3.5a)$$

$$\mathcal{B}'_{\text{op}(i)}(\tilde{I}) = \pi_{\tilde{I}}^R(\text{End}_{\mathcal{A}(I')}(\mathcal{H}_{\bar{i}}))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}} \quad (3.5b)$$

Recall the group action $U = U_{\text{op}(i,j)} = U_{i \boxtimes \bar{j}}$ of $\mathcal{G}_c$ on $\mathcal{H}_{\text{op}(i,j)}$ defined by Thm. 1.17. Recall the antiunitary map

$$\Theta_{i,\bar{j}} : \mathcal{H}_{\text{op}(i,j)} \rightarrow \mathcal{H}_{\text{op}(j,i)}$$

defined by Thm. 1.70, which satisfies

$$\Theta_{j,\bar{i}} \Theta_{i,\bar{j}} = 1$$

and hence is involutive when $i = j$.

Theorem 3.11. *Let $\mathcal{H}_i \in \text{Rep}(\mathcal{A})$. Then the nets $\mathcal{B}_{\text{op}(i)}$ and $\mathcal{B}'_{\text{op}(i)}$ satisfy the following properties for each $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$.*

(1) (Isotony) *If $\tilde{I} \subset \tilde{J}$, then*

$$\mathcal{B}_{\text{op}(i)}(\tilde{I}) \subset \mathcal{B}_{\text{op}(i)}(\tilde{J}) \quad \mathcal{B}'_{\text{op}(i)}(\tilde{I}) \subset \mathcal{B}'_{\text{op}(i)}(\tilde{J})$$

(2) (Extension property) *We have*

$$\pi_{i \boxtimes \tilde{I}, I}(\mathcal{A}(I)) \subset \mathcal{B}_{\text{op}(i)}(\tilde{I}) \cap \mathcal{B}'_{\text{op}(i)}(\tilde{I})$$

(3) (Conformal covariance) *For each $g \in \mathcal{G}_c$, the operator $U(g) = U_{\text{op}(i,i)}(g)$ satisfies*

$$U(g)\mathcal{B}_{\text{op}(i)}(\tilde{I})U(g)^* = \mathcal{B}_{\text{op}(i)}(g\tilde{I}) \quad U(g)\mathcal{B}'_{\text{op}(i)}(\tilde{I})U(g)^* = \mathcal{B}'_{\text{op}(i)}(g\tilde{I})$$

Moreover, if $g \in \mathcal{G}_c(I)$, then

$$U_{\text{op}(i,i)}(g) \in \pi_{i \boxtimes \tilde{I}, I}(\mathcal{A}(I))$$

and hence $U_{\text{op}(i,i)}(g) \in \mathcal{B}_{\text{op}(i)}(\tilde{I})$ by the extension property.

(4) (PCT symmetry) *The involutive anti-unitary operator $\Theta_{i,\tilde{I}}$ on $\mathcal{H}_{\text{op}(i,i)}$ satisfies*

$$\begin{aligned} \Theta_{i,\tilde{I}}U_{\text{op}(i,i)}(g)\Theta_{i,\tilde{I}} &= U_{\text{op}(i,i)}(\mathbf{r}g\mathbf{r}) \\ \Theta_{i,\tilde{I}}\mathcal{B}_{\text{op}(i)}(\tilde{I})\Theta_{i,\tilde{I}} &= \mathcal{B}'_{\text{op}(i)}(\tilde{I}) \end{aligned}$$

for each $g \in \mathcal{G}_c$.

(5) (Positive energy) *The self-adjoint generator of the one-parameter group $u \mapsto U_{\text{op}(i,i)}(\varrho_c(u))$ is positive.*

(6) (**Boundary-boundary Haag duality**) *$\mathcal{B}'_{\text{op}(i)}$ is the **dual net** of $\mathcal{B}_{\text{op}(i)}$ in the sense of [Gui21b, Sec. 4], that is, for each $\tilde{I} \in \tilde{\mathcal{J}}$,*

$$\mathcal{B}_{\text{op}(i)}(\tilde{I})' = \mathcal{B}'_{\text{op}(i)}(\tilde{I})$$

(7) (Irreducibility) *$\mathcal{B}_{\text{op}(i)}(\tilde{I})$ and $\mathcal{B}'_{\text{op}(i)}(\tilde{I})$ are type III factors. Moreover, $\bigcup_{\tilde{I} \in \tilde{\mathcal{J}}} \mathcal{B}_{\text{op}(i)}(\tilde{I})$ and $\bigcup_{\tilde{I} \in \tilde{\mathcal{J}}} \mathcal{B}'_{\text{op}(i)}(\tilde{I})$ together generate $\mathfrak{L}(\mathcal{H}_{\text{op}(i,i)})$.*

As a consequence of the boundary-boundary Haag duality and the isotony, we obtain the **boundary-boundary locality**: if $\tilde{J}$ is clockwise to $\tilde{I}$, then

$$[\mathcal{B}_{\text{op}(i)}(\tilde{I}), \mathcal{B}'_{\text{op}(i)}(\tilde{J})] = 0$$

Proof. Isotony: This is due to Rem. 1.54 and the obvious fact that $\text{End}_{\mathcal{A}(I'_1)}(\mathcal{H}_i) \subset \text{End}_{\mathcal{A}(I'_2)}(\mathcal{H}_i)$ and $\text{End}_{\mathcal{A}(I'_1)}(\mathcal{H}_{\bar{i}}) \subset \text{End}_{\mathcal{A}(I'_2)}(\mathcal{H}_{\bar{i}})$.

Extension property: Let $x \in \mathcal{A}(I)$. By (1.34), we have

$$\pi_{i \boxtimes \bar{i}, I}(x) = \pi_I^L(\pi_{i, I}(x))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}}$$

where $\pi_{i, I}(x)$ belongs to $\text{End}_{\mathcal{A}(I')}(\mathcal{H}_i)$. So $\pi_{i \boxtimes \bar{i}, I}(x) \in \mathcal{B}_{\text{op}(i)}(\tilde{I})$ by Def. 3.10. Similarly, (1.38) implies

$$\pi_{i \boxtimes \bar{i}, I}(x) = \pi_I^R(\pi_{\bar{i}, I}(x))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}}$$

where $\pi_{\bar{i}, I}(x) \in \text{End}_{\mathcal{A}(I')}(\mathcal{H}_{\bar{i}})$, and hence $\pi_{i \boxtimes \bar{i}, I}(x) \in \mathcal{B}'_{\text{op}(i)}(\tilde{I})$.

Conformal covariance: The first part is due to Thm. 1.56. Suppose that $g \in \mathcal{G}_c(I)$. Then $U_{\text{op}(i, i)}(g)$ equals $\pi_{i \boxtimes \bar{i}, I}(U_0(g))$ by Thm. 1.17, and hence belongs to $\pi_{i \boxtimes \bar{i}, I}(\mathcal{A}(I))$.

PCT symmetry: The first relation is due to Prop. 1.66. The second is due to Cor. 1.73.

Positive energy: Similar to the proof of Thm. 2.33, this is due to [Wei06].

Haag duality: By Def. 3.10,

$$\mathcal{B}'_{\text{op}(i)}(\tilde{\mathbb{S}}_-^1) = \pi_{\tilde{\mathbb{S}}_-^1}^R(\text{End}_{\mathcal{A}(\mathbb{S}_+^1)}(\mathcal{H}_{\bar{i}}))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}} = \left(\pi_{\tilde{\mathbb{S}}_+^1}^R(\text{End}_{\mathcal{A}(\mathbb{S}_-^1)}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{i}}} \right)'$$

where the last identity is due to Thm. 1.57. Since the last term is $\mathcal{B}_{\text{op}(i)}(\tilde{\mathbb{S}}_+^1)'$, we obtain $\mathcal{B}_{\text{op}(i)}(\tilde{I})' = \mathcal{B}'_{\text{op}(i)}(\tilde{I}')$ for $\tilde{I} = \tilde{\mathbb{S}}_+^1$. The general case follows easily from this one using the conformal covariance established above.

Irreducibility: Similar to the proof of Thm. 2.33, the fact that $\mathcal{B}_{\text{op}(i)}(\tilde{I})$ and $\mathcal{B}'_{\text{op}(i)}(\tilde{I})$ are type III factors follows from Rem. 1.55. Thus $\mathcal{B}_{\text{op}(i)}(\tilde{I}) \cup \mathcal{B}'_{\text{op}(i)}(\tilde{I})'$ generates $\mathfrak{L}(\mathcal{H}_{\text{op}(i, i)})$. By the boundary-boundary Haag duality, $\mathcal{B}_{\text{op}(i)}(\tilde{I}) \cup \mathcal{B}'_{\text{op}(i)}(\tilde{I}')$ generates $\mathfrak{L}(\mathcal{H}_{\text{op}(i, i)})$. $\square$

Remark 3.12. One should view $\mathcal{B}_{\text{op}(i)}$ and $\mathcal{B}'_{\text{op}(i)}$ as nets with domains $\tilde{\mathcal{J}}_+$ and $\tilde{\mathcal{J}}_-$, respectively, by using the equivalence $\tilde{\mathcal{J}}_+ \simeq \tilde{\mathcal{J}} \simeq \tilde{\mathcal{J}}_-$ in Def. 3.4. Now, for each $\tilde{I} \in \tilde{\mathcal{J}}$, if we view $\tilde{I}$ as in $\tilde{\mathcal{J}}_+$, and if we view its clockwise complement $\tilde{I}'$ as in $\tilde{\mathcal{J}}_-$, then one easily checks that

$$\tilde{I}' = \text{Int}\{p \in \partial_- \mathbb{R}_{\text{op}}^{1,1} : p \text{ is spacelike separated from } \tilde{I}\}$$

where the interior Int is with respect to $\partial_- \mathbb{R}_{\text{op}}^{1,1}$. In other words, $\tilde{I}'$ is the intersection of $\partial_- \mathbb{R}_{\text{op}}^{1,1}$ and the spacelike complement of $\tilde{I}$. This justifies the name “boundary-boundary Haag duality” in Thm. 3.11.

3.3 Boundary nets $\mathcal{B}_{\text{op}(i)}$ and $\mathcal{B}'_{\text{op}(j)}$ acting on the open-string state space

$$\mathcal{H}_{\text{op}(i, j)} = \mathcal{H}_i \boxtimes_{\mathcal{A}} \mathcal{H}_{\bar{j}}$$

Recall from Def. 3.9 that $\mathcal{H}_{\text{op}(i, j)} = \mathcal{H}_i \boxtimes \mathcal{H}_{\bar{j}}$.

Definition 3.13. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}(\mathcal{A})$ and $\tilde{I} \in \tilde{\mathcal{J}}$, define normal representations

$$\varpi_{\text{op}(i,j),\tilde{I}}^L : \mathcal{B}_{\text{op}(i)}(\tilde{I}) \rightarrow \mathfrak{L}(\mathcal{H}_{\text{op}(i,j)}) \quad \varpi_{\text{op}(i,j),\tilde{I}}^R : \mathcal{B}'_{\text{op}(j)}(\tilde{I}) \rightarrow \mathfrak{L}(\mathcal{H}_{\text{op}(i,j)})$$

abbreviated to $\varpi_{\tilde{I}}^L$ and $\varpi_{\tilde{I}}^R$ when no confusion arises, such that the following commuting squares commute:

$$\begin{array}{ccc} \text{End}_{\mathcal{A}(I')}(\mathcal{H}_i) & \xrightarrow{\pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}}} & \mathfrak{L}(\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}) \\ & \searrow \cong & \nearrow \varpi_{\tilde{I}}^L \\ & \pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}} & \mathcal{B}_{\text{op}(i)}(\tilde{I}) \end{array}$$

$$\begin{array}{ccc} \text{End}_{\mathcal{A}(I')}(\mathcal{H}_{\tilde{I}}) & \xrightarrow{\pi_{\tilde{I}}^R|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}}} & \mathfrak{L}(\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}) \\ & \searrow \cong & \nearrow \varpi_{\tilde{I}}^R \\ & \pi_{\tilde{I}}^R|_{\mathcal{H}_j \boxtimes \mathcal{H}_{\tilde{I}}} & \mathcal{B}'_{\text{op}(j)}(\tilde{I}) \end{array}$$

Remark 3.14. The above definition makes sense because $\pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{I}}}$ and $\pi_{\tilde{I}}^R|_{\mathcal{H}_j \boxtimes \mathcal{H}_{\tilde{I}}}$ are faithful, cf. Rem. 1.55. Note also that $\varpi_{\text{op}(i,i),\tilde{I}}^L$ and $\varpi_{\text{op}(j,j),\tilde{I}}^R$ are inclusion maps.

Theorem 3.15. Let $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}(\mathcal{A})$. Then the following properties hold for each $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$.

(1) (Isotony) If $\tilde{I} \subset \tilde{J}$, then

$$\varpi_{\tilde{J}}^L|_{\mathcal{B}_{\text{op}(i)}(\tilde{I})} = \varpi_{\tilde{I}}^L \quad \varpi_{\tilde{J}}^R|_{\mathcal{B}'_{\text{op}(j)}(\tilde{I})} = \varpi_{\tilde{I}}^R$$

(2) (Extension property) For each $x \in \mathcal{A}(I)$ and $y \in \mathcal{A}(J)$, recalling that

$$\pi_{i \boxtimes \tilde{I}}(x) \in \mathcal{B}_{\text{op}(i)}(\tilde{I}) \quad \pi_{j \boxtimes \tilde{J}}(y) \in \mathcal{B}'_{\text{op}(j)}(\tilde{J})$$

by the extension property in Thm. 3.11, we have

$$\varpi_{\tilde{I}}^L(\pi_{i \boxtimes \tilde{I},I}(x)) = \pi_{i \boxtimes \tilde{I},I}(x) \quad \varpi_{\tilde{J}}^L(\pi_{j \boxtimes \tilde{J},J}(y)) = \pi_{i \boxtimes \tilde{J},J}(y)$$

when acting on $\mathcal{H}_{\text{op}(i,j)}$.

(3) (Conformal covariance) For each $g \in \mathcal{G}_c$ and each

$$A \in \mathcal{B}_{\text{op}(i)}(\tilde{I}) \quad B \in \mathcal{B}'_{\text{op}(j)}(\tilde{I})$$

noting that $U(g)AU(g)^*$ belongs to $\mathcal{B}_{\text{op}(i)}(g\tilde{I})$ and $U(g)BU(g)^*$ belongs to $\mathcal{B}'_{\text{op}(j)}(g\tilde{I})$ by Thm. 3.11, we have

$$\begin{aligned} U(g)\varpi_{\tilde{I}}^L(A)U(g)^* &= \varpi_{g\tilde{I}}^L(U(g)AU(g)^*) && \text{on } \mathcal{H}_{\text{op}(i,j)} \\ U(g)\varpi_{\tilde{I}}^R(B)U(g)^* &= \varpi_{g\tilde{I}}^R(U(g)BU(g)^*) && \text{on } \mathcal{H}_{\text{op}(i,j)} \end{aligned}$$

- (4) (PCT symmetry) The anti-unitary map $\Theta_{i,\bar{j}} : \mathcal{H}_{\text{op}(i,j)} \rightarrow \mathcal{H}_{\text{op}(j,i)}$ (with inverse $\Theta_{j,\bar{i}} : \mathcal{H}_{\text{op}(j,i)} \rightarrow \mathcal{H}_{\text{op}(i,j)}$, cf. Rem. 1.71) satisfies

$$\begin{aligned}\Theta_{i,\bar{j}} \cdot U_{\text{op}(i,j)}(g) \cdot \Theta_{j,\bar{i}} &= U_{\text{op}(j,i)}(\mathbf{r}g\mathbf{r}) && \text{on } \mathcal{H}_{\text{op}(j,i)} \\ \Theta_{i,\bar{j}} \cdot \varpi_{\tilde{I}}^L(A) \cdot \Theta_{j,\bar{i}} &= \varpi_{\tilde{I}'}^R(\Theta_{i,\bar{i}}A\Theta_{i,\bar{i}}) && \text{on } \mathcal{H}_{\text{op}(j,i)}\end{aligned}$$

for each $g \in \mathcal{G}_c$ and $A \in \mathcal{B}_{\text{op}(i)}(\tilde{I})$, noting that $\Theta_{i,\bar{i}}A\Theta_{i,\bar{i}} \in \mathcal{B}'_{\text{op}(i)}(\mathbf{r}\tilde{I})$ by Thm. 3.11.

- (5) (Positive energy) The self-adjoint generator of the one-parameter group $u \mapsto U_{\text{op}(i,j)}(\varrho_c(u))$ is positive.
- (6) (**Boundary-boundary Haag duality**) The following von Neumann algebras on $\mathcal{H}_{\text{op}(i,j)}$ are commutants of each other:

$$\varpi_{\tilde{I}}^L(\mathcal{B}_{\text{op}(i)}(\tilde{I})) \quad \varpi_{\tilde{I}'}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{I}')) \quad (3.6)$$

The Haag duality in this theorem admits an interpretation analogous to that in Rem. 3.12. Moreover, as a consequence of the isotony and the Haag duality, we obtain the **boundary-boundary locality**: if $\tilde{J}$ is clockwise to $\tilde{I}$, then

$$[\varpi_{\tilde{I}}^L(\mathcal{B}_{\text{op}(i)}(\tilde{I})), \varpi_{\tilde{J}}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{J}))] = 0$$

Proof. Isotony: This is due to Rem. 1.54.

Extension property: By (1.34), the element $\pi_{i,I}(x) \in \text{End}_{\mathcal{A}(I')}(\mathcal{H}_i)$ is sent by $\pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{j}}}$ and $\pi_{\tilde{I}}^L|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{j}}}$ to $\pi_{i \boxtimes \tilde{i}}(x)$ and $\pi_{i \boxtimes \tilde{j}}(x)$, respectively. Thus $\varpi_{\tilde{I}}^L(\pi_{i \boxtimes \tilde{i},I}(x)) = \pi_{i \boxtimes \tilde{j},I}(x)$ by Def. 3.13. A similar argument proves $\varpi_{\tilde{J}}^L(\pi_{j \boxtimes \tilde{j},J}(y)) = \pi_{i \boxtimes \tilde{j},J}(y)$.

Conformal covariance: This is due to Thm. 1.56.

PCT symmetry: The first relation is due to Prop. 1.66. The second is due to Thm. 1.72.

Positive energy: Similar to the proof of Thm. 2.33, this is due to [Wei06].

Haag duality: By Def. 3.13, the two von Neumann algebras in (3.6) are equal to $\pi_{\tilde{I}}^L(\text{End}_{\mathcal{A}(I')}(\mathcal{H}_i))$ and $\pi_{\tilde{I}'}^R(\text{End}_{\mathcal{A}(I)}(\mathcal{H}_{\tilde{j}}))$ respectively. The latter two von Neumann algebras (acting on $\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{j}}$) are commutants of each other by Thm. 1.57. $\square$

3.4 Restricted bulk net $\mathcal{B}_{\text{cl}}|_{\mathcal{O}_{(0,\pi)}}$ acting on $\mathcal{H}_{\text{op}(i,j)}$

Fix $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}(\mathcal{A})$.

Definition 3.16. For each $O \in \mathcal{O}_{(-\pi,\pi)}$, corresponding to $(\tilde{I}, \tilde{J})$ satisfying (2.8), choose $g_+, g_- \in \mathcal{G}$ such that by viewing $g_{\pm}$ as in (1.3), and by viewing $\tilde{I}, \tilde{J}$ as subsets of the universal cover $\mathbb{R}$ of $\mathbb{R}/2\pi\mathbb{Z} = \mathbb{S}^1$, we have

$$\begin{aligned}g_+(\theta) &= \theta/2 && \text{for each } \theta \in \tilde{I} \\ g_-(\theta) &= \theta/2 - \pi && \text{for each } \theta \in \tilde{J}\end{aligned}$$

Note that $(g_+, g_-)O = \sqrt{O}$. Lift g_+, g_- to elements of $\mathcal{G}_c$, still denoted by g_+, g_- . By the conformal covariance in Thm. 2.33, we can define (spatial) isomorphisms of von Neumann algebras

$$\mathbf{Q}_{\sqrt{O}} : \mathcal{B}_{\text{cl}}(O) \rightarrow \mathcal{B}_{\text{cl}}(\sqrt{O}) \quad A \mapsto U_{\text{cl}}(g_+, g_-)AU(g_+, g_-)^*$$

This definition is independent of the choice $g_{\pm}$.

Explanation. To prove the independence of $\mathbf{Q}_{\sqrt{O}}$ on $g_{\pm}$, assume that $h_{\pm}$ satisfy the same properties as $g_{\pm}$. Then $h_+^{-1}g_+$ fixes each point of $\tilde{I}$. Thus, by (1.3), $h_+^{-1}g_+$ fixes each point of $\tilde{I} + 2\pi\mathbb{Z}$ (when $\tilde{I}$ is viewed as in $\mathbb{R}$). It follows from Def. 1.4 that $h_+^{-1}g_+ \in \mathcal{G}_c(I')$. Similarly, $h_-^{-1}g_- \in \mathcal{G}_c(J')$. Then, by choosing suitable arg-functions on I', J' , the spacelike complement O' of O corresponds to $((I', \arg_{I'}), (J', \arg_{J'}))$. By the conformal covariance in Thm. 2.33, we have

$$U_{\text{cl}}(h_+, h_-)^{-1}U_{\text{cl}}(g_+, g_-) \in \mathcal{B}_{\text{cl}}(O')$$

Therefore, by the locality in Thm. 2.33, $\text{Ad}_{U_{\text{cl}}(g_+, g_-)}$ equals $\text{Ad}_{U_{\text{cl}}(h_+, h_-)}$ when applied to $\mathcal{B}_{\text{cl}}(O)$. $\square$

Remark 3.17. Clearly, if $O_1 \subset O_2$ are in $\mathcal{O}_{(-\pi, \pi)}$, then $\mathbf{Q}_{\sqrt{O_2}}|_{\mathcal{B}_{\text{cl}}(O_1)} = \mathbf{Q}_{\sqrt{O_1}}$.

Definition 3.18. For each $O \in \mathcal{O}_{(-\pi, \pi)}$, which corresponds uniquely to $(\tilde{I}, \tilde{J})$ satisfying (2.8), choose $\tilde{K}_+, \tilde{K}_- \in \tilde{\mathcal{J}}$ such that

$$\sqrt{\tilde{I}}, \tilde{K}_+, -\sqrt{\tilde{J}}, \tilde{K}_- \quad \text{are in clockwise order} \quad (3.7)$$

Choose K_+ -unitary $\xi_0 \in \mathcal{H}_i(K_+)$ and K_- -unitary $\eta_0 \in \mathcal{H}_{\tilde{j}}(K_-)$, and define a faithful normal unitary representation

$$\varpi_{\text{op}(\mathbf{i}, \mathbf{j}), \sqrt{O}} : \mathcal{B}_{\text{cl}}(\sqrt{O}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{op}(\mathbf{i}, \mathbf{j})})$$

abbreviated to $\varpi_{\sqrt{O}}$ when no confusion arises, such that by setting

$$V(\xi_0, \eta_0) = L(\xi_0, \tilde{K}_+)R(\eta_0, \tilde{K}_-) \equiv R(\eta_0, \tilde{K}_-)L(\xi_0, \tilde{K}_+) : \mathcal{H}_0 \rightarrow \mathcal{H}_{\text{op}(\mathbf{i}, \mathbf{j})}$$

and by choosing any $\tilde{K} \in \tilde{\mathcal{J}}$ containing $\tilde{I}, \tilde{J}$ (which exists by Rem. 2.24), the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{A}(\sqrt{\tilde{I}}, -\sqrt{\tilde{J}}) & \xrightarrow{\text{Ad}_{V(\xi_0, \eta_0)}} & \mathcal{L}(\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{j}}) \\ \pi_{\tilde{K}}^L|_{\mathcal{H}_{\sqrt{O} \boxtimes \mathcal{H}_{\sqrt{O}}}} \downarrow \simeq & & \uparrow \varpi_{\sqrt{O}} \\ \mathcal{B}_{\text{cl}}(O) & \xrightarrow[\simeq]{\mathbf{Q}_{\sqrt{O}}} & \mathcal{B}_{\text{cl}}(\sqrt{O}) \end{array} \quad (3.8)$$

Note that the isomorphism from the upper-left corner to the lower-left corner is given by Def. 2.29.

Remark 3.19. The definition of $\varpi_{\sqrt{O}}$ is independent of $\tilde{K}$, as indicated in the explanation to Def. 2.29, and also by Rem. 3.17. It is independent of $\tilde{K}_1, \tilde{K}_2$ due to the isotony property for L, R operators (cf. Cor. 1.43). It is also independent of the choice of ξ_0, η_0 , as explained in the following proof.

Proof. Choose K_+ -unitary $\xi_1 \in \mathcal{H}_i(K_+)$ and K_- -unitary $\eta_1 \in \mathcal{H}_{\bar{j}}(K_-)$. Then $L(\xi_0, \tilde{K}_+)^* L(\xi_1, \tilde{K}_+)|_{\mathcal{H}_0}$ commutes with $\mathcal{A}(K'_+)$, and hence belongs to $\mathcal{A}(K_+)$ by the Haag duality. Thus

$$L(\xi_1, \tilde{K}_+)|_{\mathcal{H}_0} = L(\xi_0, \tilde{K}_+)x|_{\mathcal{H}_0}$$

for some (necessarily unitary) $x \in \mathcal{A}(K_+)$. Thus

$$V(\xi_1, \eta_0) = V(\xi_0, \eta_0)x$$

By Def. 2.28, elements of $\mathcal{A}(K_+)$ commute with those of $\mathfrak{A}(\sqrt{I}, -\sqrt{J})$. Therefore $\text{Ad}_{V(\xi_1, \eta_0)} = \text{Ad}_{V(\xi_0, \eta_0)}$ on $\mathfrak{A}(\sqrt{I}, -\sqrt{J})$. A similar argument shows $\text{Ad}_{V(\xi_1, \eta_1)} = \text{Ad}_{V(\xi_1, \eta_0)}$ on $\mathfrak{A}(\sqrt{I}, -\sqrt{J})$. Hence $\text{Ad}_{V(\xi_1, \eta_1)} = \text{Ad}_{V(\xi_0, \eta_0)}$ on $\mathfrak{A}(\sqrt{I}, -\sqrt{J})$. $\square$

Note that in Def. 3.18, we do not require $\sqrt{\tilde{K}}, \tilde{K}_+, -\sqrt{\tilde{K}}, \tilde{K}_-$ to be in clockwise order (indeed, these intervals may intersect), although after fixing $\tilde{K}$, one may choose $\tilde{K}_{\pm}$ so that they are in clockwise order.

Theorem 3.20. *The following properties hold for each $O, O_1, O_2 \in \mathcal{O}_{(-\pi, \pi)}$.*

(1) *(Isotony) If $O_1 \subset O_2$, then*

$$\varpi_{\sqrt{O_2}}|_{\mathcal{B}_{\text{cl}}(\sqrt{O_1})} = \varpi_{\sqrt{O_1}}$$

(2) **(Bulk-bulk locality)** *Suppose that O_1, O_2 are spacelike separated (equivalently, $\sqrt{O_1}, \sqrt{O_2}$ are spacelike separated), then*

$$[\varpi_{\sqrt{O_1}}(\mathcal{B}_{\text{cl}}(\sqrt{O_1})), \varpi_{\sqrt{O_2}}(\mathcal{B}_{\text{cl}}(\sqrt{O_2}))] = 0$$

(3) *(Extension property) Let O correspond to $(\tilde{I}, \tilde{J})$. For each $x \in \mathcal{A}(\sqrt{I})$ and $y \in \mathcal{A}(-\sqrt{J})$, noting that*

$$\pi_{\sqrt{0} \boxtimes \sqrt{0}, \sqrt{I}}^+(x) \in \mathcal{B}_{\text{cl}}(\sqrt{O}) \quad \pi_{\sqrt{0} \boxtimes \sqrt{0}, -\sqrt{J}}^-(y) \in \mathcal{B}_{\text{cl}}(\sqrt{O})$$

by the extension property in Thm. 2.33, we have

$$\begin{aligned} \varpi_{\sqrt{O}}(\pi_{\sqrt{0} \boxtimes \sqrt{0}, \sqrt{I}}^+(x)) &= \pi_{i \boxtimes \bar{j}, \sqrt{I}}(x) \\ \varpi_{\sqrt{O}}(\pi_{\sqrt{0} \boxtimes \sqrt{0}, -\sqrt{J}}^-(y)) &= \pi_{i \boxtimes \bar{j}, -\sqrt{J}}(y) \end{aligned}$$

(4) (Conformal covariance) Suppose that $g \in \mathcal{G}_c$ and $A \in \mathcal{B}_{\text{cl}}(\sqrt{O})$. Then, noting that $U_{\text{cl}}(g, g)AU_{\text{cl}}(g, g)^*$ belongs to $\mathcal{B}_{\text{cl}}((g, g)\sqrt{O})$ by the conformal covariance in Thm. 2.33, we have

$$U_{i\boxtimes j}(g)\varpi_{\sqrt{O}}(A)U_{i\boxtimes j}(g)^* = \varpi_{(g,g)\sqrt{O}}(U_{\text{cl}}(g, g)AU_{\text{cl}}(g, g)^*)$$

(5) (**Bulk-boundary Haag duality**) Let $\tilde{E}_+$ and $\tilde{E}_-$ be the intersection of the spacelike complement of $\sqrt{O}$ with $\partial_+\mathbb{R}_{\text{op}}^{1,1}$ and $\partial_-\mathbb{R}_{\text{op}}^{1,1}$ respectively, i.e., for each $\pm \in \{+, -\}$,

$$\tilde{E}_{\pm} = \text{Int}\{p \in \partial_{\pm}\mathbb{R}_{\text{op}}^{1,1} : p \text{ is spacelike separated from elements of } \sqrt{O}\}$$

where the interior is with respect to $\partial_{\pm}\mathbb{R}_{\text{op}}^{1,1}$. Then $\tilde{E}_+ \in \tilde{\mathcal{J}}_+$ and $\tilde{E}_- \in \tilde{\mathcal{J}}_-$. Moreover, by viewing $\tilde{E}_+, \tilde{E}_- \in \tilde{\mathcal{J}}$ via the equivalence (3.2), we have

$$\varpi_{\tilde{E}_+}^L(\mathcal{B}_{\text{op}(i)}(\tilde{E}_+)) \vee \varpi_{\tilde{E}_-}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{E}_-)) = \varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O}))'$$

The bulk-boundary Haag duality implies the **bulk-boundary locality**: If $\tilde{E}_+ \in \tilde{\mathcal{J}}_+$ and $\tilde{E}_- \in \tilde{\mathcal{J}}_-$ are spacelike separated from $\sqrt{O}$, then both $\varpi_{\tilde{E}_+}^L(\mathcal{B}_{\text{op}(i)}(\tilde{E}_+))$ and $\varpi_{\tilde{E}_-}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{E}_-))$ commutes with $\varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O}))$.

Proof. Isotony: This is obvious.

Bulk-bulk locality: Since $\varpi_{\sqrt{O}}$ is normal for each O , by the additivity of $\mathcal{B}_{\text{cl}}$ (Cor. 2.40) and Lem. 2.42, it suffices to assume that $O_1, O_2 \subset O$ for some standard $O \in \mathcal{O}_{(-\pi, \pi)}$. In that case, we have $\varpi_{\sqrt{O_i}}(\mathcal{B}_{\text{cl}}(\sqrt{O_i})) = \varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O_i}))$. Thus the commutativity follows from the locality in Thm. 2.33.

Extension property: See Lem. 3.21.

Conformal covariance: See Lem. 3.22.

Bulk-boundary Haag duality: Let $(\tilde{I}, \tilde{J})$ satisfy (2.8) and correspond to O . Choose $\tilde{K}_+, \tilde{K}_- \in \tilde{\mathcal{J}}$ such that K_+, K_- are the two components of the interior of $\mathbb{S}^1 \setminus (\sqrt{I} \cup -\sqrt{J})$, and satisfying (3.7). By viewing $\tilde{K}_+ \in \tilde{\mathcal{J}}_+$ and $\tilde{K}_- \in \tilde{\mathcal{J}}_-$, one checks (e.g. by projecting $\tilde{E}_+, \tilde{E}_-$ onto the u^- -axis, cf. (3.3b)) that

$$\tilde{E}_+ = \tilde{K}_+ \quad \tilde{E}_- = \tilde{K}_- \tag{3.9}$$

This proves the first part of the Haag duality.

By Def. 3.18, we have

$$\varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O})) = \text{Ad}_{V(\xi_0, \eta_0)}(\mathfrak{A}(\sqrt{I}, -\sqrt{J})) \tag{a}$$

By Prop. 1.50, we have

$$\text{End}_{\mathcal{A}(E'_+)}(\mathcal{H}_i) = \text{Ad}_{L(\xi_0, \tilde{E}_+)|_{\mathcal{H}_0}}(\mathcal{A}(E_+))$$

Therefore, since $\tilde{E}_-$ is clockwise to $\tilde{E}_+$, by (1.33) we have

$$\pi_{\tilde{E}_+}^L(\text{End}_{\mathcal{A}(E'_+)}(\mathcal{H}_i))|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\tilde{J}}} = \text{Ad}_{R(\eta_0, \tilde{E}_-)}(\text{End}_{\mathcal{A}(E'_+)}(\mathcal{H}_i)) = \text{Ad}_{V(\xi_0, \eta_0)}(\mathcal{A}(E_+))$$

It follows from Def. 3.13 that

$$\varpi_{\tilde{E}_+}^L(\mathcal{B}_{\text{op}(i)}(\tilde{E}_+)) = \text{Ad}_{V(\xi_0, \eta_0)}(\mathcal{A}(E_+)) \quad (\text{b})$$

A similar argument shows that

$$\varpi_{\tilde{E}_-}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{E}_-)) = \text{Ad}_{V(\xi_0, \eta_0)}(\mathcal{A}(E_-)) \quad (\text{c})$$

Combining (a), (b), (c) with the relation (cf. Def. 2.28)

$$\mathcal{A}(E_+) \vee \mathcal{A}(E_-) = \mathfrak{A}(\sqrt{I}, -\sqrt{J})'$$

the proof of the second part of the Haag duality is finished. $\square$

Lemma 3.21. *Extension property in Thm. 3.20 holds.*

Proof. We prove the relation for x . The relation for y can be proved in a similar way. Choose $g_+, g_- \in \mathcal{G}_c$ as in Def. 3.16. Let $\hat{x} \in \mathcal{A}(I)$ such that the isomorphism $\mathbf{Q}_{\sqrt{I}} : \mathcal{A}(I) \rightarrow \mathcal{A}(\sqrt{I})$ in Def. 2.5 sends $\hat{x}$ to x , that is,

$$\mathbf{Q}_{\sqrt{I}}(\hat{x}) \equiv U_0(g_+) \hat{x} U_0(g_+)^* = x \quad (\star)$$

Choose ξ_0, η_0 and $\tilde{K}$ as in Def. 3.18. In view of the commutative diagram (3.8), the proof will be finished if we can show for the element $x \in \mathcal{A}(\sqrt{I})$ that

$$\begin{array}{ccc} x & \xrightarrow{\text{Ad}_{V(\xi_0, \eta_0)}} & \pi_{i \boxtimes \tilde{J}, \sqrt{I}}(x) \\ \pi_{\tilde{K}}^L|_{\mathcal{H}_{\sqrt{0} \boxtimes \mathcal{H}_{\sqrt{0}}}} \downarrow & & \\ \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}^+(\hat{x}) & \xrightarrow{\mathbf{Q}_{\sqrt{0}}} & \pi_{\sqrt{0} \boxtimes \sqrt{0}, \sqrt{I}}^+(x) \end{array}$$

The top vertical arrow is due to (1.20). By $(\star)$ and Def. 2.7, we have

$$\pi_{\sqrt{0}, I}(\hat{x} \otimes 1) = x$$

By viewing $x \in \mathfrak{A}(\sqrt{I}, -\sqrt{J}) \subset \text{End}_{\mathcal{A}^{\otimes 2}(\tilde{K})}(\mathcal{H}_{\sqrt{0}})$ (cf. (2.10)), it follows from (1.34) that

$$\begin{aligned} \pi_{\tilde{K}}^L(x)|_{\mathcal{H}_{\sqrt{0} \boxtimes \mathcal{H}_{\sqrt{0}}}} &= \pi_{\tilde{K}}^L(\pi_{\sqrt{0}, I}(\hat{x} \otimes 1))|_{\mathcal{H}_{\sqrt{0} \boxtimes \mathcal{H}_{\sqrt{0}}}} \\ &\stackrel{\text{Rem. 1.54}}{=} \pi_{\tilde{I}}^L(\pi_{\sqrt{0}, I}(\hat{x} \otimes 1))|_{\mathcal{H}_{\sqrt{0} \boxtimes \mathcal{H}_{\sqrt{0}}}} = \pi_{\sqrt{0} \boxtimes \sqrt{0}, I}(\hat{x} \otimes 1) \end{aligned}$$

where the last term is exactly $\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x})$ by Def. 2.1. This proves the left vertical arrow. By Def. 3.16,

$$\mathbf{Q}_{\sqrt{O}}(\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x})) = \text{Ad}_{U_{\text{cl}}(g_+,g_-)}(\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x}))$$

Recall from Def. 2.31 that $U_{\text{cl}}(g_+,g_-) = U_{\sqrt{0}\boxtimes\sqrt{0}}^+(g_+)U_{\sqrt{0}\boxtimes\sqrt{0}}^-(g_-)$. By Rem. 2.3, $U_{\sqrt{0}\boxtimes\sqrt{0}}^-(g_-)$ commutes with $\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x})$. Hence

$$\begin{aligned} \mathbf{Q}_{\sqrt{O}}(\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x})) &= \text{Ad}_{U_{\sqrt{0}\boxtimes\sqrt{0}}^+(g_+)}(\pi_{\sqrt{0}\boxtimes\sqrt{0},I}^+(\hat{x})) \\ &\stackrel{\text{Cor.1.20}}{=} \pi_{\sqrt{0}\boxtimes\sqrt{0},\sqrt{I}}^+(U_0(g_+)\hat{x}U_0(g_+)^*) \stackrel{(*)}{=} \pi_{\sqrt{0}\boxtimes\sqrt{0},\sqrt{I}}^+(x) \end{aligned}$$

This proves the bottom vertical arrow. □

Lemma 3.22. *Conformal covariance in Thm. 3.20 holds.*

Proof. Step 1. It suffices to prove the lemma when O is standard, since the general case can be reduced to this case by enlarging O (cf. Rem. 2.24). Thus, we may assume that O corresponds to $(\tilde{I}, \tilde{I})$.

Moreover, we assume that there exists $E \in \mathcal{J}$ of length $< \pi/2$ such that $g \in \mathcal{G}_c(E)$, and that $\tilde{I}$ has length $< \pi/100$. The general case can be easily reduced to this special case by Lem. 1.18 and the additivity of $\mathcal{B}_{\text{cl}}$ (Prop. 2.39, applied to a collection of standard double cones covering the open interval from the top to the bottom of O).

By our assumption on the lengths, there exists $\tilde{J} \in \tilde{\mathcal{J}}$ satisfying:

- $\tilde{I} \subset \tilde{J}$.
- Either $E \subset \sqrt{J}$ or $E \subset -\sqrt{J}$.

We assume $E \subset \sqrt{J}$; the other case can be treated in a similar way. In particular,

$$g \in \mathcal{G}_c(\sqrt{J})$$

(In the second case, we have instead $g \in \mathcal{G}_c(-\sqrt{J})$.) We can now forget about E . Let $D \in \mathcal{O}_{(-\pi,\pi)}$ correspond to $(\tilde{J}, \tilde{J})$.

Step 2. Note that $(g,g)\sqrt{D} = \sqrt{D}$. By Thm. 1.17, we have $U_{i\boxtimes\tilde{J}}(g) = \pi_{i\boxtimes\tilde{J},\sqrt{J}}(U_0(g))$. Therefore, by Lem. 3.21,

$$U_{i\boxtimes\tilde{J}}(g) = \varpi_{\sqrt{D}}(\pi_{\sqrt{0}\boxtimes\sqrt{0},\sqrt{J}}^+(U_0(g))) = \varpi_{\sqrt{D}}(U_{\text{cl}}^+(g))$$

where the last term is due to Thm. 1.17. Thus, for each $A \in \mathcal{B}_{\text{cl}}(\sqrt{O}) \subset \mathcal{B}_{\text{cl}}(\sqrt{D})$,

$$U_{i\boxtimes\tilde{J}}(g)\varpi_{\sqrt{D}}(A)U_{i\boxtimes\tilde{J}}(g)^* = \varpi_{\sqrt{D}}(U_{\text{cl}}^+(g)AU_{\text{cl}}^+(g)^*)$$

Since $g \in \mathcal{G}_c(\sqrt{J})$ and hence fixes elements of $(-\sqrt{J}, \arg_{-\sqrt{J}})$ for each $\arg_{-\sqrt{J}}$ (cf. Rem. 1.9), it follows from the conformal covariance in Thm. 2.33 that

$$U_{\text{cl}}(1, g) \in \mathcal{B}_{\text{cl}}(\text{the causal complement of } \sqrt{D} \text{ in } \mathbb{R}_{\text{cl}}^{1,1})$$

Therefore, by the locality in Thm. 2.33, we have $\text{Ad}_{U_{\text{cl}}(1, g)} = \text{id}$ on $\mathcal{B}_{\text{cl}}(\sqrt{D})$. Since $U_{\text{cl}}^+(g) = U_{\text{cl}}(g, 1)$ (cf. Def. 2.31), we obtain

$$U_{i[\Box]j}(g) \varpi_{\sqrt{D}}(A) U_{i[\Box]j}(g)^* = \varpi_{\sqrt{D}}(U_{\text{cl}}(g, g) A U_{\text{cl}}(g, g)^*)$$

and hence

$$U_{i[\Box]j}(g) \varpi_{\sqrt{O}}(A) U_{i[\Box]j}(g)^* = \varpi_{(g, g)\sqrt{O}}(U_{\text{cl}}(g, g) A U_{\text{cl}}(g, g)^*)$$

by the isotony in Thm. 3.20 and the fact that $\sqrt{O} \subset \sqrt{D}$ and $(g, g)\sqrt{O} \subset \sqrt{D}$. $\square$

Theorem 3.23 (Bulk-boundary Haag duality). *Let $n \geq 1$. Let $O_1, \dots, O_n \in \mathcal{O}_{(-\pi, \pi)}$ whose closures are mutually spacelike separated, and which are contained in a common double cone $O \in \mathcal{O}_{(-\pi, \pi)}$. Let Σ be the spacelike complement of $\sqrt{O_1}, \dots, \sqrt{O_n}$ in $\mathbb{R}_{\text{op}}^{1,1}$. Let $D_1, \dots, D_{n-1} \in \mathcal{O}_{(-\pi, \pi)}$ such that $\sqrt{D_1}, \dots, \sqrt{D_{n-1}}$ are the connected components of Σ that are contained in $\mathbb{R}_{(0, \pi)}^{1,1}$. Let*

$$\tilde{E}_+ = \Sigma \cap \partial_+ \mathbb{R}_{\text{op}}^{1,1} \quad \tilde{E}_- = \Sigma \cap \partial_- \mathbb{R}_{\text{op}}^{1,1}$$

Then $\tilde{E}_+ \in \tilde{\mathcal{J}}_+$ and $\tilde{E}_- \in \tilde{\mathcal{J}}_-$, and the following two von Neumann algebras on $\mathcal{H}_{\text{op}(i, j)}$ are commutants of each other:

$$\bigvee_{j=1}^n \varpi_{\sqrt{O_j}}(\mathcal{B}_{\text{cl}}(\sqrt{O_j}))$$

$$\varpi_{\tilde{E}_+}^L(\mathcal{B}_{\text{op}(i)}(\tilde{E}_+)) \vee \varpi_{\tilde{E}_-}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{E}_-)) \vee \bigvee_{j=1}^{n-1} \varpi_{\sqrt{D_j}}(\mathcal{B}_{\text{cl}}(\sqrt{D_j}))$$

Proof. The case $n = 1$ follows from Thm. 3.20. Now assume $n > 1$. We assume that $O_1, \dots, O_n$ are listed from left to right. By shrinking O , we assume that the left corners of O_1 and O are the same, and the right corners of O_n and O are the same. Then $\tilde{E}_{\pm}$ is the intersection of the spacelike complement of $\sqrt{O}$ (in $\mathbb{R}_{\text{op}}^{1,1}$) with $\partial_{\pm} \mathbb{R}_{\text{op}}^{1,1}$, which belongs to $\tilde{\mathcal{J}}_{\pm}$ by Thm. 3.20. Moreover, by Thm. 3.20, the commutant of $\varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O}))$ is generated by $\varpi_{\tilde{E}_+}^L(\mathcal{B}_{\text{op}(i)}(\tilde{E}_+))$ and $\varpi_{\tilde{E}_-}^R(\mathcal{B}'_{\text{op}(j)}(\tilde{E}_-))$. Therefore, it suffices to show that

$$\varpi_{\sqrt{O}}(\mathcal{B}_{\text{cl}}(\sqrt{O})) \cap \left(\bigvee_{j=1}^{n-1} \varpi_{\sqrt{D_j}}(\mathcal{B}_{\text{cl}}(\sqrt{D_j})) \right)' = \bigvee_{j=1}^n \varpi_{\sqrt{O_j}}(\mathcal{B}_{\text{cl}}(\sqrt{O_j}))$$

This follows from the faithfulness of the normal representation $\varpi_{\sqrt{O}}$, together with the fact that the relative commutant of $\bigvee_{j=1}^{n-1} \mathcal{B}_{\text{cl}}(\sqrt{D_j})$ in $\mathcal{B}_{\text{cl}}(\sqrt{O})$ is equal to $\bigvee_{j=1}^n \mathcal{B}_{\text{cl}}(\sqrt{O_j})$ (as implied by Thm. 2.62 or by its proof). $\square$

3.5 Nets of boundary algebroids changing the boundary conditions

For each $\mathcal{H}_i \in \text{Rep}(\mathcal{A})$, in Sec. 3.3 we have considered the nets $\mathcal{B}_{\text{op}(i)}, \mathcal{B}'_{\text{op}(i)}$ of (left and right) boundary operators preserving the boundary condition i . From Def. 3.10, for each $\tilde{I} \in \tilde{\mathcal{J}}$ we have an isomorphisms

$$\mathcal{B}_{\text{op}(i)}(\tilde{I}) \simeq \text{End}_{\mathcal{A}(I')}(\mathcal{H}_i) \quad \mathcal{B}'_{\text{op}(i)}(\tilde{I}) \simeq \text{End}_{\mathcal{A}(I')}(\mathcal{H}_{\bar{i}})$$

More generally, for each $\mathcal{H}_i, \mathcal{H}_k \in \text{Rep}(\mathcal{A})$, we view

$$\tilde{I} \in \tilde{\mathcal{J}} \mapsto \mathcal{B}_{\text{op}(i \rightarrow k)}(\tilde{I}) = \text{Hom}_{\mathcal{A}(I')}(\mathcal{H}_i, \mathcal{H}_k)$$

as the net of von Neumann algebroids of left boundary operators changing the boundary condition from i to k . Similarly, for each $\mathcal{H}_j, \mathcal{H}_l \in \text{Rep}(\mathcal{A})$, we view

$$\tilde{I} \in \tilde{\mathcal{J}} \mapsto \mathcal{B}'_{\text{op}(j \rightarrow l)}(\tilde{I}) = \text{Hom}_{\mathcal{A}(I')}(\mathcal{H}_{\bar{j}}, \mathcal{H}_{\bar{l}})$$

as the net of von Neumann algebroids of right boundary operators changing the boundary condition from j to l .

The nets $\mathcal{B}_{\text{op}(i \rightarrow k)}$ and $\mathcal{B}'_{\text{op}(j \rightarrow l)}$ act on boundary state spaces as follows: Choose any I' -unitary $\psi \in \mathcal{H}_{\bar{b}}(I')$. For each $\mathcal{H}_b \in \text{Rep}(\mathcal{A})$, we have

$$\begin{aligned} \varpi_{\tilde{I}}^L : \mathcal{B}_{\text{op}(i \rightarrow k)}(\tilde{I}) &\rightarrow \mathfrak{L}(\mathcal{H}_{\text{op}(i,b)}, \mathcal{H}_{\text{op}(k,b)}) \\ A &\mapsto R(\psi, \tilde{I}')AR(\psi, \tilde{I}')^*|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{b}}} \end{aligned}$$

This definition is independent of the choice of ψ . Indeed, if we choose I -unitary vectors $\xi_i \in \mathcal{H}_i(I)$ and $\xi_k \in \mathcal{H}_k(I)$, then by Prop. 1.50,

$$A = L(\xi_k, \tilde{I})xL(\xi_i, \tilde{I})^*|_{\mathcal{H}_i}$$

for some $x \in \mathcal{A}(I)$. Hence, by the locality in Thm. 1.37,

$$\varpi_{\tilde{I}}^L(L(\xi_k, \tilde{I})xL(\xi_i, \tilde{I})^*|_{\mathcal{H}_i})\Big|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{b}}} = L(\xi_k, \tilde{I})xL(\xi_i, \tilde{I})^*\Big|_{\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{b}}}$$

Similarly, we define

$$\begin{aligned} \varpi_{\tilde{I}}^R : \mathcal{B}'_{\text{op}(j \rightarrow l)}(\tilde{I}) &\rightarrow \mathfrak{L}(\mathcal{H}_{\text{op}(b,j)}, \mathcal{H}_{\text{op}(b,l)}) \\ B &\mapsto L(\phi, \tilde{I})BL(\phi, \tilde{I})^*|_{\mathcal{H}_b \boxtimes \mathcal{H}_{\bar{j}}} \end{aligned}$$

where $\phi \in \mathcal{H}_b(I')$ is I' -unitary. For each $x \in \mathcal{A}(I)$ and each I -unitary $\eta_{\bar{j}} \in \mathcal{H}_{\bar{j}}(I), \eta_{\bar{l}} \in \mathcal{H}_{\bar{l}}(I)$, one checks

$$\varpi_{\tilde{I}}^R(R(\eta_{\bar{l}}, \tilde{I})xR(\eta_{\bar{j}}, \tilde{I})^*|_{\mathcal{H}_{\bar{j}}})\Big|_{\mathcal{H}_b \boxtimes \mathcal{H}_{\bar{j}}} = R(\eta_{\bar{l}}, \tilde{I})xR(\eta_{\bar{j}}, \tilde{I})^*\Big|_{\mathcal{H}_b \boxtimes \mathcal{H}_{\bar{j}}}$$

The following trick allows us to generalize directly many properties in Sec. 3.2 and 3.3 to $\mathcal{B}_{\text{op}(i \rightarrow k)}$ and $\mathcal{B}_{\text{op}(j \rightarrow l)}$: By (1.31), the fusion product of two isometric morphisms of $\mathcal{A}$ -modules is isometric. Therefore, we may view $\mathcal{H}_i \boxtimes \mathcal{H}_{\bar{j}}$ and $\mathcal{H}_k \boxtimes \mathcal{H}_{\bar{l}}$ as $\mathcal{A}$ -submodules of $\mathcal{H}_{i \oplus k} \boxtimes \mathcal{H}_{\bar{j} \oplus \bar{l}}$. By viewing

$$\mathcal{B}_{\text{op}(i \rightarrow k)}(\tilde{I}) \subset \mathcal{B}_{\text{op}(i \oplus k)}(\tilde{I}) \quad \mathcal{B}'_{\text{op}(j \rightarrow l)}(\tilde{I}) \subset \mathcal{B}'_{\text{op}(j \oplus l)}(\tilde{I})$$

the representation $\varpi_{\tilde{I}}^L$ of $\mathcal{B}_{\text{op}(i \oplus k)}(\tilde{I})$ restricts to that of $\mathcal{B}_{\text{op}(i \rightarrow k)}(\tilde{I})$, and the representation $\varpi_{\tilde{I}}^R$ of $\mathcal{B}'_{\text{op}(j \oplus l)}(\tilde{I})$ restricts to that of $\mathcal{B}'_{\text{op}(j \rightarrow l)}(\tilde{I})$. Using this observation, one can for instance generalize the bulk-boundary locality in Thm. 3.20 to $\mathcal{B}_{\text{op}(i \rightarrow k)}$, $\mathcal{B}'_{\text{op}(j \rightarrow l)}$, and the bulk net $\mathcal{B}_{\text{cl}}$.

A Consequences of (1.20) and conditions (1)-(4) of Thm. 1.37

Let $\mathcal{A}$ be a conformal net, and let $G \rightarrow \text{Aut}(\mathcal{A})$ be a group homomorphism.

A.1 G -covariance

We begin the proof of the G -covariance by proving the following special case:

Lemma A.1. *Let $\mathcal{H}_i \in \text{Rep}^G(\mathcal{A})$ and $\phi \in G$. Then for each $\tilde{I} \in \tilde{\mathcal{I}}$ and $\xi \in \mathcal{H}_i(I)$, the following identities hold as bounded linear operators $\mathcal{H}_0 \rightarrow \mathcal{H}_{\phi i}$.*

$$\Gamma_{\phi} L(\xi, \tilde{I})|_{\mathcal{H}_0} = L(\Gamma_{\phi} \xi, \tilde{I})\phi|_{\mathcal{H}_0} \quad \Gamma_{\phi} R(\xi, \tilde{I})|_{\mathcal{H}_0} = R(\Gamma_{\phi} \xi, \tilde{I})\phi|_{\mathcal{H}_0} \quad (\text{A.1})$$

Proof. By Rem. 1.35, the operator $\Gamma_{\phi} L(\xi, \tilde{I})\phi^{-1}|_{\mathcal{H}_0}$ belongs to $\text{Hom}_{\mathcal{A}(\tilde{I}')}(\mathcal{H}_0, \mathcal{H}_i)$, and hence is the unique element in that set sending Ω to

$$\Gamma_{\phi} L(\xi, \tilde{I})\phi^{-1}\Omega = \Gamma_{\phi} L(\xi, \tilde{I})\Omega = \Gamma_{\phi} \xi$$

So it equals $L(\Gamma_{\phi} \xi, \tilde{I})|_{\mathcal{H}_0}$. The second identity can be proved in the same way. $\square$

Remark A.2. In Lem. A.1, the operators $\Gamma_{\phi} L(\xi, \tilde{I})|_{\mathcal{H}_0}$ and $L(\Gamma_{\phi} \xi, \tilde{I})\phi|_{\mathcal{H}_0}$ originally have codomains $\mathcal{H}_{\phi(i \boxtimes 0)}$ and $\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_0$, respectively. To ensure that the first identity in (A.1) holds, we identify $\mathcal{H}_{\phi(i \boxtimes 0)} \simeq \mathcal{H}_{\phi i} \simeq \mathcal{H}_{\phi i \boxtimes 0}$ using the unitors in (1.11d). Similarly, to obtain the second identity in (A.1), we identify $\mathcal{H}_{\phi(0 \boxtimes i)} \simeq \mathcal{H}_{\phi i} \simeq \mathcal{H}_{0 \boxtimes \phi i}$ using the unitors in (1.11c). This observation will be needed in the proofs of (1.11c) and (1.11d) in Subsec. A.2.

Theorem A.3. *For each $\mathcal{H}_i, \mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$ and $\phi \in G$, there is a (necessarily unique) unitary isomorphism of G -twisted $\mathcal{A}$ -modules*

$$\mathbf{D}_{\phi, i, j} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)}$$

satisfying Def. 1.39.

Proof. Step 1. It suffices to construct $\mathbf{D}_{\phi,i,j}$ satisfying the first relation of (1.30), since the second one will follow immediately from Prop. 1.42.

For each $\xi_1, \xi_2 \in \mathcal{H}_i(\tilde{I})$ and $\eta_1, \eta_2 \in \mathcal{H}_j$, by Prop. 1.46,

$$\begin{aligned} \langle L(\Gamma_\phi \xi_2, \tilde{I}) \Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_1, \tilde{I}) \Gamma_\phi \eta_1 \rangle &= \langle \Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_2, \tilde{I})^* L(\Gamma_\phi \xi_1, \tilde{I}) \Gamma_\phi \eta_1 \rangle \\ &= \langle \Gamma_\phi \eta_2 | L(L(\Gamma_\phi \xi_2, \tilde{I})^* \Gamma_\phi \xi_1, \tilde{I}) \Gamma_\phi \eta_1 \rangle \end{aligned}$$

Set $x = L(\Gamma_\phi \xi_2, \tilde{I})^* L(\Gamma_\phi \xi_1, \tilde{I})|_{\mathcal{H}_0}$, which belongs to $\mathcal{A}(I)$. Then the above expression equals

$$\langle \eta_2 | \Gamma_\phi^{-1} L(x \Omega, \tilde{I}) \Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \Gamma_\phi^{-1} \pi_{\tilde{I}}(x) \Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(\phi^{-1} x \phi) \eta_1 \rangle$$

where Prop. 1.44 and (1.7) are used. By Lem. A.1,

$$\phi^{-1} x \phi = (L(\Gamma_\phi \xi_2, \tilde{I}) \phi|_{\mathcal{H}_0})^* (L(\Gamma_\phi \xi_1, \tilde{I}) \phi|_{\mathcal{H}_0}) = L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})|_{\mathcal{H}_0}$$

Thus

$$\langle L(\Gamma_\phi \xi_2, \tilde{I}) \Gamma_\phi \eta_2 | L(\Gamma_\phi \xi_1, \tilde{I}) \Gamma_\phi \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})|_{\mathcal{H}_0}) \eta_1 \rangle$$

A similar (and indeed simpler) computation shows

$$\langle L(\xi_2, \tilde{I}) \eta_2 | L(\xi_1, \tilde{I}) \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})|_{\mathcal{H}_0}) \eta_1 \rangle$$

Thus, by the density of fusion products in Thm. 1.37, there is a unitary map

$$\mathbf{D}_{\phi,i,j}^{\tilde{I}} : \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j} \rightarrow \mathcal{H}_{\phi(i \boxtimes j)} \quad L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta \mapsto \Gamma_\phi L(\xi, \tilde{I}) \eta$$

for each $\xi \in \mathcal{H}_i(\tilde{I}), \eta \in \mathcal{H}_j$.

Step 2. If $\tilde{I}_1, \tilde{I}_2 \in \tilde{\mathcal{J}}$ are close in the sense that there exists $\tilde{I}_0 \in \tilde{\mathcal{J}}$ satisfying $\tilde{I}_0 \subset \tilde{I}_1$ and $\tilde{I}_0 \subset \tilde{I}_2$, then $\mathbf{D}_{\phi,i,j}^{\tilde{I}_1}$ and $\mathbf{D}_{\phi,i,j}^{\tilde{I}_2}$ agree on vectors of the form $L(\mathcal{H}_{\phi i}(I_0), \tilde{I}_0) \mathcal{H}_{\phi j}$, and hence agree on $\mathcal{H}_{\phi i} \boxtimes \mathcal{H}_{\phi j}$ by the density of fusion products. If $\tilde{I}_1$ and $\tilde{I}_2$ are not close, one can connect them by a chain of arg-valued intervals such that each interval in the chain is close to the preceding one. It follows that $\mathbf{D}_{\phi,i,j}^{\tilde{I}_1}$ and $\mathbf{D}_{\phi,i,j}^{\tilde{I}_2}$ also agree in this general case. We have thus proved that $\mathbf{D}_{\phi,i,j}^{\tilde{I}}$ is independent of the choice of $\tilde{I}$, and hence can be denoted by $\mathbf{D}_{\phi,i,j}$.

Step 3. For each $\tilde{J} \in \tilde{\mathcal{J}}$ and $y \in \mathcal{A}(y)$, and for each $\tilde{I}$ anticlockwise to $\tilde{J}$ and each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$, we compute using (1.7) that

$$\begin{aligned} \mathbf{D}_{\phi,i,j} \pi_{\tilde{J}}(y) L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta &= \mathbf{D}_{\phi,i,j} L(\Gamma_\phi \xi, \tilde{I}) \pi_{\tilde{J}}(y) \Gamma_\phi \eta \\ &= \mathbf{D}_{\phi,i,j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \pi_{\tilde{J}}(\phi^{-1} y \phi) \eta = \Gamma_\phi L(\xi, \tilde{I}) \pi_{\tilde{J}}(\phi^{-1} y \phi) \eta \\ &= \Gamma_\phi \pi_{\tilde{J}}(\phi^{-1} y \phi) L(\xi, \tilde{I}) \eta = \pi_{\tilde{J}}(y) \Gamma_\phi L(\xi, \tilde{I}) \eta = \pi_{\tilde{J}}(y) \mathbf{D}_{\phi,i,j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\phi \eta \end{aligned}$$

This proves that $\mathbf{D}_{\phi,i,j}$ is a morphism of G -twisted $\mathcal{A}$ -modules. □

A.2 The compatibility conditions in Remark 1.30

A.2.1 Equations (1.11)

Proof of (1.11a). Choose any $\tilde{I} \in \tilde{\mathcal{J}}, \xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$, we have

$$\begin{aligned} (\mathfrak{T}_\phi(S) \boxtimes \mathfrak{T}_\phi(T))L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\eta &= (\Gamma_\phi S\Gamma_\phi^{-1} \boxtimes \Gamma_\phi T\Gamma_\phi^{-1})L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\eta \\ &= L(\Gamma_\phi S\xi, \tilde{I})\Gamma_\phi T\eta \end{aligned}$$

where Prop. 1.45 is used. Thus, by Def. 1.39,

$$\mathbf{D}_{\phi, \tilde{I}, \tilde{J}}(\mathfrak{T}_\phi(S) \boxtimes \mathfrak{T}_\phi(T))L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\eta = \Gamma_\phi L(S\xi, \tilde{I})T\eta$$

Def. 1.39 and Prop. 1.45 also imply

$$\begin{aligned} \mathfrak{T}_\phi(S \boxtimes T)\mathbf{D}_{\phi, i, j}L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\eta &= \Gamma_\phi(S \boxtimes T)\Gamma_\phi^{-1} \cdot \Gamma_\phi L(\xi, \tilde{I})\eta \\ &= \Gamma_\phi L(S\xi, \tilde{I})T\eta \end{aligned}$$

This proves (1.11a), thanks to the density of fusion products. $\square$

Proof of (1.11b). Choose any $\tilde{I}, \tilde{J} \in \tilde{\mathcal{J}}$ with $\tilde{J}$ clockwise to $\tilde{I}$. Then for each $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j(J), \chi \in \mathcal{H}_k$,

$$\begin{aligned} &\mathbf{D}_{\phi, i, j \boxtimes k}(\mathbf{1} \boxtimes \mathbf{D}_{\phi, j, k})L(\Gamma_\phi\xi, \tilde{I})R(\Gamma_\phi\eta, \tilde{J})\Gamma_\phi\chi \\ &= \mathbf{D}_{\phi, i, j \boxtimes k}L(\Gamma_\phi\xi, \tilde{I})\mathbf{D}_{\phi, j, k}R(\Gamma_\phi\eta, \tilde{J})\Gamma_\phi\chi \\ &= \mathbf{D}_{\phi, i, j \boxtimes k}L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi R(\eta, \tilde{J})\chi = \Gamma_\phi L(\xi, \tilde{I})R(\eta, \tilde{J})\chi \end{aligned}$$

where the naturality in Thm. 1.37 is used. Similarly,

$$\begin{aligned} &\mathbf{D}_{\phi, i \boxtimes j, k}(\mathbf{D}_{\phi, i, j} \boxtimes \mathbf{1})R(\Gamma_\phi\eta, \tilde{J})L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\chi \\ &= \mathbf{D}_{\phi, i \boxtimes j, k}R(\Gamma_\phi\eta, \tilde{J})\mathbf{D}_{\phi, i, j}L(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\chi \\ &= \mathbf{D}_{\phi, i \boxtimes j, k}R(\Gamma_\phi\eta, \tilde{J})\Gamma_\phi L(\xi, \tilde{I})\chi = \Gamma_\phi R(\eta, \tilde{J})L(\xi, \tilde{I})\chi \end{aligned}$$

This finishes the proof, thanks to the locality and the density of fusion products in Thm. 1.37. $\square$

Proof of (1.11c). For each $\tilde{I} \in \tilde{\mathcal{J}}, \xi \in \mathcal{H}_i, \chi \in \mathcal{H}_0$, we compute

$$(\mathbf{B}_\phi \boxtimes \mathbf{1})R(\Gamma_\phi\xi, \tilde{I})\phi\chi = R(\Gamma_\phi\xi, \tilde{I})\mathbf{B}_\phi\phi\chi = R(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\chi$$

where the naturality in Thm. 1.37 and the definition of $\mathbf{B}_\phi$ are used. By Def. 1.39,

$$\mathbf{D}_{\phi, 0, i}(\mathbf{B}_\phi \boxtimes \mathbf{1})R(\Gamma_\phi\xi, \tilde{I})\phi\chi = \mathbf{D}_{\phi, 0, i}R(\Gamma_\phi\xi, \tilde{I})\Gamma_\phi\chi = \Gamma_\phi R(\xi, \tilde{I})\chi$$

Identify $\mathcal{H}_0 \boxtimes \mathcal{H}_{\phi i} \simeq \mathcal{H}_{\phi i} \simeq \mathcal{H}_{\phi(0 \boxtimes i)}$ using the unitors. Then, by Lem. A.1 and Rem. A.2, we have

$$\mathbf{D}_{\phi, 0, i}(\mathbf{B}_\phi \boxtimes \mathbf{1})R(\Gamma_\phi\xi, \tilde{I})\phi\chi = R(\Gamma_\phi\xi, \tilde{I})\phi\chi$$

By the density of fusion products, $\mathbf{D}_{\phi, 0, i}(\mathbf{B}_\phi \boxtimes \mathbf{1})$ equals the identity. $\square$

Proof of (1.11d). This is similar to the proof of (1.11c), and hence is omitted. $\square$

A.2.2 Equations (1.12)

Proof of (1.12a). Choose any $\tilde{I} \in \tilde{\mathcal{J}}$ and $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$. By the definition of $\mathbf{A}$ (in Rem. A.2) and Prop. 1.45,

$$(\mathbf{A}_i \boxtimes \mathbf{A}_j)L(\xi, \tilde{I})\eta = L(\mathbf{A}_i\xi, \tilde{I})\mathbf{A}_j\eta = L(\Gamma_e\xi, \tilde{I})\Gamma_e\eta$$

By Def. 1.39,

$$\mathbf{D}_{e,i,j}(\mathbf{A}_i \boxtimes \mathbf{A}_j)L(\xi, \tilde{I})\eta = \Gamma_e L(\xi, \tilde{I})\eta$$

Since $L(\xi, \tilde{I})\eta \in \mathcal{H}_{i \boxtimes j}$, by the definition of $\mathbf{A}$,

$$\mathbf{A}_{i \boxtimes j}L(\xi, \tilde{I})\eta = \Gamma_e L(\xi, \tilde{I})\eta$$

The proof is thus finished by the density of fusion products in Thm. 1.37. □

Proof of (1.12b). By the definitions of $\mathbf{A}$ and $\mathbf{B}$ in Rem. A.2,

$$\mathbf{B}_e = \Gamma_e \circ e^{-1}|_{\mathcal{H}_0} = \Gamma_e|_{\mathcal{H}_0} = \mathbf{A}_0$$

This finishes the proof. □

A.2.3 Equations (1.13)

Proof of (1.13a). Choose any $\tilde{I} \in \tilde{\mathcal{J}}$ and $\xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$. Then the vector

$$\chi := L(\Gamma_\phi \Gamma_\psi \xi, \tilde{I})\Gamma_\phi \Gamma_\psi \eta$$

belongs to $\mathcal{H}_{\phi(\psi i)} \boxtimes \mathcal{H}_{\phi(\psi j)}$, the top left corner of (1.13a). By the definition of $\mathbf{C}$ in Rem. 1.30-(c), we have

$$\mathbf{C}_{\phi,\psi,i}\Gamma_\phi \Gamma_\psi \xi = \Gamma_{\phi\psi} \xi \quad \mathbf{C}_{\phi,\psi,j}\Gamma_\phi \Gamma_\psi \eta = \Gamma_{\phi\psi} \eta$$

Therefore, by Prop. 1.45, we have

$$(\mathbf{C}_{\phi,\psi,i} \boxtimes \mathbf{C}_{\phi,\psi,j})\chi = L(\Gamma_{\phi\psi} \xi, \tilde{I})\Gamma_{\phi\psi} \eta$$

By Def. 1.39,

$$\mathbf{D}_{\phi\psi,i,j}(\mathbf{C}_{\phi,\psi,i} \boxtimes \mathbf{C}_{\phi,\psi,j})\chi = \Gamma_{\phi\psi} L(\xi, \tilde{I})\eta$$

On the other hand, by Def. 1.39,

$$\mathbf{D}_{\phi,\psi i,\psi j}\chi = \Gamma_\phi L(\Gamma_\psi \xi, \tilde{I})\Gamma_\psi \eta$$

Therefore, since $\mathfrak{T}_\phi(\mathbf{D}_{\psi,i,j}) = \Gamma_\phi \circ \mathbf{D}_{\psi,i,j} \circ \Gamma_\phi^{-1}$, Def. 1.39 implies

$$\mathfrak{T}_\phi(\mathbf{D}_{\psi,i,j})\mathbf{D}_{\phi,\psi i,\psi j}\chi = \Gamma_\phi \circ \Gamma_\psi L(\xi, \tilde{I})\eta$$

By the definition of $\mathbf{C}$,

$$\mathbf{C}_{\phi,\psi,i\boxtimes j}\mathfrak{T}_\phi(\mathbf{D}_{\psi,i,j})\mathbf{D}_{\phi,\psi i,\psi j}\chi = \Gamma_{\phi\psi}L(\xi, \tilde{I})\eta$$

This finishes the proof, thanks to the density of fusion products in Thm. 1.37. $\square$

Proof of (1.13b). Choose any $\chi \in \mathcal{H}_0$. By the definition of $\mathbf{B}$ in Rem. A.2, we have $\mathbf{B}_\phi \cdot \phi\psi\chi = \Gamma_\phi \cdot \psi\chi$, and hence

$$\mathfrak{T}_\phi(\mathbf{B}_\psi) \cdot \mathbf{B}_\phi \cdot \phi\psi\chi = \mathfrak{T}_\phi(\mathbf{B}_\psi) \cdot \Gamma_\phi \cdot \psi\chi = \Gamma_\phi \mathbf{B}_\psi \cdot \psi\chi = \Gamma_\phi \Gamma_\psi \chi$$

By the definitions of $\mathbf{C}$ and $\mathbf{B}$,

$$\mathbf{C}_{\phi,\psi,0} \cdot \mathfrak{T}_\phi(\mathbf{B}_\psi) \cdot \mathbf{B}_\phi \cdot \phi\psi\chi = \mathbf{C}_{\phi,\psi,0} \cdot \Gamma_\phi \Gamma_\psi \chi = \Gamma_{\phi\psi} \chi = \mathbf{B}_{\phi\psi} \cdot \phi\psi\chi$$

This finishes the proof. $\square$

From now on, we always assume Conventions 1.29 and 1.31.

A.3 Braiding

The braiding in $\text{Rep}^G(\mathcal{A})$ is defined to be the unique operation satisfying (5) of Thm. 1.37. Moreover, as shown in [MS26a, Sec. 3], the axioms of a braiding in a G -crossed braided W^* -tensor category can be derived from (5). In this subsection, we explain how (5) follows directly from (1)-(4).

Lemma A.4. Suppose that $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$. Then for each $\tilde{I} \in \tilde{\mathcal{J}}$ and $\xi \in \mathcal{H}_i(I)$, we have

$$L(\xi, \tilde{I})|_{\mathcal{H}_0} = R(\xi, \tilde{I})\phi|_{\mathcal{H}_0}$$

Proof. This is a direct reformulation of Lem. 1.34. $\square$

Theorem A.5. For each $\phi \in G$, $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$, $\mathcal{H}_j \in \text{Rep}^G(\mathcal{A})$, there is a unique unitary map (which is also a G -twisted $\mathcal{A}$ -module isomorphism) $\mathbb{B}_{i,j} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_j$ such that

$$\mathbb{B}_{i,j}L(\xi, \tilde{I})\eta = R(\xi, \tilde{I})\Gamma_\phi\eta \quad \text{for each } \xi \in \mathcal{H}_i(I), \eta \in \mathcal{H}_j$$

Proof. The uniqueness is clear from the density of fusion products in Thm. 1.37. By the computation in the proof of Thm. A.3, for each $\xi_1, \xi_2 \in \mathcal{H}_i(I)$ and $\eta_1, \eta_2 \in \mathcal{H}_j$, we have

$$\langle L(\xi_2, \tilde{I})\eta_2 | L(\xi_1, \tilde{I})\eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I})|_{\mathcal{H}_0}) \eta_1 \rangle \quad (\text{A.2a})$$

$$\langle R(\xi_2, \tilde{I})\eta_2 | R(\xi_1, \tilde{I})\eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(R(\xi_2, \tilde{I})^* R(\xi_1, \tilde{I})|_{\mathcal{H}_0}) \eta_1 \rangle \quad (\text{A.2b})$$

Therefore, by (1.7) and Lem. A.4,

$$\begin{aligned} \langle R(\xi_2, \tilde{I})\Gamma_\phi\eta_2 | R(\xi_1, \tilde{I})\Gamma_\phi\eta_1 \rangle &= \langle \eta_2 | \Gamma_\phi^{-1} \pi_{\tilde{I}}(R(\xi_2, \tilde{I})^* R(\xi_1, \tilde{I}) |_{\mathcal{H}_0}) \Gamma_\phi \eta_1 \rangle \\ &= \langle \eta_2 | \pi_{\tilde{I}}(\phi^{-1} R(\xi_2, \tilde{I})^* R(\xi_1, \tilde{I}) \phi |_{\mathcal{H}_0}) \eta_1 \rangle = \langle \eta_2 | \pi_{\tilde{I}}(L(\xi_2, \tilde{I})^* L(\xi_1, \tilde{I}) |_{\mathcal{H}_0}) \eta_1 \rangle \\ &= \langle L(\xi_2, \tilde{I})\eta_2 | L(\xi_1, \tilde{I})\eta_1 \rangle \end{aligned}$$

Therefore, there is a unitary map $\mathbb{B}_{i,j}^{\tilde{I}} : \mathcal{H}_i \boxtimes \mathcal{H}_j \rightarrow \mathcal{H}_{\phi i} \boxtimes \mathcal{H}_j$ sending each $L(\xi, \tilde{I})\eta$ to $R(\xi, \tilde{I})\Gamma_\phi\eta$. As in Step 2 of the proof of Thm. A.3, this map is independent of $\tilde{I}$, and hence can be written as $\mathbb{B}_{i,j}$.

For each $x \in \mathcal{A}(I)$, by Prop. 1.44 and 1.46 we have

$$\begin{aligned} \mathbb{B}_{i,j} \pi_{\tilde{I}}(x) L(\xi, \tilde{I})\eta &= \mathbb{B}_{i,j} L(\pi_{\tilde{I}}(x)\xi, \tilde{I})\eta = R(\pi_{\tilde{I}}(x)\xi, \tilde{I})\Gamma_\phi\eta \\ &= \pi_{\tilde{I}}(x) R(\xi, \tilde{I})\Gamma_\phi\eta = \pi_{\tilde{I}}(x) \mathbb{B}_{i,j} L(\xi, \tilde{I})\eta \end{aligned}$$

This proves that $\mathbb{B}_{i,j}$ is a G -twisted $\mathcal{A}$ -module morphism. □

A.4 Conformal covariance

Proof of Thm. 1.37-(6). Fix $\tilde{I} \in \tilde{\mathcal{J}}$. By Lem. 1.18, it suffices to prove (1.25) when $\xi \in \mathcal{H}_i(I)$ and $g \in \mathcal{G}_{\mathcal{A}}(J)$, where $\tilde{J} \in \tilde{\mathcal{J}}$ is such that $\tilde{I}$ and $\tilde{J}$ can be covered by some $\tilde{K} \in \tilde{\mathcal{J}}$. By Rem. 1.35, we have

$$U(g)L(\xi, \tilde{I})U(g)^{-1}|_{\mathcal{H}_0} \in \text{Hom}_{\mathcal{A}(g\tilde{I})}(\mathcal{H}_0, \mathcal{H}_i)$$

and hence

$$g\xi g^{-1} := U(g)L(\xi, \tilde{I})U(g)^{-1}\Omega \in \mathcal{H}_i(gI)$$

By Thm. 1.17 and Prop. 1.44, we have

$$U(g) = \pi_{\tilde{K}}(U_0(g)) = L(U_0(g)\Omega, \tilde{K})$$

when acting on any object of $\text{Rep}^G(\mathcal{A})$. Therefore, by Prop. 1.46 and isotony (Cor. 1.43),

$$\begin{aligned} U(g)L(\xi, \tilde{I})U(g)^{-1} &= L(U_0(g)\Omega, \tilde{K})L(\xi, \tilde{K})L(U_0(g)^{-1}\Omega, \tilde{K}) \\ &= L(U_0(g)\Omega, \tilde{K})L(L(\xi, \tilde{K})U_0(g)^{-1}\Omega, \tilde{K}) = L(L(U_0(g)\Omega, \tilde{K})L(\xi, \tilde{K})U_0(g)^{-1}\Omega, \tilde{K}) \\ &= L(U(g)L(\xi, \tilde{K})U(g)^{-1}\Omega, \tilde{K}) = L(g\xi g^{-1}\Omega, \tilde{K}) \end{aligned}$$

This proves the first identity in (1.25). The second identity follows by a similar argument. □

A.5 Balancing

Proposition A.6. *Let $\phi, \omega \in G$, $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$, $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, $\tilde{I} \in \tilde{\mathcal{J}}$, $\xi \in \mathcal{H}_i(I)$, and $\eta \in \mathcal{H}_j$. Then, as elements of $\mathcal{H}_{\phi\omega i} \boxtimes \mathcal{H}_{\phi\omega j}$, we have*

$$\Gamma_{\phi\omega} L(\xi, \varrho(2\pi)\tilde{I})\eta = \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\omega \eta$$

Proof. Let $\tilde{I}_1 = \varrho(2\pi)\tilde{I}$. Assume without loss of generality that $\eta \in \mathcal{H}_j(I') = \mathcal{H}_j(I)$. Then by Prop. 1.42 and the braiding in Thm. 1.37, and noting $\Gamma_\phi \Gamma_\omega = \Gamma_{\phi\omega}$,

$$\begin{aligned} \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j} L(\Gamma_\phi \xi, \tilde{I}) \Gamma_\omega \eta &= \mathbb{B}_{\phi\omega j, \phi i} R(\Gamma_\phi \xi, \tilde{I}) \Gamma_{\phi\omega} \eta = \mathbb{B}_{\phi\omega j, \phi i} R(\Gamma_\phi \xi, \tilde{I}_1'') \Gamma_{\phi\omega} \eta \\ &= \mathbb{B}_{\phi\omega j, \phi i} L(\Gamma_{\phi\omega} \eta, \tilde{I}_1') \Gamma_\phi \xi = R(\Gamma_{\phi\omega} \eta, \tilde{I}_1') \Gamma_{\phi\omega\phi^{-1}} \Gamma_\phi \xi = R(\Gamma_{\phi\omega} \eta, \tilde{I}_1') \Gamma_{\phi\omega} \xi \\ &= L(\Gamma_{\phi\omega} \xi, \tilde{I}_1) \Gamma_{\phi\omega} \eta = \Gamma_{\phi\omega} L(\xi, \varrho(2\pi)\tilde{I})\eta \end{aligned}$$

where the G -covariance in Thm. 1.37 is used for the last equality. $\square$

Recall that for each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$, the balancing $\vartheta_i \in \text{Hom}_{\mathcal{A}}(\mathcal{H}_i, \mathcal{H}_{\phi i})$ is defined by $\Gamma_\phi \circ U_i(\varrho_{\mathcal{A}}(2\pi))$.

Theorem A.7. *For each $\mathcal{H}_i \in \text{Rep}^\phi(\mathcal{A})$ and $\mathcal{H}_j \in \text{Rep}^\omega(\mathcal{A})$, we have*

$$\mathfrak{T}_\phi(\vartheta_i) = \vartheta_{\phi i} \quad \vartheta_{i \boxtimes j} = \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j}(\vartheta_i \boxtimes \vartheta_j)$$

Proof. By Rem. 1.26,

$$\mathfrak{T}_\phi(\vartheta_i) = \Gamma_\phi \vartheta_i \Gamma_\phi^{-1}|_{\mathcal{H}_{\phi i}} = \Gamma_\phi \Gamma_\phi U_i(\varrho_{\mathcal{A}}(2\pi)) \Gamma_\phi^{-1}|_{\mathcal{H}_{\phi i}} = \Gamma_\phi U_{\phi i}(\varrho_{\mathcal{A}}(2\pi))|_{\mathcal{H}_{\phi i}} = \vartheta_{\phi i}$$

For each ξ, η as in Prop. A.6, by Prop. 1.45,

$$\begin{aligned} \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j}(\vartheta_i \boxtimes \vartheta_j) L(\xi, \tilde{I})\eta &= \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j} L(\vartheta_i \xi, \tilde{I}) \vartheta_j \eta \\ &= \mathbb{B}_{\phi\omega j, \phi i} \mathbb{B}_{\phi i, \omega j} L(\Gamma_\phi \varrho_{\mathcal{A}}(2\pi) \xi, \tilde{I}) \Gamma_\omega \varrho_{\mathcal{A}}(2\pi) \eta \\ &= \Gamma_{\phi\omega} L(\varrho_{\mathcal{A}}(2\pi) \xi, \varrho(2\pi)\tilde{I}) \varrho(2\pi) \eta = \Gamma_{\phi\omega} \varrho_{\mathcal{A}}(2\pi) L(\xi, \tilde{I}) \eta \end{aligned}$$

where the conformal covariance in Thm. 1.37 is used in the last equality. $\square$

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